

The block below carries no module guard, so docstrip copies it into EVERY generated file. That is what we want for the TeX files and wrong for the latexmkrc, which is Perl and does not understand lines opening with two percent signs – hence the negated guard around it.

The guard lines themselves are hidden behind a false conditional. docstrip reads a guard line either way, but ltxdoc, which typesets this file, would otherwise set the literal text of the guard as a stray paragraph in the body of the manual.

Nothing in this area may name a TeX conditional, not even in prose: outside the meta-comment at the top of the file, every documentation line here IS LaTeX source and is executed, and a conditional named in passing opens or closes a real one. That is not hypothetical – an earlier draft of this very comment swallowed the rest of the manual.

Three conditional primitives are deliberately absent from the lists below: the ones that close or branch a conditional. `\DoNotIndex` scans its argument with TeX’s own conditional machinery live, so a bare one of them closes the wrong conditional. The manual ended its run with “Too many }’s” and exited with status 1, which failed the docs step of the harness even though the PDF had been written. Leaving them out of the lists means they get indexed, which is harmless. Note that they cannot be named here either, not even inside `...` shorthand split across a line.

The coppe document class

Version 4.0

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Abstract

In this work, it is described the `coppe` document class as well as other files distributed by the COPPETEX project. This class is suitable for writing academic dissertations, thesis and qualifying exams according to the formatting rules of the Alberto Luiz Coimbra Institute for Graduate Studies and Research in Engineering. The minimalist set of macro commands allows its users to concentrate most of their efforts on text composition rather than on the document layout.

1 Introduction

Writing documents in L^AT_EX may be a laborious task when the authors have to prepare their manuscripts rigorously respecting formatting rules imposed by publishers. Regardless of difficulty, a lot of thesis presented to the Coordination of Graduate Studies and Research in Engineering of the Federal University of Rio de Janeiro (COPPE/UFRJ) is typeset in L^AT_EX. This demand motivated the creation of the COPPETEX project, which tries to facilitate and encourage the use of L^AT_EX within the COPPE/UFRJ scope.

The `coppe` document class is the main product of COPPETEX. It was designed to be clear and succinct. It enables the creation of dissertations, qualifying exams and thesis in a simple and automatic way. The main goal of the `coppe` class is to maintain authors strictly focused on text composition without worrying about margins sizes, line spacing, paper size, vertical and horizontal alignment, etc. The COPPETEX project comprehends also BibTEX and MakeIndex style files for creating lists of references, symbols and abbreviations. Although there aren't official guidelines to write qualifying exams, we provide this option just for convenience, as this exam is a requisite to obtain the DSc degree and, for some of the programs, the MSc degree.

In which follows, it is described the user interface of the `coppe` class. Some details about using the style files cited above are also given. We use the term *thesis* to generally refer to dissertation, qualifying exam, and thesis itself.

2 License

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copying, distributing or modifying the source code, among other acts covered by this license.

To see the full text of the GNU GPL license, go to the `COPYING` file attached to this package.

3 Support

We maintain a WhatsApp group where users can send questions, comments, and bugs to. More details can be found [here](#).

However, as the project maintenance was transferred to a new repository, bug reports, as well as new feature requests, should be directed to <https://github.com/COPPE-UFRJ/CoppeTeX>.

4 User interface

`\frontmatter` A thesis to be approved by the Academic Registry at COPPE/UFRJ must contain three-parts: *front*, *main* and *back* matters (COPPE/UFRJ Academic Registry guidelines). Each one of these parts is started by calling its corresponding macro `\frontmatter`, `\mainmatter` or `\backmatter`. The front matter of a thesis consists of front cover and face, cataloging page, dedication, acknowledgments, abstracts, table of contents, and lists of tables, algorithms, symbols and abbreviations. The main matter is just composed by chapters, while the back matter usually consists of bibliographic references, appendices and index.

You must invoke the `\frontmatter` macro immediately after the `\maketitle` one. The `\mainmatter` command comes right before the first chapter, and `\backmatter` must be typed before the list of references.

Cover (capa)

The ABNT cover suggested in the manual's Annex A. The `\maketitle` command builds it automatically, ahead of the title page: the university, the full COPPE name, and the graduate program (taken from `\department`), followed by the author, the title, and the city and year, all centred and in uppercase. The UFRJ and COPPE logos head the page. The capa is not counted.

Front face

The front face is unnumbered. There, it is not allowed to use hyphenation (COPPE/UFRJ Academic Registry guidelines). It is constructed by calling `\maketitle`. Next, it is described the commands used to enter the information required to create it.

`\author` The `\author` command was redefined. Here, it takes two arguments: the author's first names and surname, e.g., `\author{First Names}{Surname}`. The words should be typed with only first letters in uppercase.

`\title` The macros `\title` and `\foreigntitle` are used to enter the titles of your monograph in Brazilian Portuguese and English respectively (the two built-in languages of CoppeTeX 4.x). For documents whose main language is neither

brazilian nor english – e.g. spanish, french, italian, or a user-supplied language –, also call `\titlein{<lang>}{<text>}` to register the title in that language; the cover and folha de rosto print whichever title matches the main language declared in the class option. See the section “Multilingual support” below for the complete picture. The `babel` package is loaded automatically by `coppe.cls`, so you do not need to load it yourself.

`\advisor` Every COPPE student is coordinated by at least one advisor. M.Sc. and D.Sc. `\examiner` students can have at most 2 and 3 advisors, respectively. Their names must be provided by issuing the command `\advisor` as below:

```
\advisor{Title}{Advisor's Name}{Surname}{Degree}
\advisor{Title}{Second Advisor's Name}{Surname}{Degree}
\advisor{Title}{Third Advisor's Name}{Surname}{Degree}
```

The advisors are not necessarily members of the thesis examination board. Thus, it is required to enter the names of all examiners using the `\examiner` macro. The examiners' names are entered differently:

```
\examiner{Title}{First Examiner's Name Surname}{Degree}
\examiner{Title}{Second Examiner's Name Surname}{Degree}
...
\examiner{Title}{N-th Examiner's Name Surname}{Degree}
```

Remember that all names must be given before calling `\maketitle`.

`\department` The Alberto Luiz Coimbra institute is divided into 12 academic units: Biomedical Engineering (PEB), Civil Engineering (PEC), Electrical Engineering (PEE), Mechanical Engineering (PEM), Metallurgical and Materials Science Engineering (PEMM), Nuclear Engineering (PEN), Ocean Engineering (PENO), Energy Planning (PPE), Production Engineering (PEP), Chemical Engineering (PEQ), Systems Engineering and Computer Science (PESC), and Transportation Engineering (PET). You must specify your department using one of the above abbreviations, e.g., `\department{PEC}`.

`\date` This macro is used to set the month and year of defense. This information is required to create the front face, cataloging details page and abstracts. For example, October 2007 should be entered as `\date{10}{2007}`.

`\keyword` The keywords should describe the concentration areas of your work. You must provide them as follows:

```
\keyword{First Keyword}
\keyword{Second Keyword}
...
\keyword{N-th Keyword}
```

Usually, six words are enough.

Folha adicional: Coleta CAPES

Since August 2026 every thesis and dissertation must carry a *folha adicional* immediately after the folha de rosto, holding the fields the CAPES/MEC collection requires plus the ficha catalogafica (SiBI manual, 9th rev. ed. 2026, 3.1.2.1.2

and Annex H). The sheet is filled in by the graduate programme, and is neither counted nor numbered.

The macros below feed that sheet. They are institutional data, so the sheet is always typeset in Portuguese regardless of the main language of the work, as the COPPE norm requires of institutional identity.

`\areaconcentracao` The area of concentration, e.g. `\areaconcentracao{Engenharia de Sistemas e Computacao}`. It appears on the folha de rosto, on the folha de aprovacao and on the folha adicional.

`\linhapesquisa` The research line, required by the CAPES model of the folha de rosto.

`\tipoproducao` The kind of intellectual output, one of `bibliografica`, `artistica`, `tecnologica` or `tecnica`. Anything else raises a warning.

`\projetovinculado` Whether the work belongs to a research project (`sim` or `nao`) and, if so, its

`\nomeprojeto` name.

`\agenciafomento` A funding agency, given as full name and acronym; repeat once per agency:

```

\agenciafomento{Conselho Nacional de Desenvolvimento
                  Cientifico e Tecnologico}{CNPq}
\agenciafomento{Fundacao Carlos Chagas Filho de Amparo a
                  Pesquisa do Estado do Rio de Janeiro}{FAPERJ}

```

Cataloging details

This page contains cataloging information useful for librarians. Fortunately, it is automatically generated from the data you entered at the time you call `\maketitle`. It is not needed in qualifying exams, though.

Dedication (optional)

`\dedication` This macro was added for convenience. The input text is placed at the right bottom of a blank page. It is emphasized and in normal size.

Abstracts

`abstract (env.)` As stated by the COPPE/UFRJ Academic Registry, abstracts must be in one

`foreignabstract (env.)` page each, with at most 250 words. We recommended that they should be only one paragraph long. They must be defined inside the environments `abstract` and `foreignabstract`.

Lists of symbols and abbreviations (optional)

`\abbrev` The lists of symbols and abbreviations are optional, although highly recommended.

`\syml` It is a good practice to define a symbol/abbreviation in its first occurrence in the text. To define a symbol use `\syml[alphabetic symbol]{Symbol}{Symbol Definition}`, and for abbreviations `\abbrev[alphabetic symbol]{Abbreviation}{Abbreviation Definition}`. These commands are called *dummy*, since they don't output anything at the place they are executed, just an entry in the correspondent list.

`\makeloabbreviations` These lists are lexicographically sorted by using the MakeIndex program, which

`\makelosymbols` is part of any L^AT_EX implementation. For `\syml`, if the optional parameter is

`\printloabbreviations` provided, it will used as sort key. This was later, in 2024, implemented also for

`\printlosymbols`

`\abbrev`, otherwise `Symbol`, or `Abreviation` will be used as sort key, what can result in an undesirable order if it contains $\text{\LaTeX}$ commands, mathematical symbols, or mix of uppercase and lowercase. `MakeIndex` needs two commands to create a final sorted list: one which generates a list of entries and the other that indicates the position where the list will be printed out. To generate the lists of symbols and abbreviations, the `coppe` class provides the commands `\makeloabbreviations` and `\makelosymbols`, respectively. They must be called in the document preamble. The commands `\printlosymbols` and `\printloabbreviations` have to be invoked at the point where you want these lists appear, e.g., following the list of tables as showed in the example. Once you call `latex`, it will be created two files with extensions `abx` and `syx`, which contain `MakeIndex` input data. They must be processed with `makeindex` in order to get the lists correctly produced, redirecting the output to files with extension `lab` and `los` respectively:

```
makeindex -s coppe.ist -o example.lab example.abx
makeindex -s coppe.ist -o example.los example.syx
```

Note the `-s` option for specifying the style `coppe.ist`. Now, rerun `latex` twice to get the references solved and you are done.

References

`COPPE $\text{\TeX}$` produces citations and the reference list with `biblatex` and the `biber` backend, through its own bibliography style implementing the post-2020 ABNT NBR 6023 rules. The class loads this style automatically: there is no `\bibliographystyle` to set and no `BIB $\text{\TeX}$` `.bst` file to choose (the former `coppe-plain` and `coppe-unsrt` `BIB $\text{\TeX}$` styles have been retired).

By default, citations follow the author–date style, e.g., “Chartier (2002)”. Passing the `numbers` class option switches to the numeric style, in which citations are bracketed numbers and the reference list is numbered in order of citation (unsorted).

Declare your bibliography database in the preamble with `\addbibresource` and print the list where it should appear with `\printbibliography`:

```
\addbibresource{example.bib} % in the preamble
...
\printbibliography           % where the list goes
```

Entry types and fields may be written in English (`@book`, `title`, `author`) or with their unaccented Portuguese synonyms (`@livro`, `titulo`, `autor`).

Run in sequence `L $\text{\TeX}$` , `biber`, and twice again `L $\text{\TeX}$` to resolve the citations and references. The style is `natbib` compatible (via the `biblatex natbib=true` option), so you may freely issue `\citet` and `\citep`, as well as the other `natbib` commands.

Reference entry types

Every ABNT document category maps to a `biblatex` entry type, each with an unaccented Portuguese synonym (a worked example of every type is in the sample manuscript’s reference list):

Academic @book (@livro); @article (@artigo); @artigodejornal (newspaper); @incollection (@partedelivro); @inbook (@emlivro); @inreference (@verbete); @proceedings (@anaais); @inproceedings (@trabalhodeevento); @thesis (@tese/@dissertacao, with the presets @dissertacaomestrado, @tesedoutorado, @tesephD, @tcc); @report (@relatorio); @periodical (@periodico); @manual; @misc (@diverso); @online.

Special (NBR 6023 §4.2.5–4.2.14) @patent (@patente); @legislation (@legislacao/@atonormat @jurisdiction (@jurisprudencia, with a relator field); @legal (@documentojuridico); @standard (@norma); @music (@partitura); @map (@mapa); @image (@iconografico); @artwork (@obradearte); @video (@audiovisual); @movie (@filme); @letter (@correspondencia); @dataset (@conjuntodedados).

Games and TV @game (@jogo); @boardgame (@jogodetabuleiro); @videogame (@jogoeletronico); @tvshow (@programadetv).

When an entry has no author or editor, it is entered by title, with the first significant word (and a leading article) in capitals.

Appendix and Annex (Optional)

`\appendix` Appendices and annexes are optional chapters that are part of the back matter.

`\annex` The `\appendix` command is a standard L^AT_EX command used before all appendices. COPPET_{EX} introduces the `\annex` command, which should be used only in the back matter (i.e., after the `\backmatter` command) and only after all existing `\appendix` chapters. This restriction is due to implementation constraints.

Therefore, the order for backmatter is:

```
\backmatter
\printbibliography
\appendix
\chapter{An Appendix}
\chapter{Another Appendix}
\annex
\chapter{First Annex}
\chapter{Second Annex}
```

Annex was introduced in v3.5

Final colophon page (`\coppetexfinalpage`)

`\coppetexfinalpage` Optional closing page modelled on the printer's *colophon*: it records, in Portuguese, the tooling and edition used to typeset the thesis. Call it once, after every other piece of content, immediately before `\end{document}`. The macro

- forces a verso (even) page, so in a two-sided print the colophon sits opposite the rear cover;
- types the body at the foot of the page (`\vfill`);
- renders the text in Portuguese unconditionally (Brazilian babel), even when the thesis main language is English, Spanish, French or Italian — the colophon is institutional metadata, not body content.

Three slot macros are user-overridable. Redefine them with `\renewcommand` before `\coppetexfinalpage` if the document deviates from the defaults:

`\coppefinalengine` the T_EX engine identifier. Default: auto-detected via `iftex` (`\hologo{pdfLaTeX}`, `\hologo{LuaLaTeX}` or `\hologo{XeLaTeX}`).

`\coppefinalfont` the text body font. Default: *Latin Modern* (`\lmodern`), which is what the class loads. Redefine if you loaded `mathptmx`, an OpenType math font, etc.

`\coppefinalmanual` the reference manual title (typeset in italics). Default: the current COPPE/UFRJ norm. Redefine if your work follows an earlier edition.

The macro also prints the COPPE_TE_X class version (captured into `\coppe@version` at class-load time, so it survives any later package that may redefine the global `\fileversion`) and the GitHub repository URL, <https://github.com/COPPE-UFRJ/CoppeTeX>.

5 Class options

There are some options users can specify in order to customize the appearance of the output produced by the `coppe` class. These options can be passed to `coppe` as follows: `\documentclass[option1, option2]{coppe}`. In which follows, we give a brief description of all supported options.

dsc, msc, dsceexam, msceexam The `coppe` class is able to produce thesis, dissertations, and qualifying exams, which are enabled by the `dsc`, `msc`, `msceexam`, and `dsceexam` options, respectively.

doublespacing The default line spacing is one-and-a-half. For enabling double spacing between lines, use the `doublespacing` option.

numbers The default citation style is the author–date scheme. For numbered citations, specify the `numbers` option to the `coppe` class; the reference list is then numbered in order of citation (unsorted). The matching `biblatex` style is selected automatically in either case.

brazilian, english, spanish, french, italian Pick the document’s *main* language. The class loads Babel with that language as the main one (the last in the Babel option list); inside the document `\selectlanguage{...}` and `\begin{otherlanguage}{...}` keep working as usual. The default, when no language option is given, is **brazilian**.

- **brazilian** and **english** are *built-in*: their strings ship inside `coppe.cls` and need no extra file.
- **spanish, french, italian** are shipped as *language packs*: two files per language (`coppe-lang-<lang>.def` and `<lang>-coppe.lbx`) that must sit next to `coppe.cls` (or in your project directory). They are auto-loaded by the class. Picking one of them as the main language without the matching pack files installed raises a clear class error.

- Any other Babel-known language can be plugged in by writing the same two files and either passing the language name as a class option or calling `\usecoppelanguage{<lang>}` in the preamble.

See the section “Multilingual support” below for the three-slot architecture, the `\titlein / \brazilianabstract` API, and how to create a new language pack.

pdfa Produce the deposited file as **PDF/A-2b**, which 2.2(d) of the manual requires of the digital version. The class loads `pdfx` at level **a-2b** and writes the XMP metadata from the document’s own `\title`, `\author`, `\keyword` and `\department`. Level **a-2b** rather than **a-1b**: **a-1b** forbids transparency, which `tcolorbox` and several graphics packages produce. The metadata reaches the PDF on the SECOND run, because `pdfx` reads it from a file the class writes at the end of the first one – which is no burden, since three runs are needed for cross-references anyway.

assinaturas Give every member of the board a signature rule of its own on the folha de aprovacao, as Annex D of the manual shows, instead of the compact list. The spacing adapts to the size of the board.

coorientador List the coadvisors on the three abstract pages, under the advisors, in the same shape the folha de rosto uses. Off by default: the abstract page listing only the advisors is a COPPE tradition, and 3.1.2.1.4 does not require either form.

rascunhoficha While writing, fill the folha adicional with a ficha catalografica typeset by the class itself, so the page is not an empty frame. **Never valid for deposit:** the real ficha comes from the SiBI generator or the Programa’s library, and goes in with `\fichacatalografica`.

twoside, oneside The 2026 edition dropped the verso margins and the deposit is digital, so the class sets one-sided pages with a 3 cm left margin on every folha. **twoside** restores the mirrored layout for anyone who is going to PRINT and bind the work; **oneside** is the default and is there to be explicit.

semmorewrites, morewrites T_EX has sixteen output streams and a work that asks for every list runs out of them; the failure, “No room for a new `\write`”, lands on whichever list came last and explains nothing. Under pdfT_EX the class therefore loads the `morewrites` package by itself when it is installed, which removes the ceiling. **semmorewrites** turns that off, for the rare case where the package disagrees with something else in the document – it buffers what is written until shipout, so a package that writes a file and reads it back in the same run can misbehave. **morewrites** means “I insist”: a missing package becomes an error instead of a warning. Under LuaLaT_EX, which has 128 streams, none of this is loaded and none of it is needed.

listasnosumario Put the pre-textual lists – figures, tables, quadros, programas, abreviaturas, simbolos – back into the sumario. **Contrary to 3.1.2.1.6**, which tells the author to follow the manual’s own sumario, and that one opens at “1 INTRODUCAO”: the lists are pre-textual elements and come BEFORE the sumario, so listing them inside it is listing what the reader

has already passed. The option exists because every CoppeTeX document written before this release had them there, and a work in its final weeks should not have its sumario change under it.

numeraisromanos Number the pre-textual folhas in lowercase roman numerals.

Contrary to 2.7, which has the count running from the folha de rosto with nothing printed until the textual part. Kept only so that documents written under the older norm still compile; do not use it for a new work.

5.1 Changing document identification

\freeconfig The user could *optionally* use the command **freeconfig** to modify the parameters that print the document identification. The command **freeconfig** needs all those parameters, which are degree initials, degree name, title, foreign title, local doctype, and foreign doctype as in the following example:

```
\freeconfig{Dr.}{Philosophiae Doctor}{PhD}{Doutor}{Dissertation}{Tese}
```

5.2 Multilingual support

CoppeTeX 4.x reserves three fixed language slots:

main the language of the body, captions, TOC labels and the **abstract** environment. Picked by the class option (**brazilian** by default; one of **english**, **spanish**, **french**, **italian** also accepted out of the box; any other Babel language possible with a user-supplied pack).

foreign the language of the **foreignabstract** environment. **english** by default for every main language; switches automatically to **brazilian** when **main = english**.

third optional. Loaded on demand with **\usecoppelanguage{<lang>}** in the preamble, useful when the body quotes or cites in a third language (e.g. a Spanish-main thesis citing a French source). Does not have a class option of its own.

\titlein
brazilianabstract (env.)
\usecoppelanguage

Two pieces of user-facing API come with the multilingual model:

- **\titlein{<lang>}{<text>}** records the title in any language. **\title{...}** still writes the Brazilian-Portuguese title and **\foreigntitle{...}** still writes the English one (back-compat); for a main language outside the **brazilian/english** pair, also call **\titlein{<lang>}{...}**. The cover and folha de rosto print whichever title matches **\coppe@mainlang**.
- The **brazilianabstract** environment is the optional third abstract used when the main language is neither **brazilian** nor **english** – typically a Spanish/French/Italian thesis that also wants a Portuguese resumo for UFRJ examiners. It wraps its body in **\begin{otherlanguage}{brazilian}** and uses the Portuguese template strings. A document with **main = brazilian** can simply keep using **abstract**; **brazilianabstract** is redundant then.
- **\usecoppelanguage{<lang>}** loads an extra language pack in the preamble without making it the main or foreign language.

Institutional names stay Portuguese

The cover, folha de rosto and ficha catalográfica are a Brazilian institutional template (NBR 14724 + COPPE manual). The strings `universityname`, `cityname`, `statename`, `countryname`, the departments declared with `\department` and the surrounding “apresentada/submetida ao...” wording are deliberately kept in Portuguese for every main language. Only the title (printed via `\coppe@selecttitle`) follows the main language.

Writing a new language pack

A language pack is just two files dropped next to `coppe.cls` (or anywhere on the TEXINPUTS path):

`coppe-lang-<lang>.def` class-level strings. Populate the dispatcher table with `\copperdefstring{<lang>}{<key>}{<value>}` for every key already populated for `brazilian/english` in `coppe.cls` (`appendix`, `annex`, `frame`, `listframe`, `source`, `program`, `listprogram`, `references`, `in:`, `art`, `platform`, `directedby`, `producedby`, `listabbreviation`, `listsymbol`, `glossary`, `advisor`, `advisors`, `dept`, `approvedmale`, `approvedfemale`, `universityname`, `cityname`, `statename`, `countryname`, `monthname1...monthname12`, `algorithm2eopt`, `babelname`). Finish with

```
\AtBeginDocument{\DeclareLanguageMapping{<lang>}{<lang>-coppe}}
```

so the `biblatex` mapping is registered after the bibliography engine is loaded by `coppe.cls` (which happens *after* the language pack is read).

`<lang>-coppe.lbx` `biblatex` localization strings. `\InheritBibliographyExtras{<lang>}` then `\DeclareBibliographyStrings` for the keys `availablefrom`, `urlseen`, `in`, `editor`, `editors`, `depositor`, `coppescale`, `mscdiss`, `dscthesis`, `phdthesis`, `tcc`. The shipped `spanish-coppe.lbx`, `french-coppe.lbx` and `italian-coppe.lbx` are the reference templates to copy from.

Tip: the `algorithm2eopt` key holds whatever `algorithm2e` calls the language. Mind the spelling: `algorithm2e` ships `portuguese`, `english`, `spanish`, `french` and `italiano` – not `italian`.

Example: a Spanish-main thesis

```
\documentclass[spanish,dsc]{coppe}
\addbibresource{example.bib}
\begin{document}
  \title{T'\i tulo em portugu\~es} % Brazilian abstract slot
  \foreigntitle{English title} % English abstract slot
  \titlein{spanish}{T'\i tulo en espa\~nol} % cover/folha
  \author{Juan}{P\~erez}
  \advisor{Prof.}{Mar'\i a}{Garc'\i a}{D.Sc.}
  \department{PESC}\date{01}{2026}
  \keyword{multilenguaje}
  \maketitle
  \frontmatter
```

```

\begin{abstract}          Resumen principal en espa\~nol.      \end{abstract}
\begin{foreignabstract} English foreign abstract.              \end{foreignabstract}
\begin{brazilianabstract} Resumo em portugu\~es para a banca. \end{brazilianabstract}
\tableofcontents
\mainmatter
\chapter{Introducci\~on} Cuerpo en espa\~nol\dots
\end{document}

```

6 Files in your project folder

In a normal install where CoppeTeX is part of your `TEXINPUTS` path (e.g. inside the local `texmf` tree of MiKTeX or TeX Live), you do not need to think about this list at all — the files travel with the class. This section is for two situations where you do:

- shipping the class together with your manuscript (a self-contained submission package); or
- installing CoppeTeX by hand, dropping the files next to `coppe.cls` in the project directory.

Always required

These files must be findable at every compilation, regardless of the main language or any class option:

`coppe.cls` the document class itself.

`coppe.dbx` custom `biblatex` datamodel (declares the ABNT-specific entry types `@game`, `@videogame`, `@boardgame`, `@tvshow` and `@map`, plus a few custom fields).

`coppe.bbx` and `coppe.cbx` `biblatex` bibliography and citation styles for the author–date scheme (the default).

`brazilian-coppe.lbx` and `english-coppe.lbx` `biblatex` localisation tables for Portuguese and English. Both are required *even when neither language is the main one*, because `coppe.bbx` registers

```

\DeclareLanguageMapping{brazilian}{brazilian-coppe}
\DeclareLanguageMapping{english}{english-coppe}

```

unconditionally, so `biblatex` tries to read both LBX files at every compilation.

Conditional on class options

`coppe-numeric.bbx` and `coppe-numeric.cbx` only when the `numbers` class option is given (`\documentclass[numbers]{coppe}`). Without it, `biblatex` never tries to read them.

`coppe.ist` only when the glossary, list-of-symbols, or list-of-abbreviations features are used. It is the `makeindex` style file invoked by the `makeindex -s coppe.ist ...` step described in subsection *Lists of symbols and abbreviations* above.

Conditional on the main language

The main language is chosen by the matching class option (`brazilian`, `english`, `spanish`, `french`, `italian`). The two built-in languages ship their strings inside `coppe.cls` and so need no extra file. The three language packs add two files each — one class-level (`coppe-lang-<lang>.def`) and one `biblatex`-level (`<lang>-coppe.lbx`):

Main language	Extra files needed
<code>brazilian</code> (default)	— (built into <code>coppe.cls</code>)
<code>english</code>	— (built into <code>coppe.cls</code>)
<code>spanish</code>	<code>coppe-lang-spanish.def</code> , <code>spanish-coppe.lbx</code>
<code>french</code>	<code>coppe-lang-french.def</code> , <code>french-coppe.lbx</code>
<code>italian</code>	<code>coppe-lang-italian.def</code> , <code>italian-coppe.lbx</code>

Loading a third language at preamble time via `\usecoppelanguage{<lang>}` brings in the same pair of files for that language. A user-supplied language pack follows the same scheme: drop `coppe-lang-<lang>.def` and `<lang>-coppe.lbx` next to `coppe.cls`.

Cover assets

The institutional title page produced by `\maketitle` needs two PDF logo files at compile time:

`ufrj-logo.pdf` and `coppe-logo.pdf` the UFRJ shield and the COPPE/UFRJ wordmark. Both ship with the CoppeTeX distribution; they must be findable from the project directory (or on `TEXINPUTS`).

7 Required L^AT_EX packages

The class is built on the standard `book` class (`12pt`, `a4paper`, `oneside`) and brings in the stock CTAN packages listed below with `\RequirePackage`. On a full T_EX Live or MiK_TE_X install they are all present already, so this list matters only for a minimal install — or when you simply want to know which commands are already in scope. You do *not* need to `\usepackage` any of these yourself; every command and environment they provide is available in your manuscript. Options in brackets are the ones the class passes.

Engine and encoding

`iftex` detects the running engine, so the class can branch its setup between `pdfTEX` and the Unicode engines (LuaT_EX/XeT_EX).

`inputenc (utf8)` UTF-8 input; loaded *only under pdfT_EX* (the Unicode engines read UTF-8 natively).

`fontenc (T1)` T1 font encoding, for correctly composed accented glyphs and hyphenation.

Mathematics and symbols

`amsmath` the AMS mathematics environments.

`amssymb` the AMS symbol fonts; loaded *only under pdfTeX*. Under a Unicode engine load `unicode-math` (or `amssymb`) yourself — the class warns at end of document if neither is present.

Page layout and spacing

`geometry` A4 one-sided margins (3/3/2/2 cm) and header height.

`fancyhdr` running headers and footers.

`titlesec` chapter and section title formatting.

`setspace` one-and-a-half line spacing (double with the `doublespacing` option).

Lists, tables and rules

`enumitem` customisable list spacing and labels.

`tabularx` tables with automatic-width columns.

`booktabs` professional horizontal rules (`\toprule`, `\midrule`, `\bottomrule`).

`longtable` tables that break across pages.

Floats, graphics and captions

`graphicx` image inclusion.

`float` float placement control (e.g. the `[H]` specifier).

`caption` caption typography (the ABNT caption-on-top layout).

`subcaption` subfigures and subtables.

Code, verbatim and algorithms

`listings` source-code listings.

`fancyvrb` extended verbatim.

`algorithm2e` (`linesnumbered`, `lined`, `algochapter`, `ruled`) typeset algorithms; the `natural-language` option tracks the main language.

Bibliography, quotations and language

`babel` multilingual typesetting; the language options are computed from the main and foreign language slots.

`csquotes` context-sensitive quotation marks (the recommended companion of `biblatex`).

`biblatex` (`backend=biber`) the bibliography engine and the ABNT `coppe` styles (`coppe` or, with the `numbers` option, `coppe-numeric`). The backend is `biber`, not `bibTeX`.

Hyperlinks

`hyperref` clickable cross-references, PDF bookmarks and document metadata.

Low-level utilities

`etoolbox` e-TeX programming toolkit.

`ltxcmds` low-level command helpers (`\ltx@ifpackageloaded, ...`).

`ifthen` the `\ifthenelse` conditional.

`hyphenat` hyphenation control (`\nohyphens`, used on the cover).

`lastpage` `\pageref{LastPage}` for the cataloging page count.

`hologo` TeX-family logos (`\hologo{LaTeX}, ...`).

`xcolor` colour support (used by `listings` and `hyperref`).

8 Quick, useful tips

Pictures. The default picture format of L^AT_EX is the Encapsulated PostScript (EPS). If you use pdfL^AT_EX, the default format becomes the PDF, but you can equally load PNG files. For such, you must enter the name of your image file without extension, e.g., `\includegraphics{filename}`, and pdfL^AT_EX will firstly look for a file called `filename.pdf` and after for file `filename.png`. For producing high quality pictures with embedded fonts we recommend the Ipe drawing software available [here](#).

Fonts. The default font in L^AT_EX is the Computer Modern. If you would like to try its enhanced version, consider using the `lmodern` package. To use Times, it is recommended to load the package `mathptmx`, rather than the deprecated `times`. There is also an enhanced Times version available with the `tgtermes` package. You can still use the Arial font face with the package `uarial`.

Hyperref. When working with PDF's, there is the possibility to add extra information to the file as the author's name, document title, subject, keywords, etc. This is easily done with the `hyperref` package, which is loaded by the `coppe` class itself (since v4.0): the user must *not* load it again in the manuscript preamble. Author, title, subject and keyword metadata are filled in automatically by `\maketitle`.

Math (amssymb vs. unicode-math). The class always loads `amsmath`. The companion AMS symbol package `amssymb`, however, is loaded *only* under pdfL^AT_EX: it is a legacy 8-bit package built around the `msam/msbm` Type 1 fonts and clashes with `unicode-math`, the modern OpenType-math package used on Unicode engines (LuaL^AT_EX, XeL^AT_EX). On those engines the class skips `amssymb` and schedules an *end-of-document* check: if by then the author has loaded neither `unicode-math`

nor `amssymb` (which they could have loaded by hand, though it is discouraged), a `ClassWarning` reminds them to add

```
\usepackage{unicode-math}
\setmathfont{Latin Modern Math} % or STIX Two Math, ...
```

to the manuscript preamble. The class deliberately does *not* auto-load `unicode-math`, because picking an OpenType math font is a typographic choice that belongs to the author. **Never load `amssymb` together with `unicode-math`:** it would re-establish the 8-bit AMS fonts on top of the OpenType math font and split symbols across two incompatible encodings.

Printing. To get your work correctly printed, you must ensure that any page scaling option (e.g., fit or shrink to printable area) isn't enabled. This kind of option often comes in print dialogs of document visualization softwares.

longquote Quotation A quotation longer than three lines is INDENTED BY 4 cm from the left margin – not set with a 4 cm left margin, which is a different and much narrower thing. 2.2(b) of the manual says “recuo de 4 cm da margem esquerda”, so on an A4 page with a 3 cm margin the quoted text starts 7 cm from the edge of the paper. It carries no quotation marks and is set in the smaller font, single spaced. The `coppe` class provides the `longquote` environment, which does all of that; the indentation is `\recuolongquote`, should a Programa ever need to change it.

9 A simple example

```
1 \example
2 \documentclass[dsc]{coppe}
3
4 \usepackage{rotating} % rodando coisas, como tabelas
5 \usepackage[most]{tcolorbox} % caixas de texto
6 \usepackage{tikz} % gráficos vetoriais (o tcolorbox já o traz)
7 % Sob LuaLaTeX/XeLaTeX, descomente as duas linhas a seguir para obter os
8 % símbolos AMS (\mathbb, \hbar, \varnothing, ...) via unicode-math. Sob
9 % pdfLaTeX, a classe já carrega amssymb automaticamente; deixe comentado.
10 \usepackage{unicode-math}
11 \setmathfont{Latin Modern Math} % ou STIX Two Math, TeX Gyre Termes Math...
12 \DeclareMathOperator{\RMSE}{RMSE} % operadores próprios (cap. de matemática)
13 \DeclareMathOperator{\diag}{diag}
14 % A classe coppe já carrega: amsmath, booktabs, longtable, subcaption,
15 % hyperref, biblatex, listings, fancyvrb, algorithm2e -- não recarregue esses
16 % pacotes aqui (em particular, hyperref deve ficar a cargo da classe para que
17 % a ordem com setspace e biblatex permaneça correta).
18
19 \addbibresource{example.bib} % base de referências (biblatex/biber)
20
21 \makelossymbols
22 \makeloabbreviations
23 % Algumas coisas opcionais que não cabem no coppetex.sty
24 \renewcommand{\familydefault}{\sfdefault} % para usar fontes SEM serifa no default
25 \renewcommand{\familydefault}{\rmdefault} % para usar fontes COM serifa no default
```



```

26
27
28 \begin{document}
29   \title{Titulo da Tese}
30   \foreigntitle{Thesis Title}
31   \author{Nome do Autor}{Sobrenome}
32   \advisor[UFRJ]{Prof.}{Nome do Primeiro Orientador}{Sobrenome}{D.Sc.}
33   \advisor[UFRJ]{Prof.}{Nome do Segundo Orientador}{Sobrenome}{Ph.D.}
34   % A classe aceita de um a tres \advisor; se houver apenas um, o rotulo na
35   % capa e na folha de rosto sai como "Orientador" (singular), com dois ou
36   % mais sai como "Orientadores" (plural). Para o caso 1/3 orientadores,
37   % veja tests/test_brazilian_one_advisor.tex e tests/test_brazilian_three_advisors.tex.
38
39   \examiner[UFRJ]{Prof.}{Nome do Primeiro Examinador Sobrenome}{D.Sc.}
40   \examiner[UFRJ]{Prof.}{Nome do Segundo Examinador Sobrenome}{Ph.D.}
41   \examiner[UFF]{Prof.}{Nome do Terceiro Examinador Sobrenome}{D.Sc.}
42   \examiner[UNICAMP]{Prof.}{Nome do Quarto Examinador Sobrenome}{Ph.D.}
43   \examiner[USP]{Prof.}{Nome do Quinto Examinador Sobrenome}{Ph.D.}
44   \department{PESC}
45   \date{06}{2026}
46
47   \keyword{Primeira palavra-chave}
48   \keyword{Segunda palavra-chave}
49   \keyword{Terceira palavra-chave}
50
51   % Dados da folha adicional (Coleta CAPES), obrigatoria desde agosto de 2026.
52   % A folha e preenchida pelo Programa; estes campos apenas a compoem.
53   \areaconcentracao{Engenharia de Sistemas e Computa\c c\~ao}
54   \linhapesquisa{Engenharia de Dados e Conhecimento}
55   \tipoproducao{bibliografica} % bibliografica | artistica | tecnologica | tecnica
56   \projetovinculado{sim} % sim | nao
57   \nomeprojeto{Nome do Projeto de Pesquisa ao qual o trabalho se vincula}
58   \agenciafomento{Conselho Nacional de Desenvolvimento Cientifico e
59                   Tecnologico}{CNPq}
60   \agenciafomento{Fundacao Carlos Chagas Filho de Amparo a Pesquisa do
61                   Estado do Rio de Janeiro}{FAPERJ}
62   % A ficha oficial vem de fichacatalografica.sibi.ufrj.br; tendo o arquivo,
63   % descomente a linha abaixo e a moldura vazia da folha adicional some.
64   % \fichacatalografica{ficha.pdf}
65
66   \maketitle
67
68   \frontmatter
69   \dedication{A alguem cujo valor é digno desta dedicatória.}
70
71   \chapter*{Agradecimentos}
72
73   Gostaria de agradecer a todos os que contribu'iram para este trabalho:
74   orientadores, colegas, familiares e \‘as ag\~encias de fomento. Lorem ipsum
75   dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt
76   ut labore et dolore magna aliqua.
77
78   \begin{abstract}
79

```

```

80 Apresenta-se, nesta tese, um exemplo de resumo. Lorem ipsum dolor sit amet,
81 consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et
82 dolore magna aliqua. Ut enim ad minim veniam, quis nostrud exercitation
83 ullamco laboris nisi ut aliquip ex ea commodo consequat. Duis aute irure dolor
84 in reprehenderit in voluptate velit esse cillum dolore eu fugiat nulla
85 pariatur. Excepteur sint occaecat cupidatat non proident, sunt in culpa qui
86 officia deserunt mollit anim id est laborum. Sed ut perspiciatis unde omnis
87 iste natus error sit voluptatem accusantium doloremque laudantium, totam rem
88 aperiam, eaque ipsa quae ab illo inventore veritatis et quasi architecto beatae
89 vitae dicta sunt explicabo, nemo enim ipsam voluptatem quia voluptas sit
90 aspernatur aut odit aut fugit.
91
92 \end{abstract}
93
94 \begin{foreignabstract}
95
96 In this work, we present an example abstract. Lorem ipsum dolor sit amet,
97 consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et
98 dolore magna aliqua. Ut enim ad minim veniam, quis nostrud exercitation
99 ullamco laboris nisi ut aliquip ex ea commodo consequat. Duis aute irure dolor
100 in reprehenderit in voluptate velit esse cillum dolore eu fugiat nulla
101 pariatur. Excepteur sint occaecat cupidatat non proident, sunt in culpa qui
102 officia deserunt mollit anim id est laborum. Sed ut perspiciatis unde omnis
103 iste natus error sit voluptatem accusantium doloremque laudantium, totam rem
104 aperiam, eaque ipsa quae ab illo inventore veritatis et quasi architecto beatae
105 vitae dicta sunt explicabo, nemo enim ipsam voluptatem quia voluptas sit
106 aspernatur aut odit aut fugit.
107
108 \end{foreignabstract}
109
110 % Ordem pre-textual (NBR 14724/6027, R55): listas (ilustracoes, tabelas,
111 % abreviaturas e siglas, simbolos) ANTES do sumario; sumario por ultimo.
112 \listoffigures
113 \listoftables
114 \listofquados
115 \listofprogramas
116 \listofalgorithms
117 \printloabbreviations
118 \printlosymbols
119 \tableofcontents
120
121 \mainmatter
122 \chapter{Introdução}
123
124 Este é um documento exemplo para o uso da classe CoppeTeX, destinado a ajudar os alunos do
125
126 A classe \verb|coppe| foi criada originalmente por Vicente Helano e George Ainsworth, poré
127 Em 2026, Geraldo Xexéo, com forte ajuda das ferramentas Claude 4.7 Opus e Max e 4.8 Opus M
128
129 Provavelmente Geraldo Xexéo, professor do Programa de Engenharia de Sistemas e Computação
130
131 A versão mais atual dessa classe é mantida no GitHub, no repositório \url{https://github.c
132
133 Esse documento segue a norma de formatação de teses e dissertações da COPPE. Ele também po

```

134

135 Há um grupo de WhatsApp para perguntas e discussões do estilo: `\href{https://github.com/COPPE-UFRJ/CoppeTeX/}`

136

137 Bugs devem ser reportados na `\href{https://github.com/COPPE-UFRJ/CoppeTeX/issues}`{página d

138

139 Esse documento é usado como exemplo de coisas que podem ser feitas. Ele está configurado p

140

141 É importante de notar que essa classe não foi construída sobre a classe `\LaTeX \` para a A

142

143 Essa classe implementa citações ABNT modernas e vários tipos de entrada descritos na norma

144

145 Este documento não substitui, mas complementa, o documento que descreve a classe.

146

147

148

149 `\chapter{Configurações Iniciais}`

150

151 A primeira coisa a fazer é escolher o tipo de documento. Isso é feito como uma opção no c

152

153 Como pode ser visto nesse documento, muita coisa pode ser configurada, o que gerará o tra

154

155 Recomendo ler o documento “The `\verb*|coppe|` document class” para entender melhor todas

156

157 `\section{Linguagem principal do texto}`

158

159 Essa classe considera que o texto principal está em português e algumas partes específicas

160 `\begin{itemize}`

161 `\item` A opção `\verb|english|` deve ser usada no comando `\verb|\documentclass|`.

162 `\item` A bibliografia segue automaticamente o idioma do Babel (português ou inglês); não é p

163 `\end{itemize}`

164

165 A variação de linguagem, em inglês ou português apenas, já é suportada pela classe `\CoppeT`

166

167

168 `\section{Por que usar o \LaTeX}`

169

170 Há uma grande discussão entre usuário de Word e `\LaTeX`, principalmente, quanto ao uso dess

171

172 Nós escolhemos o `\LaTeX` por alguns motivos: grande facilidade de seguir um estilo sem se p

173

174 As principais desvantagens são: idiossincrasias que podem gastar tempo, pouco controle sob

175

176 `\section{Como e onde usar o \LaTeX}`

177

178 Existem muitos tutoriais de `\LaTeX`, mas basicamente, em 2024, ele é usado em dois ambiente

179 `\begin{enumerate}`

180 `\item` Na sua máquina, instalando uma versão completa como o `\hologo{MiKTeX}`, típico do Win

181 `\item` Usar um ambiente na rede, como o `Overleaf`.

182 `\end{enumerate}`

183

184 Em todo caso, recomendo fortemente que, ao mesmo tempo, mantenha versões no Git e faça o b

185

186

187 `\chapter{Algumas Regras da COPPE}`

188

189 Todas abreviaturas e símbolos devem ser definida antes de utilizada. Isso é facilmente fei

190

191 É imprescindível definir os símbolos, tal como o

192 conjunto dos números reais $\mathbb{R}$ e o conjunto vazio $\emptyset$.

193 `\syml{\mathbb{R}}{Conjunto dos números reais}`

194 `\syml{\emptyset}{Conjunto vazio}`. Usamos esse exemplo aqui justamente para mostrar como

195

196 Para as listas de abreviaturas e símbolos funcionarem no Overleaf é necessário rodar o `\ve`

197

198 Como as listas de símbolos e de abreviaturas usamo o mesmo comando usado para criar índice

199

200 `\section{Citações}`

201 `\label{sec:citacoes}`

202

203 Citações curtas podem ser feitas `\enquote{o comando quote}` ou direto com “duas crases e do

204

205 `\begin{longquote}`

206 Um exemplo de citação longa nas regras da ABNT (4cm de recuo e fonte menor)

207 feita com o ambiente `\verb*|longquote|` The primary objective of this

208 investigation was to determine the feasibility of detecting corrosion in

209 aluminum Naval aircraft components with neutron radiographic interrogation

210 and the use of standard corrosion penetrameters. Secondary objectives

211 included the determination of the effect of object thickness on image quality,

212 the defining of minimum levels of detectability and a preliminary investigation

213 of a means whereby the degree of corrosion could be quantified with neutron

214 radiographic data. `\citep{article-example}`

215 `\end{longquote}`

216

217 Citações devem apontar as referências. Para isso, está disponível o ótimo pacote `\verb*|na`

218

219 Em todo caso, `\textbf{deve se tomar enorme atenção com as citações, para evitar ocorrer em`

220

221 Quando se cita uma obra por meio de outra (`\textit{apud}`), pela ABNT entra na lista de ref

222

223 `\chapter{Floats}`

224

225 Grande parte dos problemas de iniciantes, e veteranos, em $\text{\LaTeX}$ é da localização dos `\textit`

226

227 A regra geral de posicionamento é que uma figura ou quadro só pode aparecer a partir da mesm

228

229 `\textbf{Pela norma atual (manual2024), a legenda}` `\verb|\caption|` `\textbf{de toda ilustração`

230

231 Quadros são ilustrações cujos dados são delimitados por linhas em `\emph{todas}` as bordas (ao

232

233

234 `\section{Tabelas e Figuras Padrão}`

235

236 Vamos ver uma tabela padrão, como a `\autoref{tab:exemplo_numeros}`.

237

238 `\begin{table}[ht]`

239 `\centering` % Centraliza a tabela

240 `\caption{Exemplo de Tabela de Números}`

241 `\label{tab:exemplo_numeros}`

```

242 \begin{tabular}{ccc} % Define a quantidade de colunas
243 \hline % Linha superior
244 \textbf{Col. 1} & \textbf{Col. 2} & \textbf{Col. 3} \\ % Cabeçalhos
245 \hline % Linha média
246 1 & 2 & 3 \\ % Primeira linha de dados
247 \hline
248 4 & 5 & 6 \\ % Segunda linha de dados
249 \hline
250 7 & 8 & 9 \\ % Terceira linha de dados
251 \hline
252 10 & 11 & 12 \\ % Quarta linha de dados
253 \hline % Linha inferior
254 \end{tabular}
255 \fonte{Elabora\c{c}\~ao pr\'opria.} % fonte obrigatoria, abaixo do float
256 \end{table}
257
258
259
260 Já a \autoref{fig:exemplo_figura} é uma figura padrão, com controle da largura.
261
262 \begin{figure}[ht]
263 \centering % Centraliza a figura
264 \caption{Exemplo de Figura com Legenda Acima} % Legenda em cima (norma atual)
265 \label{fig:exemplo_figura} % Etiqueta para referência cruzada
266 \includegraphics[width=0.5\textwidth]{coppe-logo.pdf} % Inclui a imagem com metade da largura
267 \source{COPPE/UFRJ.} % fonte obrigatoria, abaixo da figura
268 \end{figure}
269
270 Quadros têm a mesma regra (legenda acima, fonte abaixo), mas com linhas em todas as bordas.
271
272 \begin{quadro}[ht]
273 \centering
274 \caption{Exemplo de Quadro}
275 \label{qua:exemplo}
276 \begin{tabular}{|l|l|}
277 \hline
278 \textbf{Ilustração} & \textbf{Delimitação} \\
279 \hline
280 Tabela & Linhas apenas no topo e na base. \\
281 \hline
282 Quadro & Linhas em todas as bordas. \\
283 \hline
284 \end{tabular}
285 \fonte{Elabora\c{c}\~ao pr\'opria.}
286 \end{quadro}
287
288
289
290
291
292 \section{Tabelas mais elegantes}
293
294 Atualmente a tendência é usar tabelas mais leves, como \autoref{tab:exemplo_numerosbom}. Iss
295

```

```

296 \begin{table}[ht]
297 \centering % Centraliza a tabela
298 \caption{Exemplo de Tabela de Números mais elegantes}
299 \label{tab:exemplo_numerosbom}
300 \begin{tabular}{ccc} % Define a quantidade de colunas
301 \toprule % Linha superior
302 \textbf{Col. 1} & \textbf{Col. 2} & \textbf{Col. 3} \\ % Cabeçalhos
303 \midrule % Linha média
304 1 & 2 & 3 \\ % Primeira linha de dados
305 4 & 5 & 6 \\ % Segunda linha de dados
306 7 & 8 & 9 \\ % Terceira linha de dados
307 10 & 11 & 12 \\ % Quarta linha de dados
308 \bottomrule % Linha inferior
309 \end{tabular}
310 \fonte{Elabora\c{c}\~ao pr'opria.}
311 \end{table}
312
313 \section{Tabelas Longas ou Largas}
314
315 Se sua tabela é muito longa ou larga, existem várias opções.
316 \begin{itemize}
317 \item alterar o tamanho da letra
318 \item Usar o longtable
319 \item rodar a tabela, fazendo ela em \textit{landscape}
320 \item fazer a tabela dentro de um minibox
321 \end{itemize}
322
323
324 \subsection{Tabelas largas demais}
325
326 É comum em teses que as tabelas sejam largas demais. Há várias formas de resolver isso.
327
328 A \autoref{tab:tabela_largafns} é larga demais, e nela isso é resolvido diminuindo a fonte p
329
330 \begin{table}[ht]
331 \centering % Centraliza a tabela
332 \caption{Exemplo de Tabela Larga com Fonte Menor}
333 \label{tab:tabela_largafns}
334 \footnotesize % Aplica uma fonte menor para a tabela
335 \begin{tabular}{ccccccc} % Aumente o número de colunas conforme necessário
336 \toprule
337 \textbf{Col. 1} & \textbf{Col. 2} & \textbf{Col. 3} & \textbf{Col. 4} & \textbf{Col. 5} & \textbf{Col. 6} & \textbf{Col. 7} \\
338 \midrule
339 Dado 1.1 & Dado 1.2 & Dado 1.3 & Dado 1.4 & Dado 1.5 & Dado 1.6 & Dado 1.7 & Dado 1.8 \\
340 Dado 2.1 & Dado 2.2 & Dado 2.3 & Dado 2.4 & Dado 2.5 & Dado 2.6 & Dado 2.7 & Dado 2.8 \\
341 Dado 3.1 & Dado 3.2 & Dado 3.3 & Dado 3.4 & Dado 3.5 & Dado 3.6 & Dado 3.7 & Dado 3.8 \\
342 \bottomrule
343 \end{tabular}
344 \fonte{Elabora\c{c}\~ao pr'opria.}
345 \end{table}
346
347 O comando \verb|\resizebox{width}{height}{content}| permite ajustar o tamanho de qualquer co
348
349 \begin{table}[ht]

```

```

350 \centering
351 \caption{Exemplo de Tabela Redimensionada}
352 \label{tab:examplerb}
353 \resizebox{\textwidth}{!}{%
354 \begin{tabular}{llll}
355 \toprule
356 Coluna 1 & Coluna 2 & Coluna 3 & Coluna 4 \\
357 \midrule
358 Dados 1 & Dados 2 & Dados 3 & Dados 4 \\
359 Dados 5 & Dados 6 & Dados 7 & Dados 8 \\
360 \bottomrule
361 \end{tabular}}%
362 }
363 \fonte{Elabora\c{c}\~ao pr\'opria.}
364 \end{table}
365
366
367 Para rodar uma tabela muito larga em 90 graus no LaTeX, voc\ea pode usar o pacote \verb*|rotat
368
369 Aqui est\aa um exemplo de como usar o ambiente \verb*|sidewaystable| para girar uma tabela. Pr
370
371 \begin{sidewaystable}
372 \centering
373 \caption{Sua Legenda Aqui}
374 \label{tab:sua_tabela}
375 \begin{tabular}{llll}
376 \toprule
377 Coluna 1 & Coluna 2 & Coluna 3 \\
378 \midrule
379 Item 1 & Item 2 & Item 3 \\
380 Item 4 & Item 5 & Item 6 \\
381 \bottomrule
382 \end{tabular}
383 \fonte{Elabora\c{c}\~ao pr\'opria.}
384 \end{sidewaystable}
385
386 Se a tabela for muito longa, o ambiente \verb|longtable| \xc3 o ideal. Ele fornece comandos par
387
388 % Exemplo de tabela longa que se estende por v\arias p\aginas
389 \begin{longtable}{|c|c|c|}
390 % primeiro cabe\c7alho (\xc3 o caption)
391 \caption{Exemplo de Tabela Longa}\label{tab:longa} \\
392 \hline \textbf{Col. 1} & \textbf{Col. 2} & \textbf{Col. 3} \\ \hline
393 \endfirsthead
394 % cabe\c7alho normal
395 \multicolumn{3}{c}%
396 {\table\thetable} -- continua\c7\aa da p\agina anterior} \\
397 \hline \textbf{Col. 1} & \textbf{Col. 2} & \textbf{Col. 3} \\ \hline
398 \endhead
399 % p\ea normal
400 \hline \multicolumn{3}{|r|}{Continua na pr\oxima p\agina} \\ \hline
401 \endfoot
402 \hline
403 % \xc3timo p\ea

```

```

404 \multicolumn{3}{|r|}{{Continua na próxima página}}\\
405 \hline \hline
406 \endlastfoot
407
408 % Conteúdo da tabela
409 1 & 2 & 3 \\
410 4 & 5 & 6 \\
411 1 & 2 & 3 \\
412 4 & 5 & 6 \\
413 1 & 2 & 3 \\
414 4 & 5 & 6 \\
415 1 & 2 & 3 \\
416 4 & 5 & 6 \\
417 1 & 2 & 3 \\
418 4 & 5 & 6 \\
419 1 & 2 & 3 \\
420 4 & 5 & 6 \\
421 1 & 2 & 3 \\
422 4 & 5 & 6 \\
423 1 & 2 & 3 \\
424 4 & 5 & 6 \\
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426 1 & 2 & 3 \\
427 4 & 5 & 6 \\
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429 4 & 5 & 6 \\
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431 4 & 5 & 6 \\
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433 4 & 5 & 6 \\
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443 4 & 5 & 6 \\
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445 4 & 5 & 6 \\
446 1 & 2 & 3 \\
447 4 & 5 & 6 \\
448 1 & 2 & 3 \\
449 4 & 5 & 6 \\1 & 2 & 3 \\
450 4 & 5 & 6 \\
451 1 & 2 & 3 \\
452 4 & 5 & 6 \\
453 1 & 2 & 3 \\
454 1 & 2 & 3 \\
455 4 & 5 & 6 \\
456 1 & 2 & 3 \\
457 4 & 5 & 6 \\

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458 1 & 2 & 3 \\
 459 4 & 5 & 6 \\
 460 1 & 2 & 3 \\
 461 4 & 5 & 6 \\
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 505 4 & 5 & 6 \\
 506 1 & 2 & 3 \\
 507 4 & 5 & 6 \\
 508 1 & 2 & 3 \\
 509 4 & 5 & 6 \\
 510 1 & 2 & 3 \\
 511 4 & 5 & 6 \\

```

512 1 & 2 & 3 \\
513 4 & 5 & 6 \\
514 1 & 2 & 3 \\
515 4 & 5 & 6 \\
516
517 % Repetir linhas semelhantes conforme necessário para estender a tabela por 3 páginas
518 \end{longtable}
519 \fonte{Elabora\c{c}\~ao pr\'opria.}% a longtable não é float: a fonte vem após o ambiente
520
521 \section{Subfiguras (subcaption)}
522
523 Para colocar várias imagens (ou tabelas) lado a lado, cada uma com sua própria legenda, use
524
525 \begin{figure}[ht]
526   \centering
527   \begin{subfigure}[b]{0.42\textwidth}
528     \centering\includegraphics[width=\linewidth]{example-image-a}
529     \caption{Primeira vista}\label{fig:sub-a}
530   \end{subfigure}
531   \hfill
532   \begin{subfigure}[b]{0.42\textwidth}
533     \centering\includegraphics[width=\linewidth]{example-image-b}
534     \caption{Segunda vista}\label{fig:sub-b}
535   \end{subfigure}
536   \caption{Figura com duas subfiguras}
537   \label{fig:subs}
538   \fonte{Elabora\c{c}\~ao pr\'opria.}
539 \end{figure}
540
541 Referencie a figura inteira com \verb|\autoref{fig:subs}|. Cada parte tem
542 rótulo próprio: \verb|\autoref{fig:sub-a}| leva à \autoref{fig:sub-a}.
543
544 \section{Imagens (graphicx)}
545
546 A classe já carrega o \verb|graphicx|. Insira imagens com \verb|\includegraphics|; as opções
547 \begin{itemize}
548   \item \verb|[width=0.8\textwidth]| ou \verb|[scale=0.5]| --- tamanho;
549   \item \verb|[angle=90]| --- rotação;
550   \item \verb|[trim=l b r t, clip]| --- recorte (corta as bordas \emph{esquerda, baixo, direi}
551 \end{itemize}
552 Formatos aceitos com \hologo{pdfLaTeX}/\hologo{LuaLaTeX}: \texttt{pdf}, \texttt{png}, \texttt{t}
553
554 \section{Introdução ao tikz}
555
556 0 \verb|tikz| (disponível também via \verb|tcolorbox|) desenha gráficos vetoriais no próprio
557
558 \begin{figure}[ht]
559   \centering
560   \begin{tikzpicture}
561     \node[draw, rounded corners] (a) at (0,0) {Início};
562     \node[draw, rounded corners] (b) at (4,0) {Fim};
563     \draw[->, thick] (a) -- (b) node[midway, above] {processo};
564   \end{tikzpicture}
565   \caption{Diagrama simples em tikz}

```

```

566 \label{fig:tikz}
567 \fonte{Elabora\c{c}\~ao pr\'opria.}
568 \end{figure}
569
570 Para gr\'aficos de dados (curvas, barras), veja o \verb|pgfplots|, constru\'ido sobre o \verb|tikz|
571
572 \section{Float ou direto no texto?}
573
574 Colocar a tabela ou imagem dentro de \verb|figure|/\verb|table| (um \emph{float}) deixa o \verb|caption|
575
576 \chapter{Using BibLaTeX}
577 \label{cap:biblatex}
578
579 Esta classe gera as refer\~encias com BibLaTeX e o \emph{backend} \texttt{biber}, no padr\~ao
580
581 \textbf{\'E} obrigat\'orio compilar com \texttt{biber} (e n\~ao com \hologo{BibTeX}). A sequ\~encia
582
583 Os comandos do \texttt{natbib} (\verb|\citet|, \verb|\citep|) continuam dispon\'iveis porque
584
585 \begin{table}[h]
586 \caption{Exemplos de cita\c{c}\~oes utilizando o comando padr\~ao}
587 \texttt{\textbackslash cite} do LaTeX e
588 o comando \texttt{\textbackslash citep},
589 dispon\'ivel pela op\c{c}\~ao \texttt{natbib} do BibLaTeX.
590 \label{tab:citation}
591 \centering
592 {\footnotesize
593 \begin{tabular}{|c|c|c|}
594 \hline
595 Tipo da Publica\c{c}\~ao & \verb|\cite| & \verb|\citep|\\
596 \hline
597 Livro & \cite{book-example} & \citep{book-example}\\
598 Artigo & \cite{article-example} & \citep{article-example}\\
599 Relat\'orio & \cite{techreport-example} & \citep{techreport-example}\\
600 Relat\'orio & \cite{techreport-exampleIn} & \citep{techreport-exampleIn}\\
601 Anais de Congresso & \cite{inproceedings-example} & \\
602 & \citep{inproceedings-example}\\
603 S\'eries & \cite{incollection-example} & \citep{incollection-example}\\
604 Em Livro & \cite{inbook-example} & \citep{inbook-example}\\
605 Disserta\c{c}\~ao de mestrado & \cite{mastersthesis-example} & \\
606 & \citep{mastersthesis-example}\\
607 Tese de doutorado & \cite{phdthesis-example} & \citep{phdthesis-example}\\
608 \hline
609 \end{tabular}}
610 \fonte{Elabora\c{c}\~ao pr\'opria.}
611 \end{table}
612
613 \begin{table}[h]
614 \caption{Exemplos de cita\c{c}\~oes utilizando o comando padr\~ao}
615 \texttt{\textbackslash cite} do LaTeX e
616 o comando \texttt{\textbackslash citet},
617 dispon\'ivel pela op\c{c}\~ao \texttt{natbib} do BibLaTeX. Al\'em disso, usando o booktabs.}
618 \label{tab:citation1}
619 \centering

```

```

620 {\footnotesize
621 \begin{tabular}{ccc}
622 \toprule
623 Tipo da Publicação & \verb|\cite| & \verb|\citet|\n
624 \midrule
625 Livro & \cite{book-example} & \citet{book-example}\n
626 Artigo & \cite{article-example} & \citet{article-example}\n
627 Relatório & \cite{techreport-example} & \citet{techreport-example}\n
628 Relatório & \cite{techreport-exampleIn} & \citet{techreport-exampleIn}\n
629 Anais de Congresso & \cite{inproceedings-example} &
630 \citet{inproceedings-example}\n
631 Séries & \cite{incollection-example} & \citet{incollection-example}\n
632 Em Livro & \cite{inbook-example} & \citet{inbook-example}\n
633 Dissertação de mestrado & \cite{mastersthesis-example} &
634 \citet{mastersthesis-example}\n
635 Tese de doutorado & \cite{phdthesis-example} & \citet{phdthesis-example}\n
636 \bottomrule
637 \end{tabular}}
638 \fonte{Elabora\c{c}\~ao pr\'opria.}
639 \end{table}
640
641 \section{Comandos de citação}
642
643 Os principais comandos são: \verb|\cite| (referência simples), \verb|\citet| (autor por ex
644
645 \subsection{Parâmetro opcional: página e “tradução nossa”}
646
647 Todos os comandos aceitam um parâmetro opcional para a localização (página, seção) e obser
648
649 \subsection{Citação de citação (\emph{apud})}
650
651 Para citar uma obra através de outra, use \verb|\citetapud| e \verb|\citepapud| (demonstra
652
653 \subsection{Múltiplas bibliografias}
654
655 Para separar as referências (por tipo, capítulo ou tema), use o ambiente \verb|refsection|
656
657 \section{Tipos de entrada bibliográfica}
658
659 Cada tipo de documento da ABNT corresponde a um tipo de entrada do BibLaTeX, com um sinôni
660
661 \subsection{Tipos acadêmicos principais}
662 \begin{itemize}
663 \item Livro / monografia --- \verb|@book| (\verb|@livro|)
664 \item Artigo de periódico --- \verb|@article| (\verb|@artigo|); matéria de jornal --- \verb|@journal|
665 \item Capítulo de livro --- \verb|@incollection| (\verb|@partedelivro|), \verb|@inbook|
666 \item Evento no todo (anais) --- \verb|@proceedings| (\verb|@anais|); trabalho em evento
667 \item Tese/dissertação --- \verb|@thesis| (\verb|@tese|, \verb|@dissertacao|); com tipo
668 \item Relatório --- \verb|@report| (\verb|@relatorio|, \verb|@relatoriotecnico|); periód
669 \item Manual --- \verb|@manual|; diversos --- \verb|@misc| (\verb|@diverso|); página/sit
670 \end{itemize}
671
672 \subsection{Tipos especiais (ABNT §4.2.5--4.2.14)}
673 \begin{itemize}

```

```

674 \item Patente --- \verb|@patent| (\verb|@patente|)
675 \item Legislação / ato normativo --- \verb|@legislation| (\verb|@legislacao|, \verb|@ato
676 \item Norma técnica --- \verb|@standard| (\verb|@norma|); partitura --- \verb|@music| (\
677 \item Documento cartográfico (mapa) --- \verb|@map| (\verb|@mapa|, \verb|@cartografico|)
678 \item Audiovisual --- \verb|@video| (\verb|@audiovisual|); filme --- \verb|@movie| (\verb
679 \end{itemize}
680
681 \subsection{Jogos e TV (tipos próprios da CoppeTeX)}
682 \begin{itemize}
683 \item Jogo --- \verb|@game| (\verb|@jogo|); de tabuleiro --- \verb|@boardgame| (\verb|@j
684 \item Programa de TV / série --- \verb|@tvshow| (\verb|@programadetv|, \verb|@serietv|)
685 \end{itemize}
686
687 Os campos também aceitam sinônimos em português (sem acentos): \verb|autor|, \verb|titulo|
688
689 \chapter{Alguns outros exemplo úteis}
690
691 \begin{tcolorbox}[title=Meu Textbox]
692 Este é o conteúdo do meu textbox. Você pode adicionar qualquer texto aqui, bem como incluir
693 \end{tcolorbox}
694
695 \begin{tcolorbox}
696 Este é o conteúdo do meu textbox sem título. Você pode adicionar qualquer texto aqui, bem co
697 \end{tcolorbox}
698
699 \begin{figure}[ht]
700 \centering
701 \begin{tikzpicture}
702 \node[anchor=south west,inner sep=0] (image) at (0,0) {\includegraphics[width=0.5\textwidth]{
703 \begin{scope}[x={(image.south east)},y={(image.north west)}]
704 % Definindo o textbox dentro da figura
705 \node[anchor=north west, text width=0.3\textwidth, fill=white, opacity=0.7, text=
706 \begin{tcolorbox}[colback=red!5!white,colframe=red!75!black,title=Textbox de
707 Este textbox fala sobre como inserir um textbox dentro de uma figura usa
708 \end{tcolorbox}
709 };
710 \end{scope}
711 \end{tikzpicture}
712 \caption{Figura com Textbox}
713 \label{fig:figura_com_textbox1}
714 \fonte{Elabora\c{c}\~ao pr\'opria.}
715 \end{figure}
716
717
718 \begin{figure}[ht]
719 \centering
720 \begin{tcolorbox}
721 Este é o conteúdo do meu textbox sem título. Você pode adicionar qualquer texto aqui, bem co
722 \end{tcolorbox}
723 \caption{Figura com Textbox simples}
724 \label{fig:figura_com_textbox}
725 \fonte{Elabora\c{c}\~ao pr\'opria.}
726 \end{figure}
727

```

```

728 \section{Listagens de código e algoritmos}
729
730 O CoppeTeX traz embutido o suporte a listagens de programas (pacote \verb|listings|) e a alg
731
732 \begin{python}[caption={Exemplo de programa em Python},label=prog:exemplo]
733 def saudacao(nome):
734     # imprime uma saudação acentuada
735     print(f"Olá, {nome}!")
736
737 saudacao("COPPE")
738 \end{python}
739 \source{Elaboração própria.}
740
741 Para inserir código no meio do texto use \verb|\pythoninline|, como em \pythoninline{saudaca
742
743 \begin{Verbatim}[frame=lines,numbers=left,numbersep=0.5em]
744 Olá, COPPE!
745 \end{Verbatim}
746
747 Há estilos prontos também para \verb|java|, \verb|xml|, \verb|html| e \verb|prolog|.
748 Há também \verb|\pythoninline| e \verb|\javainline| para código no meio do
749 texto, e o ambiente \verb|Verbatim| (com as opções acima) para a saída de programas. Para al
750
751 \begin{algorithm}[H]
752 \caption{Soma dos elementos de um vetor}
753 \label{alg:soma}
754 \Dados{vetor  $v$  com  $n$  elementos}
755 \Resultado{soma  $s$  dos elementos de  $v$ }
756  $s \leftarrow 0$ ;
757 \Para{ $i \leftarrow 1$  \Ate{ }  $n$ }{
758      $s \leftarrow s + v[i]$ ;
759 }
760 \Retorna  $s$ ;
761 \end{algorithm}
762
763 \chapter{Escrevendo matemática}
764
765 A classe carrega \texttt{amsmath} em qualquer motor. Sob \hologo{pdfLaTeX} ela também carr
766
767 \section{Alinhamento (align)}
768 O ambiente \verb|align| alinha equações pelo \verb|&| e numera cada linha (\verb|\\| quebr
769 \begin{align}
770 f(x) &= (x+1)^2 \\
771      &= x^2 + 2x + 1 .
772 \end{align}
773
774 \section{Matrizes (matrix)}
775 O \texttt{amsmath} traz \verb|pmatrix|, \verb|bmatrix|, \verb|vmatrix|, entre outros:
776 \begin{equation}
777 A = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} \\
778 \quad \quad \quad \\
779 \det A = \begin{vmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{vmatrix} \\
780 \end{equation}
781

```

```

782 \section{Subequações (subequations)}
783 Para numerar um grupo de equações sob o mesmo número, com letras (a), (b):
784 \begin{subequations}
785 \begin{align}
786   a &= b + c , \\
787   d &= e - f .
788 \end{align}
789 \end{subequations}
790
791 \section{Estruturas alinhadas (array)}
792 O \verb|array| monta estruturas dentro do modo matemático, como definições por partes:
793 \begin{equation}
794   |x| = \left\{ \begin{array}{l}
795     x, \quad \text{se } x \geq 0 , \\
796     -x, \quad \text{caso contrário.}
797   \end{array} \right.
798 \end{equation}
799
800 \section{Operadores (\textbackslash DeclareMathOperator)}
801 Para operadores com a tipografia correta (romano, espaçamento adequado), declare-os no pre
802 \begin{verbatim}
803 \usepackage{amsmath}
804 \DeclareMathOperator{\RMSE}{RMSE}
805 \DeclareMathOperator{\diag}{diag}
806 \end{verbatim}
807 Assim:
808 \begin{itemize}
809   \item \verb|$\RMSE(y,\hat{y})$| produz  $\RMSE(y,\hat{y})$ ;
810   \item \verb|$\diag(\lambda_1,\lambda_2)$| produz
811      $\diag(\lambda_1,\lambda_2)$ .
812 \end{itemize}
813
814 \section{Unicode e LuaLaTeX (unicode-math)}
815 \label{sec:unicode-math}
816 Ao migrar para \hologo{LuaLaTeX} (ou \hologo{XeLaTeX}), \textbf{não carregue} \texttt{amss
817 \begin{verbatim}
818 \usepackage{amsmath}           % a classe já carrega
819 \usepackage{unicode-math} % NÃO use \usepackage{amssymb} junto
820 \setmathfont{Latin Modern Math} % ou STIX Two Math
821 \end{verbatim}
822 O código clássico continua valendo ( $\mathbb{R}$ ,  $\int_a^b f(x)dx$ ,  $\alpha$ ) e você ai
823
824 \chapter{Referências cruzadas, notas e índices}
825 \label{cap:refs}
826
827 Este capítulo reúne os recursos para \emph{apontar} para coisas: rótulos, notas de rodapé,
828
829 \section{Rótulos e referências cruzadas}
830
831 Marque um elemento com \verb|\label{tipo:nome}| e aponte para ele com \verb|\ref| (só o nú
832
833 \section{Notas de rodapé}
834
835 Use \verb|\footnote{texto}| \footnote{Como esta nota, que a classe imprime em espaçamento s

```



```

890 \syml[marco]{Marco}{A palavra Marco corrigida}
891
892 A Lista de Abreviações segue, a partir de 2024, a mesma regra, e aqui seguem alguns exemplos
893 \abbrev{GoT}{Game of Thrones}
894 \abbrev[GOT]{GoT}{Game of Thrones ordenado como GOT}
895 \abbrev[iot]{IoT}{IoT ordenado como iot}
896 \abbrev[IoT]{IoT}{IoT ordenado como IoT}
897 \abbrev[IOT]{IoT}{IoT ordenado como IOT}
898 \abbrev{IoT}{IoT com ordenação default}
899 \abbrev[ITU]{ITU}{ITU mesmo}
900
901 \newsigla{ufrj}{UFRJ}{Universidade Federal do Rio de Janeiro}
902 \newsigla{cpgp}{CPGP}{Comissão de Pós-Graduação e Pesquisa de Engenharia}
903
904 As siglas (manual2024, seção 2.8) são expandidas na primeira ocorrência e registradas automaticamente
905
906
907
908
909 \chapter{Conclusões}
910
911 \backmatter
912
913 % Tipos exóticos da ABNT (NBR 6023): patente, legislação, norma, mapa,
914 % partitura --- declarados via sinônimos em português em example.bib.
915 \nocite{patente-example,legislacao-example,norma-example,mapa-example,partitura-example}
916 \nocite{jurisprudencia-example}
917 \nocite{videogame-example,filme-example,semautor-example,semautorart-example}
918 \nocite{jornalcaderno-example,jornal-example}
919 \printbibliography
920
921
922
923 \appendix
924
925 \chapter{Um apêndice}
926
927 Segundo a norma da ABNT (Associação Brasileira de Normas Técnicas), a definição e utilização de apêndices
928
929 Apêndice: O apêndice é um texto ou documento elaborado pelo autor do trabalho com o objetivo de complementar
930
931
932 \chapter{Outro apêndice}
933
934 \annex
935
936
937 \chapter{Um Anexo}
938 Segundo a norma da ABNT (Associação Brasileira de Normas Técnicas), a definição e utilização de anexos
939
940
941
942 Anexo: O anexo, por sua vez, consiste em um texto ou documento não elaborado pelo autor, que complementa o
943

```

```

944 No modelo \CoppeTeX os anexos devem obrigatoriedade vir depois dos apêndices e usam o comand
945
946
947 \chapter{Outro Anexo}
948
949
950 \coppetexfinalpage% colofão institucional (página par, face do contracapa)
951
952 \end{document}
953
954 \example

```

10 Implementation

10.1 The ‘coppe.cls’ file

```

955 \*class
956 \def\filename{coppe.dtx}
957 \def\fileversion{v4.1}
958 \def\filedate{2026/09/09}
959 \NeedsTeXFormat{LaTeX2e}[1995/12/01]
960 \ProvidesClass{coppe}[\filedate\ \fileversion\ COPPE Dissertations and Thesis]
961 % Save the class version and date into private macros immediately, before any
962 % required package can redefine the very common \fileversion / \filedate
963 % global names (LaTeX has no convention reserving them for the host class).
964 % \coppe@version / \coppe@date are what user-facing macros (e.g.
965 % \coppetexfinalpage) should reference.
966 \let\coppe@version\fileversion
967 \let\coppe@date\filedate
968 % Document date stored in private slots so \date can set them without ever
969 % touching TeX's \month/\year primitives (which would break \today). They
970 % default to the current system month/year, so a document without \date
971 % still renders sensibly on the cover and folha de rosto.
972 \edef\coppe@docmonth{\the\month}
973 \edef\coppe@docyear{\the\year}
974 \LoadClass[12pt,a4paper,oneside]{book}
975 % setspace, hyperref and the bibliography engine (biblatex+biber) are loaded at
976 % the END of the class (after every class hook is defined), in that order:
977 % setspace -> hyperref -> biblatex. hyperref must come late so all the macros
978 % it patches (TOC, refs, lists) are already defined, and biblatex must come
979 % after hyperref so its citation links and back-references work.
980 \RequirePackage{hyphenat}
981 \RequirePackage{lastpage}
982 \RequirePackage{ifthen}
983 \RequirePackage{graphicx}
984 \RequirePackage{tabularx}
985 \RequirePackage{booktabs} % regras de tabela profissionais (toprule/midrule/bottomrule)
986 \RequirePackage{longtable} % tabelas que quebram entre páginas
987 \RequirePackage{etoolbox}
988 \RequirePackage{ltxcmds}
989 \RequirePackage{hologo} % TeX-family logos: \hologo{TeX}, \hologo{LaTeX}, ...
990 % Render \TeX, \LaTeX and friends via hologo (\CoppeTeX is a custom logo, kept).
991 \def\TeX{\hologo{TeX}}
992 \def\LaTeX{\hologo{LaTeX}}

```

```

993 \def\LaTeXe{\hologo{LaTeXe}}
994 \def\BibTeX{\hologo{BibTeX}}
995 \def\pdfLaTeX{\hologo{pdfLaTeX}}
996 \def\pdfTeX{\hologo{pdfTeX}}
997 \def\LuaLaTeX{\hologo{LuaLaTeX}}
998 \def\XeLaTeX{\hologo{XeLaTeX}}
999 \def\MiKTeX{\hologo{MiKTeX}}
1000 % Engine-aware UTF-8 input: pdfLaTeX needs inputenc(utf8); LuaLaTeX (and XeTeX)
1001 % are UTF-8 native, so inputenc must NOT be loaded there (it would clash). Source
1002 % files may use accented Latin characters (âéíóú) directly. Under pdfLaTeX we also
1003 % suggest LuaLaTeX, which handles Unicode/fonts better (at the cost of speed).
1004 \RequirePackage{iftex}
1005 \ifPDFTeX
1006   \RequirePackage[utf8]{inputenc}
1007   \AtEndOfClass{\ClassWarningNoLine{coppe}{%
1008     Although slower, you should try compiling with LuaLaTeX}}
1009 \fi
1010 % lmodern MUST come with the T1 encoding. Without it, T1 falls back to the EC
1011 % fonts, which exist only as METAFONT bitmaps in most installations, and the
1012 % whole document is typeset in Type3 bitmap fonts. Measured before this change:
1013 % example.pdf carried 87 Type3 fonts against 6 Type1, and coppe.pdf 133 against
1014 % 2. The consequences are not cosmetic:
1015 %
1016 % * the text renders blurred at any zoom and prints poorly;
1017 % * copying and searching text in the PDF is unreliable;
1018 % * PDF/A -- required for deposit by 2.2(d) of the 2026 manual -- is not
1019 %   attainable with bitmap text.
1020 %
1021 % Latin Modern is the vector successor of Computer Modern, metrically and
1022 % visually near-identical, and ships with every current distribution, so the
1023 % page layout does not move.
1024 \RequirePackage{lmodern}
1025 \RequirePackage[T1]{fontenc}
1026 % Math support. amsmath is universal and works on every engine, so the class
1027 % brings it in unconditionally (it must precede hyperref, which is loaded at
1028 % the end of the class). amssymb is the legacy AMS symbol package, built
1029 % around the 8-bit Type 1 fonts msam/msbm; on a Unicode engine (LuaLaTeX or
1030 % XeLaTeX) it clashes with unicode-math, which already provides all of its
1031 % symbols through Unicode math codes. So amssymb is loaded ONLY under
1032 % pdfLaTeX; Unicode-engine users get a ClassWarning pointing them at
1033 % unicode-math (the class does not auto-load unicode-math because choosing
1034 % an OpenType math font is a typographic decision that belongs to the author).
1035 \RequirePackage{amsmath}
1036 \ifPDFTeX
1037   \RequirePackage{amssymb}
1038 \else
1039   % Unicode engine: defer the check to end of document. Only warn if the
1040   % user did NOT add either unicode-math (the modern OpenType-math route)
1041   % or amssymb (the legacy route, which they would have loaded manually).
1042   % This avoids nagging authors who already followed the documented
1043   % \usepackage{unicode-math} workflow.
1044   \AtEndDocument{%
1045     \ifpackageloaded{unicode-math}{\}%
1046     \ifpackageloaded{amssymb}{\}%

```

```

1047 \ClassWarningNoLine{coppe}{%
1048     You are compiling with a Unicode engine (LuaLaTeX or XeLaTeX) and\MessageBreak
1049     neither unicode-math nor amssymb is loaded. AMS-style symbols\MessageBreak
1050     (\string\mathbb, \string\hbar, \string\varnothing, ...) will be undefined.\MessageBreak
1051     Add to your preamble:\MessageBreak
1052     \space\space\string\usepackage{unicode-math}\MessageBreak
1053     \space\space\string\setmathfont{Latin Modern Math}\MessageBreak
1054     Do NOT load amssymb together with unicode-math -- it would clash\MessageBreak
1055     with the OpenType math font setup}}}%
1056 \fi
1057 % Margins per the SiBI manual, 9th rev. ed. (2026), 2.3: left 3cm, top 3cm,
1058 % right 2cm, bottom 2cm -- ONE-SIDED. The 2025 edition also specified verso
1059 % margins (right 3cm / left 2cm) and recommended printing the body on both
1060 % sides; the 2026 revision DELETED the verso margins entirely, following CEPG
1061 % Resolution 246/2023, which ended the printed deposit. A digital-only document
1062 % has no gutter, so the mirrored layout (previously inner=outer=2cm plus a 1cm
1063 % bindingoffset) no longer has any normative basis and is now opt-in via the
1064 % \texttt{twoside} class option, handled after \ProcessOptions below.
1065 %
1066 % headheight=15pt is given here (not patched later) so geometry's own page
1067 % arithmetic stays self-consistent: fancyhdr needs >= 14.49998pt of header
1068 % height to host the running head, and the standard 12pt default triggers
1069 % its "headheight is too small" warning. Letting geometry know up front
1070 % keeps \topmargin a derived value -- we never touch it by hand.
1071 % (geometry has no \texttt{oneside} key -- only \texttt{twoside}; with the key
1072 % absent it follows the class, which \LoadClass set to one-sided above.)
1073 % headsep is given explicitly to place the folio. 2.7 measures it the way it
1074 % measures the right margin -- to the digits themselves: "a 2cm da borda
1075 % superior, ficando o ultimo algarismo a 2cm da borda direita". The right edge
1076 % already landed at 2.02cm; the top of the digits sat at 1.87cm, too high.
1077 % geometry derives \topmargin from top - 1in - headheight - headsep, so
1078 % shortening headsep lowers the header and leaves the body top untouched:
1079 % topmargin + headheight + headsep is 13.088pt either way, and 3cm is where the
1080 % body still starts. The value is CALIBRATED, not computed -- the folio sits a
1081 % little lower than the arithmetic alone predicts, because fancyhdr's box does
1082 % not put the digits flush with the top of \headheight. Two measured points,
1083 % taken from the rendered page rather than from the PDF's font metrics:
1084 % headsep 18.07pt -> digits at 1.8669cm headsep 14.29pt -> 2.0616cm
1085 % which put 2.0000cm at 15.49pt while the folio was still set at the body size.
1086 % It is \footnotesize now (2.2(b) puts pagination in the smaller font), the
1087 % digits are shorter, and their top therefore sits lower for the same baseline:
1088 % re-measured, 15.49pt gave 2.0659cm and the value moved to 16.87pt. That is
1089 % precisely why this is a measured constant and not a computed one -- change
1090 % the folio's size or headheight and it has to be measured again, not scaled.
1091 % Verified on the rendered page: the top of the digits and their right edge come
1092 % out at the same 2.019cm, and the right margin is 2cm by construction, so that
1093 % shared 0.019cm is the ruler's own bias and not a residual error.
1094 \RequirePackage[a4paper,headheight=15pt,headsep=16.87pt,%
1095 top=3cm,bottom=2cm,left=3cm,right=2cm]{geometry}
1096 % The first paragraph of every section must be indented like any other. LaTeX's
1097 % default is the Anglo-Saxon convention -- no indent after a heading -- which is
1098 % not what Brazilian typography (nor the SiBI manual's own examples) uses.
1099 % indentfirst was mentioned only as a commented suggestion in the example
1100 % preamble, so every document built with this class came out with the first

```

```

1101 % paragraph of each section flush left. Loading it in the class fixes it for
1102 % everybody instead of asking each author to remember.
1103 %
1104 % Loaded BEFORE titlesec so titlesec's \@afterindent handling sees the patched
1105 % \@afterindentfalse.
1106 % Long unbreakable tokens -- \verb|...| in running prose, command names, URLs --
1107 % cannot be hyphenated, so TeX overfills the line rather than stretching the
1108 % spaces. example.tex alone produced 17 overfull boxes this way, the worst 86pt,
1109 % which is the "text does not always respect the margins" a recent review
1110 % reported. \emergencystretch gives TeX a final pass in which it may stretch
1111 % interword spaces instead of running into the margin. It cannot save a single
1112 % token longer than the measure, but it clears the ordinary cases.
1113 \setlength\emergencystretch{3em}
1114 \RequirePackage{indentfirst}
1115 \RequirePackage{titlesec}
1116 % --- Section heading formats (manual2024 2.6 / proposta P22) -----
1117 % chapter (numbered): "N TITLE", ALL CAPS, bold, flush left (no "Capitulo N").
1118 % APENDICE/ANEXO chapters print "APENDICE A -- Title" (R75/R76); normal chapters
1119 % keep "N Title". \appendix and \annex raise the flag (see below).
1120 \newif\if@coppeappendix \@coppeappendixfalse
1121 \titleformat{\chapter}[hang]
1122   {\normalfont\Large\bfseries}
1123   {\if@coppeappendix\MakeUppercase{\appendixname}\ \thechapter\ \textendash\else\thechapter\
1124   {1ex}\{\MakeUppercase}
1125 \titlespacing*{\chapter}{0pt}{10pt}{1.5\baselineskip}
1126 % 2.6 says the numeric indicative of a section precedes its title flush left,
1127 % and that titles WITHOUT a numeric indicative are centred -- and then lists
1128 % them, "apendice(s)" and "anexo(s)" among them. A letter is not a numeric
1129 % indicative, and the manual sets its own Annex A centred. The chapter format is
1130 % therefore re-issued when \appendix or \annex is called: same label, same
1131 % travessao, but a centred block instead of a flush-left hang. Numbered chapters
1132 % are untouched -- they DO carry a numeric indicative and stay flush left.
1133 \newcommand\coppe@appendixheadings{%
1134   \titleformat{\chapter}[block]
1135     {\normalfont\Large\bfseries\filcenter}
1136     {\MakeUppercase{\appendixname}\ \thechapter\ \textendash}
1137     {1ex}\{\MakeUppercase}%
1138   \titlespacing*{\chapter}{0pt}{10pt}{1.5\baselineskip}}
1139 \let\coppe@orig@appendix\appendix
1140 \renewcommand\appendix{%
1141   \coppe@orig@appendix\@coppeappendixtrue\coppe@appendixheadings}
1142 % Sumario (TOC) entries for APENDICE/ANEXO chapters read "APENDICE A -- Title"
1143 % / "ANEXO A -- Title" (R75/R76), not the bare "A". The label word is chosen by
1144 % which macro is written (so it is correct at .toc-render time even though the
1145 % appendixname has changed back); the chapter letter is baked in at write time.
1146 \newif\if@coppeannex \@coppeannexfalse
1147 \newcommand\coppe@tocapp[1]{%
1148   \MakeUppercase{\coppestring{appendix}}%
1149   \nobreakspace#1\nobreakspace\textendash\nobreakspace}
1150 \newcommand\coppe@tocanx[1]{%
1151   \MakeUppercase{\coppestring{annex}}%
1152   \nobreakspace#1\nobreakspace\textendash\nobreakspace}
1153 \def\@chapter[#1]#2{%
1154   \ifnum \c@secnumdepth >\m@ne

```

```

1155 \if@mainmatter
1156 \refstepcounter{chapter}%
1157 \typeout{\@chapapp\space\thechapter.}%
1158 \if@coppeappendix
1159 \if@coppeannex
1160 \addcontentsline{toc}{chapter}{\protect\coppe@tocanx{\thechapter}#1}%
1161 \else
1162 \addcontentsline{toc}{chapter}{\protect\coppe@tocapp{\thechapter}#1}%
1163 \fi
1164 \else
1165 \addcontentsline{toc}{chapter}{\protect\numberline{\thechapter}#1}%
1166 \fi
1167 \else
1168 \addcontentsline{toc}{chapter}{#1}%
1169 \fi
1170 \else
1171 \addcontentsline{toc}{chapter}{#1}%
1172 \fi
1173 \chaptermark{#1}%
1174 \addtocontents{lof}{\protect\addvspace{10\p@}}%
1175 \addtocontents{lot}{\protect\addvspace{10\p@}}%
1176 \if@twocolumn
1177 \topnewpage[\@makechapterhead{#2}]%
1178 \else
1179 \makechapterhead{#2}%
1180 \afterheading
1181 \fi}
1182 % chapter (unnumbered): centred ALL CAPS bold (errata, lists, resumo, sumario,
1183 % referencias, glossario, ... -- unnumbered headings are centred, P29-P30).
1184 \titleformat{name=\chapter,numberless}[hang]
1185 {\normalfont\Large\bfseries\filcenter}{\Opt}{\MakeUppercase}
1186 \titlespacing*{name=\chapter,numberless}{0pt}{10pt}{1.5\baselineskip}
1187 % section: ALL CAPS, normal weight -- the \bfseries here contradicted this very
1188 % comment until v4.1, and made the secondary level as heavy as the primary one,
1189 % defeating the "destaque gradativo" of 2.6. The gradation is now
1190 % chapter = ALL CAPS bold, section = ALL CAPS regular, subsection = Title Case
1191 % bold, subsubsection = Title Case bold italic, as read from the manual.
1192 \titleformat{\section}
1193 {\normalfont\large}{\thesection}{1ex}{\MakeUppercase}
1194 % subsection: bold (author writes Title Case).
1195 \titleformat{\subsection}
1196 {\normalfont\bfseries}{\thesubsection}{1ex}{}
1197 % subsubsection: bold italic.
1198 \titleformat{\subsubsection}
1199 {\normalfont\bfseries\itshape}{\thesubsubsection}{1ex}{}
1200 % paragraph = quinary section: italic, the author capitalising only the first
1201 % word. 2.6 recommends going as deep as the quinary, and the annotation on the
1202 % 2025 manual (specs/ANOTACOES.md) describes exactly this shape.
1203 %
1204 % book makes \paragraph a RUN-IN heading, so the text continued on the same
1205 % line -- 2.6 requires "o texto deve ser iniciado em outra linha". The positive
1206 % "after" spacing in \titlespacing turns it into a displayed heading.
1207 \titleformat{\paragraph}[hang]
1208 {\normalfont\itshape}{\theparagraph}{1ex}{}

```

```

1209 \titlespacing*{\paragraph}{0pt}{2.5ex plus 1ex minus .2ex}{1.5ex plus .2ex}
1210 % book defaults secnumdepth to 2, so \subsubsection and \paragraph came out
1211 % UNNUMBERED however carefully \titleformat named \thesubsubsection. 2.6 has
1212 % the numbering following the whole chain down to the quinary, and the manual's
1213 % own sumario is four levels deep, so tocdepth follows.
1214 \setcounter{secnumdepth}{4}
1215 \setcounter{tocdepth}{4}
1216 % 2.6: "no sumario, as secões devem ser grafadas conforme apresentadas no corpo
1217 % do trabalho". The body sets chapter and section in ALL CAPS (via the
1218 % \MakeUppercase in their \titleformat), but that case change belongs to the
1219 % printed heading alone -- the .toc keeps the title as typed, so the sumario
1220 % came out in ordinary case and contradicted the body.
1221 %
1222 % \l@chapter and \l@section are the two TOC line formatters; wrapping their
1223 % entry argument is enough, and leaves \l@subsection and below untouched, which
1224 % is what we want since those levels are NOT uppercased in the body either.
1225 % The entry carries \numberline{N} plus the title; \MakeUppercase passes the
1226 % digits through unchanged and leaves robust commands (including hyperref's
1227 % link wrappers) intact.
1228 \let\coppe@orig@lchapter\l@chapter
1229 \renewcommand\l@chapter[2]{\coppe@orig@lchapter{\MakeUppercase{#1}}{#2}}
1230 \let\coppe@orig@lsection\l@section
1231 \renewcommand\l@section[2]{\coppe@orig@lsection{\MakeUppercase{#1}}{#2}}
1232 % --- Captions (manual2024 2.10-2.11 / proposta P6): smaller font, ABNT
1233 % "Figura 1 -- título" travessao separator (sources handled by \source). ---
1234 \RequirePackage{caption}
1235 % justification=centering makes MULTI-LINE captions centre like the one-line
1236 % ones. Without it, caption's singlelinecheck centres a caption that fits on one
1237 % line and justifies one that wraps, so in a document with mixed caption lengths
1238 % a couple of tables looked left-aligned while the rest looked centred -- which
1239 % is what a recent review noticed about tables 5.1 and 5.2. 2.10 does not
1240 % prescribe the alignment; it does not tolerate two alignments either.
1241 \captionsetup{font=footnotesize,labelsep=endash,justification=centering}
1242 % subcaption (NÃO subfig/subfigure) for subfigures / subtables; load right after
1243 % caption so it picks up the ABNT captionsetup above.
1244 \RequirePackage{subcaption}
1245 % --- Floats: caption on TOP and a mandatory "Fonte:" line at the bottom.
1246 % manual2024 2.10-2.11 (R46/R51: identification on top; R48/R52: the source
1247 % at the bottom is mandatory). The float package's plaintop style forces the
1248 % caption above the body for figure, table and the new quadro float. -----
1249 \RequirePackage{float}
1250 \floatstyle{plaintop}
1251 \restylefloat{figure}
1252 \restylefloat{table}
1253 % New "quadro" float -- "Quadro" (pt) / "Frame" (en) (R53) -- with its own list
1254 % file (.loq). A quadro's data is bordered on all sides (drawn by the author),
1255 % unlike a tabela (rules on top and bottom only).
1256 \providecommand{\quadroname}{\coppestring{frame}}
1257 \providecommand{\listquadroname}{\coppestring{listframe}}
1258 \newfloat{quadro}{htbp}{loq}[chapter]
1259 \floatname{quadro}{\quadroname}
1260 \providecommand{\quadroautorefname}{\quadroname}
1261 % Format the Lista de Quadros entries exactly like the Lista de Figuras ones
1262 % (float's own \l@quadro lays the number/title out incorrectly).

```

```

1263 \let\l@quadro\l@figure
1264 \newcommand\listofquadros{%
1265   \chapter*{\listquadroname}%
1266   \coppe@listtoc{\listquadroname}%
1267   \@mkboth{\MakeUppercase\listquadroname}{\MakeUppercase\listquadroname}%
1268   \@starttoc{loq}}
1269 % \source / \fonte: the illustration source (ABNT-mandatory). Typeset below the
1270 % float as a smaller, single-spaced, centred line; it does NOT advance any
1271 % caption counter. \cpsource/\cpfonte are clash-safe aliases (the rename rule);
1272 % the label is "Fonte" (pt) / "Source" (en).
1273 \newcommand{\cpsourcenam}{\coppestring{source}}
1274 \newcommand{\cpsource}[1]{%
1275   \par\addvspace{\smallskipamount}%
1276   {\footnotesize\singlespacing\centering\cpsourcenam: #1\par}}
1277 \let\cpfonte\cpsource
1278 \@ifundefined{source}{\let\source\cpsource}{}%
1279 \@ifundefined{fonte}{\let\fonte\cpsource}{}%
1280 % \newcoppefloat{env}{caption name}{list title}: declare an extra caption-on-top
1281 % float (e.g. "picture") that supports \source and gains a \listof<env>s command.
1282 \newcommand{\newcoppefloat}[3]{%
1283   {\floatstyle{plaintop}\newfloat{#1}{htbp}{lo#1}}%
1284   \floatname{#1}{#2}%
1285   \expandafter\newcommand\csname listof#1\endcsname{%
1286     \chapter*{#3}\coppe@listtoc{#3}%
1287     \@mkboth{\MakeUppercase{#3}}{\MakeUppercase{#3}}%
1288     \@starttoc{lo#1}}}}
1289 % --- Footnotes (manual2024 / R16, R8, R21): single spacing and a 5 cm rule
1290 % (filete) from the left margin; footnotesize is the book default. -----
1291 \let\coppe@orig@footnotetext\@footnotetext
1292 \long\def\@footnotetext#1{\coppe@orig@footnotetext{\singlespacing#1}}
1293 \renewcommand\footnoterule{\kern-3\p@\hrule\@width 5cm\kern 2.6\p@}
1294 % --- alineas: list keyed by lowercase letters a) b) c); subitems with "--"
1295 % (manual2024 2.6 / proposta P24). -----
1296 \RequirePackage{enumitem}
1297 \newlist{alineas}{enumerate}{2}
1298 \setlist{alineas,1}{label=\alph*},leftmargin=1.8em,itemsep=0pt}
1299 \setlist{alineas,2}{label=--,leftmargin=1.8em,itemsep=0pt}
1300 % --- Page numbering: arabic in the top OUTER corner on textual pages
1301 % (manual2024 2.7 / proposta P25-P26). Pre-textual numbering is set by
1302 % \frontmatter (unnumbered by default; roman with the numeraisromanos
1303 % class option). -----
1304 % Every list goes through \@starttoc, which allocates a stream the first time
1305 % it sees an extension. That is where the ceiling is hit in practice, so that
1306 % is where the class checks -- one \ifnum before the allocation, and an error
1307 % that names the three ways out instead of the kernel's bare "No room".
1308 %
1309 % \count17 is the write allocation counter, and 15 is the last usable stream.
1310 % Under LuaTeX there are 128 and the counter is managed differently, so the
1311 % check is skipped there; the same is true when morewrites is in charge.
1312 % The wrap waits for \begin{document} on purpose: hyperref redefines
1313 % \@starttoc as well, and it is loaded at the very end of this class, so a
1314 % \let taken here would be thrown away.
1315 \newcommand\coppe@checkwrites[1]{%
1316   \ifdefined\directlua\else

```



```

1317 \if@coppemorewrites\else
1318 \@ifundefined{tf@#1}{%
1319 \ifnum\count17>14
1320 \ClassError{coppe}{Acabaram os 16 fluxos de escrita do TeX}{%
1321 Este trabalho pede mais listas do que o pdfTeX consegue manter
1322 abertas ao mesmo tempo.\MessageBreak
1323 Tres saidas, nesta ordem de preferencia:\MessageBreak
1324 (1) compile com lualatex, que tem 128 fluxos;\MessageBreak
1325 (2) passe a opcao 'morewrites' a classe coppe;\MessageBreak
1326 (3) remova uma lista que o trabalho nao usa.\MessageBreak
1327 A lista que faltou agora foi a de extensao '#1', mas o problema
1328 nao e dela: e a soma de todas.}%
1329 \fi}{}%
1330 \fi
1331 \fi}
1332 % The wrapping itself lives in a macro of its own rather than inside the hook:
1333 % \AtBeginDocument stores its argument by expansion, and a parameter written
1334 % there does not survive it -- the document died with "Parameters must be
1335 % numbered consecutively" at \begin{document}. One level of nesting, one
1336 % doubling, and the hook only has to call it.
1337 \newcommand\coppe@wrapstarttoc{%
1338 \let\coppe@orig@starttoc@starttoc
1339 \def\@starttoc##1{\coppe@checkwrites{##1}\coppe@orig@starttoc{##1}}
1340 \AtBeginDocument{\coppe@wrapstarttoc}
1341 \RequirePackage{fancyhdr}
1342 % \headheight is set to 15pt by geometry up above (see the geometry call),
1343 % which is what fancyhdr needs (>= 14.49998pt) for the running head.
1344 % 2.2(b) lists what is set in "fonte menor e uniforme": long quotations,
1345 % footnotes, PAGINATION, and the captions of illustrations and tables. The
1346 % folio was the one item on that list still coming out at the body size of 12pt.
1347 % It is \footnotesize now, which is the same smaller size the other three
1348 % already use -- "uniforme" is part of the requirement, not just "menor".
1349 % Pre-textual lists reach the sumario only under the listasnosumario option
1350 % (see the option's declaration for the reason). Post-textual material calls
1351 % \addcontentsline directly and is unaffected.
1352 \newcommand\coppe@listtoc[1]{%
1353 \if@coppe@listastoc\addcontentsline{toc}{chapter}{#1}\fi}
1354 % \coppe@glosspre marks the two PRE-textual uses of the glossary environment --
1355 % the list of abbreviations and the list of symbols -- apart from the genuine
1356 % post-textual Glossario, which shares the same environment but must appear in
1357 % the sumario.
1358 \newif\if@coppeglosspre \@coppeglossprefalse
1359 \newcommand\coppe@folio{\footnotesize\thepage}
1360 \fancypagestyle{coppe}{\fancyhf{}\fancyhead[R0,LE]{\coppe@folio}%
1361 \renewcommand{\headrulewidth}{0pt}}
1362 % Blank pages inserted by \cleardoublepage (two-sided) must be empty: no page
1363 % number may leak onto them (e.g. the verso right after the cover).
1364 \renewcommand{\cleardoublepage}{%
1365 \clearpage
1366 \if@twoside
1367 \ifodd\c@page\else
1368 \hbox{}\thispagestyle{empty}\newpage
1369 \if@twocolumn\hbox{}\newpage\fi
1370 \fi

```

```

1371 \fi}
1372 \def\CoppeTeX{\rm C\kern-.05em{\sc o\kern-.025em p\kern-.025em
1373 p\kern-.025em e}}\kern-.08em
1374 T\kern-.1667em\lower.5ex\hbox{E}\kern-.125emX\spacefactor1000}

1375 \newboolean{maledoc}
1376 \setboolean{maledoc}{false}
1377 %
1378 % Class options.
1379 % Multilingual model (CoppeTeX 4.x, branch nlinguas). The class holds three
1380 % fixed language slots:
1381 % * main -- the language of the body, captions, and the main abstract.
1382 % Set by a class option: brazilian (default), english,
1383 % spanish, french, italian, or any user-supplied language
1384 % that ships a coppe-lang-<name>.def file.
1385 % * foreign -- the language used by the foreignabstract environment and
1386 % by \foreign@... registers. Defaults to english (or to
1387 % brazilian when main=english) for back-compat.
1388 % * third -- optional, loaded on demand with \usecoppelanguage{<name>}
1389 % in the preamble (for citations or quotations in a third
1390 % language). Has no class option of its own.
1391 % Note: the legacy \if@english boolean (defined and toggled here in
1392 % v3.x for backward compatibility) was removed in v4.0 -- nothing
1393 % inside the class or in the biblatex style files (.bbx/.cbx/.dbx/.ltx)
1394 % tests it. Code that needs to branch on the main language should test
1395 % \ifx\coppe@mainlang\coppe@englishstr instead.
1396 \def\coppe@mainlang{brazilian}
1397 \def\coppe@foreignlang{english}
1398 \DeclareOption{brazilian}{%
1399 \def\coppe@mainlang{brazilian}\def\coppe@foreignlang{english}}
1400 \DeclareOption{english}{%
1401 \def\coppe@mainlang{english}\def\coppe@foreignlang{brazilian}}
1402 \DeclareOption{spanish}{%
1403 \def\coppe@mainlang{spanish}\def\coppe@foreignlang{english}}
1404 \DeclareOption{french}{%
1405 \def\coppe@mainlang{french}\def\coppe@foreignlang{english}}
1406 \DeclareOption{italian}{%
1407 \def\coppe@mainlang{italian}\def\coppe@foreignlang{english}}
1408 \DeclareOption{msc}{%
1409 \newcommand{\@degree}{M.Sc.}
1410 \newcommand{\@degreename}{Mestrado}
1411 \newcommand{\local@degname}{Mestre}
1412 \newcommand{\foreign@degname}{Master}
1413 \newcommand\local@doctype{Disserta{\c c}{\~ a}o}
1414 \newcommand\foreign@doctype{Dissertation}
1415 }
1416 \DeclareOption{dsccexam}{%
1417 \newcommand{\@degree}{D.Sc.}
1418 \newcommand{\@degreename}{Doutorado}
1419 \newcommand{\local@degname}{Doutor}
1420 \newcommand{\foreign@degname}{Doctor}
1421 \setboolean{maledoc}{true}
1422 \newcommand\local@doctype{Exame de Qualifica{\c c}{\~ a}o}
1423 \newcommand\foreign@doctype{Qualifying Exam}
1424 }

```

```

1425 \DeclareOption{mscexam}{%
1426 \newcommand{\@degree}{M.Sc.}
1427 \newcommand{\@degreename}{Mestrado}
1428 \newcommand{\local@degname}{Mestre}
1429 \newcommand{\foreign@degname}{Master}
1430 \setboolean{maledoc}{true}
1431 \newcommand\local@doctype{Exame de Qualifica{\c c}{\~ a}o}
1432 \newcommand\foreign@doctype{Qualifying Exam}
1433 }
1434 \DeclareOption{dsc}{%
1435 \newcommand{\@degree}{D.Sc.}
1436 \newcommand{\@degreename}{Doutorado}
1437 \newcommand{\local@degname}{Doutor}
1438 \newcommand{\foreign@degname}{Doctor}
1439 \newcommand\local@doctype{Tese}
1440 \newcommand\foreign@doctype{Thesis}
1441 }
1442 \newif\if@coppenumeric\@coppenumericfalse
1443 \DeclareOption{numbers}{\@coppenumerictrue}
1444 % numeraisromanos: show roman numerals on the pre-textual pages. Default: the
1445 % pre-textual pages are counted but not numbered (manual2024 2.7).
1446 \newif\if@copperoman\@copperomanfalse
1447 \DeclareOption{numeraisromanos}{\@copperomantrue}

```

Default line spacing is one-and-a-half; the doublespacing option switches to double spacing. Since `setspace` is loaded at the end of the class (after `hyperref`), the actual `\onehalfspacing`/`\doublespacing` call is deferred: the option only flips a flag, which the late-load block reads.

```

1448 \newif\if@coppedoublespace\@coppedoublespacefalse
1449 \DeclareOption{doublespacing}{\@coppedoublespacetrue}
1450 % Two-sided layout is opt-in since v4.1: the 2026 manual dropped the verso
1451 % margins and the deposit is digital (CEPG 246/2023). Given \texttt{twoside},
1452 % the mirrored 3cm inner / 2cm outer layout is restored after \ProcessOptions.
1453 % With no \fichacatalografica file given, the folha adicional leaves a framed
1454 % blank where the ficha goes. The rascunhoficha option fills that blank with the
1455 % class's own LaTeX-composed ficha instead -- useful while writing, never valid
1456 % for deposit: the official ficha comes from the SiBI generator.
1457 % Annex D of the manual draws one signature rule per board member, with the
1458 % identification underneath. The COPPE folha de aprovacao has listed the names
1459 % compactly, without rules, for years. The compact list stays the default and
1460 % the rules are opt-in, so existing theses do not change shape overnight.
1461 % PDF/A output. Opt-in for now: 2.2(d) of the 2026 manual requires the deposited
1462 % file to be PDF/A, but the constraints (embedded fonts, colour profile, no
1463 % encryption, restricted transparency) can trip up a document that pulls in
1464 % arbitrary images or packages, and breaking existing theses to enforce it would
1465 % be worse than letting the author switch it on when depositing.
1466 \newif\if@coppepdfa \@coppepdfafalse
1467 \DeclareOption{pdfa}{\@coppepdfatrue}
1468 \newif\if@coppeassin \@coppeassinfalse
1469 \DeclareOption{assinaturas}{\@coppeassintrue}
1470 % As paginas de resumo listam os ORIENTADORES abaixo do titulo. O coorientador
1471 % nao aparece ali, e isso e tradicao da COPPE, nao exigencia do manual: a
1472 % 3.1.2.1.4 descreve o resumo e nao lista quem assina a orientacao. Como ha
1473 % Programas que preferem ve-lo, a opcao acrescenta os coorientadores nas tres

```

```

1474 % paginas de resumo, no mesmo formato da folha de rosto. Sem a opcao nada muda.
1475 \newif\if@coppecoadvresumo \@coppecoadvresumofalse
1476 \DeclareOption{coorientador}{\@coppecoadvresumotrue}
1477 % 3.1.2.1.6 says the sumario enumerates the divisions "na mesma ordem e grafia
1478 % em que a materia nele se sucede" and adds "inclusive os elementos
1479 % pos-textuais" -- and then settles the question by pointing at its own sumario
1480 % as the model. That sumario opens at "1 INTRODUCAO": the lists of figures, of
1481 % tables, of abbreviations and of symbols come BEFORE the sumario and are not
1482 % listed in it, which is also the rule of NBR 6027 for pre-textual elements.
1483 % The class was adding all of them, so every thesis carried six entries the
1484 % manual's own example does not have. They come out. Post-textual elements --
1485 % Referencias, Glossario, Apendices, Anexos, Indice -- stay, as required.
1486 %
1487 % Programas that want the old behaviour ask for it by name.
1488 \newif\if@coppelistastoc \@coppelistastocfalse
1489 \DeclareOption{listasnosumario}{\@coppelistastoctrue}
1490 \newif\if@copperascunho \@copperascunhofalse
1491 \DeclareOption{rascunhoficha}{\@copperascunhotrue}
1492 \newif\if@coppetwoside \@coppetwosidefalse
1493 \DeclareOption{twoside}{\@coppetwosidetrue}
1494 \DeclareOption{oneside}{\@coppetwosidefalse}
1495 % --- Fluxos de escrita -----
1496 % TeX has SIXTEEN \write streams and no more. Every list costs one for the
1497 % whole run: the .toc, the .lof, the .lot, and one for each list this class
1498 % adds (.loq for quadros, .lol for programas, .loa for algoritmos), plus the
1499 % .abx and .syx behind the lists of abbreviations and of symbols, the .idx of
1500 % the index, hyperref's .out, biblalex's .bcf and the kernel's .aux and friends.
1501 % A work that asks for all of them sits exactly at the ceiling.
1502 %
1503 % The kernel's failure in that situation -- "No room for a new \write" -- names
1504 % neither the cause nor the way out, and lands on whichever list happens to be
1505 % requested last, which is usually \tableofcontents and has nothing to do with
1506 % it. The class therefore checks before the allocation and says what is
1507 % happening (see \coppe@starttoc below).
1508 %
1509 % There are three ways out and the class chooses none of them for the author:
1510 %
1511 % * compile with LuaLaTeX. It has 128 streams; nothing else has to change,
1512 %   and the class supports it.
1513 % * let the class load 'morewrites', which multiplexes the streams so the
1514 %   ceiling disappears under pdfLaTeX.
1515 % * drop a list the work does not use. Each one returns a stream.
1516 %
1517 % The DEFAULT is the second one, and it is on. An error telling the author to
1518 % pass an option is advice for someone who already knows: a student two days
1519 % from the deposit reads "No room for a new \write" and concludes the class is
1520 % broken. So under pdfTeX the class loads morewrites by itself.
1521 %
1522 % Three things keep that from becoming a burden:
1523 %
1524 % * it is loaded ONLY IF INSTALLED. Missing, the class carries on without it
1525 %   and warns; morewrites does not become a hard dependency of a work that
1526 %   would never have reached the ceiling anyway.
1527 % * 'senmorewrites' turns it off, for whoever hits a conflict. morewrites

```

```

1528 % patches \newwrite, \openout and \write and buffers what is written until
1529 % shipout, so a package that writes a file and reads it back in the SAME run
1530 % can misbehave. That is the real cost of this default, and it is why the
1531 % escape exists.
1532 % * 'morewrites' still exists and now means "I insist": with it, a missing
1533 % package is an error rather than a warning.
1534 %
1535 % Under LuaTeX none of this applies and nothing is loaded.
1536 \newif\if@coppemorewrites \@coppemorewritestrue
1537 \newif\if@coppemwforced \@coppemwforcedfalse
1538 \DeclareOption{morewrites}{\@coppemorewritestrue\@coppemwforcedtrue}
1539 \DeclareOption{semmorewrites}{\@coppemorewritesfalse}
1540 \ProcessOptions\relax
1541 % morewrites has to come before anything else allocates, so it goes first
1542 % thing after the options are read.
1543 \if@coppemorewrites
1544   \ifdefined\directlua
1545     % Under LuaTeX there is nothing to solve: 128 streams already. The flag is
1546     % cleared too, so the \starttoc guard further down knows the ceiling is not
1547     % in play here either.
1548     \ClassInfo{coppe}{morewrites dispensado: o LuaTeX ja tem 128 fluxos}%
1549     \@coppemorewritesfalse
1550   \else
1551     \IfFileExists{morewrites.sty}
1552     {\RequirePackage{morewrites}}
1553     {\if@coppemwforced
1554       \ClassError{coppe}{Opcao 'morewrites' pedida, mas morewrites.sty nao
1555         esta instalado}{Instale o pacote morewrites, ou compile com lualatex,
1556         que nao precisa dele.}%
1557       \else
1558         \ClassWarningNoLine{coppe}{0 pacote morewrites nao esta instalado.\MessageBreak
1559           Sem ele o pdfTeX so aguenta 16 fluxos de escrita, e um trabalho\MessageBreak
1560           com todas as listas passa disso. Se a compilacao parar por falta\MessageBreak
1561           de fluxo, instale morewrites ou compile com lualatex}%
1562         \@coppemorewritesfalse
1563       \fi}%
1564   \fi
1565 \fi
1566 % \LoadClass and geometry ran before \ProcessOptions (see above), so the
1567 % two-sided layout can only be applied here. geometry reads left/right as
1568 % inner/outer once twoside is on, so the 3cm/2cm pair set above becomes the
1569 % mirrored gutter with no further arithmetic. \@twosidetrue is what book's own
1570 % twoside option sets, and is what \cleardoublepage and fancyhdr test.
1571 % In one-sided mode fancyhdr treats every page as odd, so the \fancyhead[RO,LE]
1572 % declarations elsewhere already put the folio in the top right corner of every
1573 % folha, as 2.7 requires.
1574 \if@coppetwoside
1575   \@twosidetrue
1576   \geometry{twoside}%
1577 \fi
1578 % Built-in language names (used by the loader and the dispatcher below).
1579 \def\coppe@brazilianstr{brazilian}
1580 \def\coppe@englishstr{english}
1581 % Load Babel with the foreign language first and the main language last

```

```

1582 % (Babel treats the LAST option in the list as the main language). With
1583 % pt-main this produces [english,brazilian]{babel} and with en-main
1584 % [brazilian,english] -- identical to the historical \if@english calls.
1585 \edef\coppe@babelopts{\coppe@foreignlang,\coppe@mainlang}
1586 % Some Babel modules change LaTeX beyond translating captions, and one of
1587 % those changes breaks the norm. With Spanish as the MAIN language,
1588 % babel-spanish redefines \numberline to put a period after the section
1589 % number, so the sumario reads "2.1. SECCION SECUNDARIA" while the section
1590 % heading itself -- which the class formats -- reads "2.1 SECCION SECUNDARIA".
1591 % The 2.6 separates the indicative from the title by a space (NBR 6024) and
1592 % the 3.1.2.1.6 wants the sumario to reproduce the text; the period fails
1593 % both. 'es-nosectiondot' turns off that single adjustment and leaves the rest
1594 % of babel-spanish (quoting, shorthands, roman numerals, captions) alone.
1595 % \coppe@babelextra@<lang> is the general hook: it applies to the MAIN
1596 % language only, which is where these layout modules act, and it is kept out
1597 % of \coppe@babelopts because the language-pack loader below iterates that
1598 % list and would look for coppe-lang-es-nosectiondot.def.
1599 \@namedef{coppe@babelextra@spanish}{,es-nosectiondot}
1600 \let\coppe@babelload\coppe@babelopts
1601 \@ifundefined{coppe@babelextra@\coppe@mainlang}{%
1602   \edef\coppe@babelload{\coppe@babelload\@nameuse{coppe@babelextra@\coppe@mainlang}}
1603   \expandafter\RequirePackage\expandafter[\coppe@babelload]{babel}
1604 % --- Multilingual string dispatcher -----
1605 % \copperdefstring{<lang>}{<key>}{<value>} adds one entry to the table for
1606 % language <lang>. The three retrievers below look strings up either at the
1607 % current babel language (\coppestring -- follows \selectlanguage), or pinned
1608 % to the main slot (\coppemainstring) or the foreign slot (\coppeforeignstring).
1609 \providecommand{\copperdefstring}[3]{%
1610   \expandafter\gdef\csname coppe@str@#1@#2\endcsname{#3}}
1611 \providecommand{\coppestring}[1]{\@nameuse{coppe@str\@language @#1}}
1612 \providecommand{\coppemainstring}[1]{\@nameuse{coppe@str\@coppe@mainlang @#1}}
1613 \providecommand{\coppeforeignstring}[1]{\@nameuse{coppe@str\@coppe@foreignlang @#1}}
1614 % Load an extra language pack at any point in the preamble (third-language
1615 % slot for citations or quotations beyond the main+foreign pair).
1616 \providecommand{\usecoppelanguage}[1]{%
1617   \def\coppe@@cur{#1}%
1618   \ifx\coppe@@cur\coppe@@brazilianstr\else
1619     \ifx\coppe@@cur\coppe@@englishstr\else
1620       \InputIfFileExists{coppe-lang-#1.def}{}
1621       {\ClassWarning{coppe}{Language pack coppe-lang-#1.def not found}}%
1622     \fi\fi}
1623 % \titlein{<lang>}{<text>}: record the title in any language. Will be
1624 % consulted by the cover/folha renderers once they are migrated in step 2.
1625 % In step 1 it is just a no-op store -- the historical \title / \foreigntitle
1626 % still drive every visible site, so behavior for pt/en is unchanged.
1627 \providecommand{\titlein}[2]{%
1628   \expandafter\gdef\csname coppe@title@#1\endcsname{#2}}
1629 % --- Built-in string tables: brazilian + english. The values mirror the
1630 % literals currently scattered through the class; step-1 only populates the
1631 % tables -- no site reads them yet, so the rendered output is unchanged. ----
1632 \copperdefstring{brazilian}{appendix}{\Ap~endice}
1633 \copperdefstring{english}{appendix}{Appendix}
1634 \copperdefstring{brazilian}{annex}{\Anexo}
1635 \copperdefstring{english}{annex}{Annex}

```

1636	\copperdefstring{brazilian}{frame}	{Quadro}
1637	\copperdefstring{english}{frame}	{Frame}
1638	\copperdefstring{brazilian}{listframe}	{Lista de Quadros}
1639	\copperdefstring{english}{listframe}	{List of Frames}
1640	\copperdefstring{brazilian}{source}	{Fonte}
1641	\copperdefstring{english}{source}	{Source}
1642	\copperdefstring{brazilian}{program}	{Programa}
1643	\copperdefstring{english}{program}	{Program}
1644	\copperdefstring{brazilian}{listprogram}	{Lista de Programas}
1645	\copperdefstring{english}{listprogram}	{List of Programs}
1646	\copperdefstring{brazilian}{references}	{Refer\~encias}
1647	\copperdefstring{english}{references}	{References}
1648	\copperdefstring{brazilian}{in:}	{em}
1649	\copperdefstring{english}{in:}	{in}
1650	\copperdefstring{brazilian}{art}	{Ilustra\c{c}\~ao}
1651	\copperdefstring{english}{art}	{Art}
1652	\copperdefstring{brazilian}{platform}	{Plataforma}
1653	\copperdefstring{english}{platform}	{Platform}
1654	\copperdefstring{brazilian}{directedby}	{Dire\c{c}\~ao}
1655	\copperdefstring{english}{directedby}	{Directed by}
1656	\copperdefstring{brazilian}{producedby}	{Produ\c{c}\~ao}
1657	\copperdefstring{english}{producedby}	{Produced by}
1658	\copperdefstring{brazilian}{listabbreviation}	{Lista de Abreviaturas e Siglas}
1659	\copperdefstring{english}{listabbreviation}	{List of Abbreviations and Acronyms}
1660	\copperdefstring{brazilian}{listsymbol}	{Lista de S{\` i}mbolos}
1661	\copperdefstring{english}{listsymbol}	{List of Symbols}
1662	\copperdefstring{brazilian}{glossary}	{Gloss{\` a}rio}
1663	\copperdefstring{english}{glossary}	{Glossary}
1664	\copperdefstring{brazilian}{advisor}	{Orientador}
1665	\copperdefstring{english}{advisor}	{Advisor}
1666	\copperdefstring{brazilian}{advisors}	{Orientadores}
1667	\copperdefstring{english}{advisors}	{Advisors}
1668	\copperdefstring{brazilian}{dept}	{Programa}
1669	\copperdefstring{english}{dept}	{Department}
1670	\copperdefstring{brazilian}{approvedmale}	{Aprovado por}
1671	\copperdefstring{brazilian}{approvedfemale}	{Aprovada por}
1672	\copperdefstring{brazilian}{approvedonmale}	{Aprovado em}
1673	\copperdefstring{brazilian}{approvedonfemale}	{Aprovada em}
1674	\copperdefstring{brazilian}{keywordsname}	{Palavras-chave}
1675	\copperdefstring{brazilian}{volumename}	{Volume}
1676	\copperdefstring{brazilian}{coadvisor}	{Coorientador}
1677	\copperdefstring{brazilian}{coadvisors}	{Coorientadores}
1678	\copperdefstring{brazilian}{ofname}	{de}
1679	\copperdefstring{english}{approvedmale}	{Approved by}
1680	\copperdefstring{english}{approvedfemale}	{Approved by}
1681	\copperdefstring{english}{approvedonmale}	{Approved on}
1682	\copperdefstring{english}{approvedonfemale}	{Approved on}
1683	\copperdefstring{english}{keywordsname}	{Keywords}
1684	\copperdefstring{english}{volumename}	{Volume}
1685	\copperdefstring{english}{coadvisor}	{Co-advisor}
1686	\copperdefstring{english}{coadvisors}	{Co-advisors}
1687	\copperdefstring{english}{ofname}	{of}
1688	% Institutional names: always Portuguese on the cover/folha (UFRJ template).	
1689	\copperdefstring{brazilian}{universityname}	{Universidade Federal do Rio de Janeiro}

```

1690 \copperdefstring{english} {universityname}{Universidade Federal do Rio de Janeiro}
1691 \copperdefstring{brazilian}{cityname} {Rio de Janeiro}
1692 \copperdefstring{english} {cityname} {Rio de Janeiro}
1693 \copperdefstring{brazilian}{statename} {RJ}
1694 \copperdefstring{english} {statename} {RJ}
1695 \copperdefstring{brazilian}{countryname} {Brasil}
1696 \copperdefstring{english} {countryname} {Brasil}
1697 % Month names.
1698 \copperdefstring{brazilian}{monthname1} {Janeiro}
1699 \copperdefstring{brazilian}{monthname2} {Fevereiro}
1700 \copperdefstring{brazilian}{monthname3} {Mar{\c c}o}
1701 \copperdefstring{brazilian}{monthname4} {Abril}
1702 \copperdefstring{brazilian}{monthname5} {Maio}
1703 \copperdefstring{brazilian}{monthname6} {Junho}
1704 \copperdefstring{brazilian}{monthname7} {Julho}
1705 \copperdefstring{brazilian}{monthname8} {Agosto}
1706 \copperdefstring{brazilian}{monthname9} {Setembro}
1707 \copperdefstring{brazilian}{monthname10}{Outubro}
1708 \copperdefstring{brazilian}{monthname11}{Novembro}
1709 \copperdefstring{brazilian}{monthname12}{Dezembro}
1710 \copperdefstring{english}{monthname1} {January}
1711 \copperdefstring{english}{monthname2} {February}
1712 \copperdefstring{english}{monthname3} {March}
1713 \copperdefstring{english}{monthname4} {April}
1714 \copperdefstring{english}{monthname5} {May}
1715 \copperdefstring{english}{monthname6} {June}
1716 \copperdefstring{english}{monthname7} {July}
1717 \copperdefstring{english}{monthname8} {August}
1718 \copperdefstring{english}{monthname9} {September}
1719 \copperdefstring{english}{monthname10}{October}
1720 \copperdefstring{english}{monthname11}{November}
1721 \copperdefstring{english}{monthname12}{December}
1722 % algorithm2e package option for this language (empty -> english default).
1723 \copperdefstring{brazilian}{algorithm2eopt}{portuguese}
1724 \copperdefstring{english} {algorithm2eopt}{}
1725 % Babel package name for the foreignabstract \begin{otherlanguage}{...}.
1726 \copperdefstring{brazilian}{babelname}{brazilian}
1727 \copperdefstring{english} {babelname}{english}
1728 % Auto-load language packs for any non-built-in language listed in babelopts.
1729 % Built-ins (brazilian, english) ship their string tables inside coppe.cls
1730 % (defined just above) and need no extra file; others (spanish/french/italian
1731 % /...) ship as coppe-lang-<name>.def alongside the class. The dispatcher
1732 % must already be defined when these run, so the loader sits AFTER the tables.
1733 \@for\coppe@@lang:=\coppe@babelopts\do{%
1734 \edef\coppe@@cur{\coppe@@lang}%
1735 \ifx\coppe@@cur\coppe@@brazilianstr\else
1736 \ifx\coppe@@cur\coppe@@englishstr\else
1737 \InputIfFileExists{coppe-lang-\coppe@@cur.def}
1738 {\ClassInfo{coppe}{Loaded language pack coppe-lang-\coppe@@cur.def}}
1739 {\ClassError{coppe}
1740 {Missing language pack 'coppe-lang-\coppe@@cur.def'.}
1741 {Install it next to coppe.cls, or pick a built-in language.}}}%
1742 \fi\fi}
1743 % Quotation marks: csquotes gives the language-aware \enquote; \citacao is its

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1744 % Portuguese alias for short, inline quotations (proposta P3).
1745 \RequirePackage{csquotes}
1746 \newcommand{\citacao}[1]{\enquote{#1}}
1747 % --- Code listings & algorithms (TODO3 "tratar listagens"): the listings
1748 % package gives the "Programa" listing (caption on top, ABNT R46) with
1749 % \listofprogramas; algorithm2e gives algorithms. Names are babel-aware and
1750 % the language styles below are based on the user's listagens.tex. -----
1751 \RequirePackage{xcolor}
1752 \RequirePackage{listings}
1753 \RequirePackage{fancyvrb}
1754 \DeclareFixedFont{\ttb}{T1}{txtt}{bx}{n}{10}
1755 \DeclareFixedFont{\ttm}{T1}{txtt}{m}{n}{10}
1756 \lstset{captionpos=t,extendedchars=true,basicstyle=\ttfamily,inputencoding=utf8,%
1757 literate=%
1758 {ã}{\~{a}}1 {ä}{\~{a}}1 {å}{\~{a}}1 {à}{\~{a}}1 {é}{\~{e}}1%
1759 {ê}{\~{e}}1 {ë}{\~{e}}1 {î}{\~{i}}1 {í}{\~{i}}1 {ô}{\~{o}}1%
1760 {ó}{\~{o}}1 {õ}{\~{o}}1 {û}{\~{u}}1 {ú}{\~{u}}1 {ç}{\c{c}}1%
1761 {Ç}{\c{C}}1 {À}{\~{A}}1 {Â}{\~{A}}1 {Á}{\~{A}}1 {Ê}{\~{E}}1%
1762 {Ë}{\~{E}}1 {Î}{\~{I}}1 {Ï}{\~{I}}1 {Ô}{\~{O}}1 {Õ}{\~{O}}1%
1763 {Û}{\~{U}}1%
1764 }
1765 % Babel-aware listing names, set after listings' own babel setup.
1766 \AtBeginDocument{%
1767 \renewcommand{\lstlistingname}{\coppestring{program}}%
1768 \renewcommand{\lstlistlistingname}{\coppestring{listprogram}}}
1769 % \listofprogramas in the coppe list style (heading + ToC + running head).
1770 \newcommand{\listofprogramas}{%
1771 \chapter*{\lstlistlistingname}%
1772 \coppe@listtoc{\lstlistlistingname}%
1773 \mkboth{\MakeUppercase\lstlistlistingname}{\MakeUppercase\lstlistlistingname}%
1774 \@starttoc{lol}}
1775 % Per-language styles + environments (python, java, xml, html, prolog).
1776 % The red hook that marks a wrapped line is built ONCE, into a box, and the
1777 % styles below insert the box. Writing the colour inline -- postbreak=\mbox{%
1778 % \textcolor{red}{\hookrightarrow}\space} -- worked for years and broke the
1779 % moment the pdfa option was switched on: pdfx puts xcolor into a converting
1780 % mode (\selectcolormodel plus \convertcolorsU/D), and a colour command run
1781 % from inside listings' line-breaking machinery then fails with "Missing
1782 % number, treated as zero" at the first wrapped line. Every listing in the
1783 % document dies, so the class's own example could not be compiled as PDF/A at
1784 % all. A box is immune: it is built here, in ordinary horizontal mode where
1785 % xcolor works normally, and listings only ever copies it.
1786 %
1787 % The \sbox waits for \begin{document} because xcolor's colours are not
1788 % defined while the class is still loading.
1789 \newsavebox\coppe@lstarrow
1790 \AtBeginDocument{\sbox\coppe@lstarrow{\textcolor{red}{\hookrightarrow}\space}}
1791 \newcommand\pythonstyle{\lstset{language=Python,basicstyle=\ttm,numbers=left,%
1792 numberstyle=\small\ttfamily,numbersep=0.5em,commentstyle=\color{purple},%
1793 morekeywords={self,as},keywordstyle=\ttb\color{rgb}{0,0,0.5},%
1794 emph={MyClass,__init__},emphstyle=\ttb\color{rgb}{0.6,0,0},%
1795 stringstyle=\color{rgb}{0,0.5,0},frame=tb,showstringspaces=false,%
1796 extendedchars=true,firstnumber=auto,inputencoding=utf8,breaklines=true,%
1797 escapechar=',postbreak=\usebox\coppe@lstarrow}}

```

```

1798 \lstnewenvironment{python}[1][\pythonstyle\lstset{#1}%
1799 \csname\@lst @SetFirstNumber\endcsname}{\csname\@lst @SaveFirstNumber\endcsname}
1800 \newcommand\pythoninline[1]{\pythonstyle\lstinline!#1!}
1801 \newcommand\pythonexternal[2][\pythonstyle\lstinputlisting{#1}{#2}}
1802 \newcommand\xmlstyle{\lstset{language=XML,basicstyle=\ttm,numbers=left,%
1803 numberstyle=\small\ttfamily,numbersep=0.5em,%
1804 morekeywords={xs:schema,xs:element,xs:sequence,xs:complexType},%
1805 keywordstyle=\ttb\color{rgb}{0,0,0.5},%
1806 emph={name,type,attributeFormDefault,elementFormDefault,xmlns:xs},%
1807 emphstyle=\ttb\color{rgb}{0.6,0,0},stringstyle=\color{rgb}{0,0.5,0},frame=tb,%
1808 showstringspaces=false,extendedchars=true,inputencoding=utf8,breaklines=true,%
1809 postbreak=\usebox\coppe@lstarrow}}
1810 \lstnewenvironment{xml}[1][\xmlstyle\lstset{#1}]{%
1811 \newcommand\htmlstyle{\lstset{language=HTML,basicstyle=\ttm,numbers=left,%
1812 numberstyle=\small\ttfamily,numbersep=0.5em,%
1813 keywordstyle=\ttb\color{rgb}{0,0,0.5},emphstyle=\ttb\color{rgb}{0.6,0,0},%
1814 stringstyle=\color{rgb}{0,0.5,0},frame=tb,showstringspaces=false,%
1815 extendedchars=true,inputencoding=utf8,breaklines=true,%
1816 postbreak=\usebox\coppe@lstarrow}}
1817 \lstnewenvironment{html}[1][\htmlstyle\lstset{#1}]{%
1818 \newcommand\prologstyle{\lstset{language=Prolog,basicstyle=\ttm,numbers=left,%
1819 numberstyle=\small\ttfamily,numbersep=0.5em,commentstyle=\color{purple},%
1820 keywordstyle=\ttb\color{rgb}{0,0,0.5},emph={MyClass,__init__},%
1821 emphstyle=\ttb\color{rgb}{0.6,0,0},stringstyle=\color{rgb}{0,0.5,0},frame=tb,%
1822 showstringspaces=false,extendedchars=true,firstnumber=auto,inputencoding=utf8,%
1823 breaklines=true,escapechar=',%
1824 postbreak=\usebox\coppe@lstarrow}}
1825 \lstnewenvironment{prolog}[1][\prologstyle\lstset{#1}%
1826 \csname\@lst @SetFirstNumber\endcsname}{\csname\@lst @SaveFirstNumber\endcsname}
1827 \newcommand\javastyle{\lstset{language=Java,basicstyle=\ttm,numbers=left,%
1828 numberstyle=\small\ttfamily,numbersep=0.5em,commentstyle=\color{purple},%
1829 morekeywords={self,as},keywordstyle=\ttb\color{rgb}{0,0,0.5},%
1830 emphstyle=\ttb\color{rgb}{0.6,0,0},stringstyle=\color{rgb}{0,0.5,0},frame=tb,%
1831 showstringspaces=false,extendedchars=true,firstnumber=auto,inputencoding=utf8,%
1832 breaklines=true,escapechar=',%
1833 postbreak=\usebox\coppe@lstarrow}}
1834 \lstnewenvironment{java}[1][\javastyle\lstset{#1}%
1835 \csname\@lst @SetFirstNumber\endcsname}{\csname\@lst @SaveFirstNumber\endcsname}
1836 \newcommand\javainline[1]{\javastyle\lstinline!#1!}
1837 \newcommand\javaexternal[2][\javastyle\lstinputlisting{#1}{#2}}
1838 % fancyvrb's Verbatim environment is available for program output; the
1839 % \DefineVerbatimEnvironment{gxoutput}/{gxoutputs} aliases that used to live
1840 % here were author-personal (gx prefix) and were removed in v4.0. Use, for
1841 % example,
1842 % \begin{Verbatim}[frame=lines,numbers=left,numbersep=0.5em]
1843 % ...
1844 % \end{Verbatim}
1845 % (or \DefineVerbatimEnvironment{...}{Verbatim}{...} in the manuscript preamble).
1846 \renewcommand\theFancyVerbLine{\small\ttfamily\arabic{FancyVerbLine}}
1847 % Algorithms: algorithm2e, language driven by the main-slot \coppemainstring
1848 % lookup (key "algorithm2eopt"). Empty means use algorithm2e's default
1849 % language (English) -- needed because algorithm2e errors on an empty option.
1850 \edef\coppe@algotlang{\coppemainstring{algorithm2eopt}}
1851 \ifx\coppe@algotlang\empty

```

```

1852 \RequirePackage[linesnumbered,lined,algochapter,ruled]{algorithm2e}
1853 \else
1854 \expandafter\RequirePackage\expandafter[\coppe@algotlang,linesnumbered,lined,algochapter,ruled]{algorithm2e}
1855 \fi
1856 % algorithm2e composes its own caption and therefore ignores the class-wide
1857 % \captionsetup{font=footnotesize,labelsep=endash}. Left alone it printed
1858 % "Algoritmo 6.1:Titulo" in bold at body size, while every other illustration
1859 % printed "Figura 6.1 -- Titulo" in footnotesize -- 2.10 asks for the
1860 % designating word, the number, a TRAVESSAO and the title, uniformly.
1861 %
1862 % The separator and the caption fonts are therefore set to match. The 'ruled'
1863 % layout (rules above and below) is kept: 2.10 prescribes where the caption and
1864 % the source go, not whether the float is delimited, and the rules read here
1865 % like the top-and-bottom delimitation a table carries.
1866 \SetAlgoCaptionSeparator{\textendash}
1867 \SetAlCapFnt{\footnotesize}
1868 \SetAlCapNameFnt{\footnotesize}
1869 \SetAlCapSty{textnormal}
1870 \SetAlCapNameSty{textnormal}
1871 \SetKwFor{Para}{para}{faça}{fim do para}
1872 \SetKwFor{Enquanto}{enquanto}{faça}{fim do enquanto}
1873 % setspace + hyperref + biblatex are loaded at the very end of the class
1874 % (see the block just before %</class>), so their hooks see every macro the
1875 % class defines, and the load order setspace -> hyperref -> biblatex is
1876 % respected.

```

`\department` This macro is used to set the author's affiliation. There are twelve options which correspond to all academic units at COPPE/UFRJ. It defines the current and the foreign names of these units.

```

1877 \newcommand\coppe@deptcode{}
1878 \newcommand\meta@deptname{}
1879 \newcommand\department[1]{%
1880 % The programme CODE is kept alongside the resolved name, and each name is
1881 % declared TWICE. \local@deptname keeps the TeX accent escapes with braces
1882 % (Computa{\c c}{\~ a}o) that the cover and the folha de rosto typeset;
1883 % \meta@deptname is the same name in literal UTF-8, and is the one that goes
1884 % into the XMP. pdfx copies whatever it is given straight through, so feeding it
1885 % the escaped form produced "Computa{c}o" in dc:description -- which is why the
1886 % subject line used to fall back to the bare programme code. Both forms are
1887 % written out here rather than derived from one another because there is no
1888 % reliable way to strip TeX accent syntax at the point where the .xmpdata is
1889 % written, and a table of thirteen entries is cheaper than that machinery.
1890 \gdef\coppe@deptcode{#1}%
1891 \ifthenelse{\equal{#1}{PEB}}{
1892   {\global\def\local@deptname{Engenharia Biom{\` e}dica}
1893   \global\def\meta@deptname{Engenharia Biomédica}
1894   \global\def\foreign@deptname{Biomedical Engineering}}{}
1895 \ifthenelse{\equal{#1}{PEC}}{
1896   {\global\def\local@deptname{Engenharia Civil}
1897   \global\def\meta@deptname{Engenharia Civil}
1898   \global\def\foreign@deptname{Civil Engineering}}{}
1899 \ifthenelse{\equal{#1}{PEE}}{
1900   {\global\def\local@deptname{Engenharia El{\` e}trica}
1901   \global\def\meta@deptname{Engenharia Elétrica}

```

```

1902 \global\def\foreign@deptname{Electrical Engineering}}{}
1903 \ifthenelse{\equal{#1}{PEM}}{
1904   {\global\def\local@deptname{Engenharia Mec{\` a}nica}
1905   \global\def\meta@deptname{Engenharia Mecânica}
1906   \global\def\foreign@deptname{Mechanical Engineering}}{}
1907 \ifthenelse{\equal{#1}{PEMM}}{
1908   {\global\def\local@deptname{Engenharia Metal{\` u}rgica e de Materiais}
1909   \global\def\meta@deptname{Engenharia Metalúrgica e de Materiais}
1910 \global\def\foreign@deptname{Metallurgical and Materials Engineering}}{}
1911 \ifthenelse{\equal{#1}{PEN}}{
1912   {\global\def\local@deptname{Engenharia Nuclear}
1913   \global\def\meta@deptname{Engenharia Nuclear}
1914   \global\def\foreign@deptname{Nuclear Engineering}}{}
1915 \ifthenelse{\equal{#1}{PEN0}}{
1916   {\global\def\local@deptname{Engenharia Oce{\` a}nica}
1917   \global\def\meta@deptname{Engenharia Oceânica}
1918   \global\def\foreign@deptname{Ocean Engineering}}{}
1919 \ifthenelse{\equal{#1}{PPE}}{
1920   {\global\def\local@deptname{Planejamento Energ{\` e}tico}
1921   \global\def\meta@deptname{Planejamento Energético}
1922   \global\def\foreign@deptname{Energy Planning}}{}
1923 \ifthenelse{\equal{#1}{PEP}}{
1924   {\global\def\local@deptname{Engenharia de Produ{\` c}{\` a}o}
1925   \global\def\meta@deptname{Engenharia de Produção}
1926   \global\def\foreign@deptname{Production Engineering}}{}
1927 \ifthenelse{\equal{#1}{PEQ}}{
1928   {\global\def\local@deptname{Engenharia Qu{\` i}mica}
1929   \global\def\meta@deptname{Engenharia Química}
1930   \global\def\foreign@deptname{Chemical Engineering}}{}
1931 \ifthenelse{\equal{#1}{PESC}}{
1932   {\global\def\local@deptname{Engenharia de Sistemas e Computa{\` c}{\` a}o}
1933   \global\def\meta@deptname{Engenharia de Sistemas e Computação}
1934   \global\def\foreign@deptname{Systems Engineering and Computer Science}}{}
1935 \ifthenelse{\equal{#1}{PET}}{
1936   {\global\def\local@deptname{Engenharia de Transportes}
1937   \global\def\meta@deptname{Engenharia de Transportes}
1938   \global\def\foreign@deptname{Transportation Engineering}}{}
1939 \ifthenelse{\equal{#1}{PENT}}{
1940   {\global\def\local@deptname{Engenharia de Nanotecnologia}
1941   \global\def\meta@deptname{Engenharia de Nanotecnologia}
1942   \global\def\foreign@deptname{Nanotechnology Engineering}}{}
1943 }

```

`\title` Used to enter the title in Brazilian Portuguese. Mirrors into `\coppe@title@brazilian` so that the per-language title slot used by the cover/folha renderers (`\coppe@selecttitle`) picks the right text for a pt-main thesis.

```

1944 \renewcommand\title[1]{%
1945   \global\def\local@title{#1}%
1946   \expandafter\gdef\csname coppe@title@brazilian\endcsname{#1}%
1947 }

```

`\foreigntitle` Used to enter the foreign title. Also mirrors into `\coppe@title@english` so that the cover renders the right text for an english-main thesis.

```

1948 \newcommand\foreigntitle[1]{%
1949   \global\def\foreign@title{#1}%
1950   \expandafter\gdef\csname coppe@title@english\endcsname{#1}%
1951 }

```

`\coppe@selecttitle` Expands to the title stored for the current main language. For pt-main this is `\local@title`; for en-main `\foreign@title`; for any other main language the user must call `\titlein{<lang>}{...}` to populate it.

```

1952 \newcommand\coppe@selecttitle{\@nameuse{coppe@title@\coppe@mainlang}}
1953 % Subtitle, mirroring the title machinery slot for slot: \subtitle feeds the
1954 % brazilian slot, \foreignsubtitle the english one, \subtitlein any other.
1955 % 3.1.1 and 3.1.2.1.1(c) require the subtitle on the capa and on the folha de
1956 % rosto, subordinated to the title by a colon.
1957 \newcommand\subtitle[1]{%
1958   \expandafter\gdef\csname coppe@subtitle@brazilian\endcsname{#1}}
1959 \newcommand\foreignsubtitle[1]{%
1960   \expandafter\gdef\csname coppe@subtitle@english\endcsname{#1}}
1961 \newcommand\subtitlein[2]{%
1962   \expandafter\gdef\csname coppe@subtitle@#1\endcsname{#2}}
1963 \newcommand\coppe@selectsubtitle{\@nameuse{coppe@subtitle@\coppe@mainlang}}
1964 % Title as the capa and the folha de rosto print it. Annexes A and B show the
1965 % title in caps and the subtitle in ordinary case after a colon, so the case
1966 % change is applied to the title alone.
1967 \newcommand\coppe@titleblock{%
1968   \MakeUppercase{\coppe@selecttitle}%
1969   \@ifundefined{coppe@subtitle@\coppe@mainlang}{: \coppe@selectsubtitle}}
1970 % Number of volumes (3.1.1 and 3.1.2.1.1(d)): printed only when there is more
1971 % than one, since a single-volume work says nothing about volumes.
1972 % Coorientador (3.1.2.1.1(f)). Same shape as \advisor, including the optional
1973 % institution; the label switches to the plural on the second call, as the
1974 % advisor label already did.
1975 \newcount\@coadvisor\@coadvisor0
1976 \newcommand\coadvisor[5][1]{%
1977   \global\@namedef{CoppeCoadvisorInst:\expandafter\the\@coadvisor}{#1}
1978   \global\@namedef{CoppeCoadvisorTitle:\expandafter\the\@coadvisor}{#2}
1979   \global\@namedef{CoppeCoadvisorName:\expandafter\the\@coadvisor}{#3}
1980   \global\@namedef{CoppeCoadvisorSurname:\expandafter\the\@coadvisor}{#4}
1981   \global\@namedef{CoppeCoadvisorDegree:\expandafter\the\@coadvisor}{#5}
1982   \global\advance\@coadvisor by 1
1983 }
1984 \newcommand\local@coadvisorstring{%
1985   \ifnum\@coadvisor>1
1986     \coppemainstring{coadvisors}%
1987   \else
1988     \coppemainstring{coadvisor}%
1989   \fi}
1990 % Mesma coisa no idioma estrangeiro, para a pagina de abstract.
1991 \newcommand\foreign@coadvisorstring{%
1992   \ifnum\@coadvisor>1
1993     \coppeforeignstring{coadvisors}%
1994   \else
1995     \coppeforeignstring{coadvisor}%
1996   \fi}
1997 \newcommand\coppe@nvolumes{1}

```

```

1998 \newcommand\coppe@thisvolume{1}
1999 \newcommand\volumes[1]{\gdef\coppe@nvolumes{#1}}
2000 \newcommand\volume[1]{\gdef\coppe@thisvolume{#1}}
2001 \newcommand\coppe@volumeline{%
2002   \ifnum\coppe@nvolumes>1
2003     \par\vspace*{4mm}%
2004     \coppemainstring{volumename}\space\coppe@thisvolume\space
2005     \coppemainstring{ofname}\space\coppe@nvolumes
2006 % The closing \par matters: on the capa the next thing is \vspace{\stretch{3}},
2007 % which does not end a paragraph, so without it the city ran on into the volume
2008 % line ("Volume 2 de 3 RIO DE JANEIRO").
2009   \par
2010   \fi}

```

\advisor Defines globally the title, name and academic degree of the advisors.

```

2011 \newcount\@advisor\@advisor0
2012 % The optional first argument is the member's institution, required by
2013 % 3.1.2.1.3(e). Omitting it keeps every pre-4.1 call working unchanged.
2014 \newcommand\advisor[5][\{%
2015   \global\@namedef{CoppeAdvisorInst:\expandafter\the\@advisor}{#1}
2016   \global\@namedef{CoppeAdvisorTitle:\expandafter\the\@advisor}{#2}
2017   \global\@namedef{CoppeAdvisorName:\expandafter\the\@advisor}{#3}
2018   \global\@namedef{CoppeAdvisorSurname:\expandafter\the\@advisor}{#4}
2019   \global\@namedef{CoppeAdvisorDegree:\expandafter\the\@advisor}{#5}
2020   \global\advance\@advisor by 1
2021   \ifnum\@advisor>1
2022     \renewcommand\local@advisorstring{Orientadores}
2023     \renewcommand\foreign@advisorstring{Advisors}
2024   \fi
2025 }

```

\examiner

```

2026 \newcount\@examiner\@examiner0
2027 % Until v4.1 this stored only "#1 #2" and DISCARDED the third argument: the
2028 % titulacao every document passed was silently thrown away and never printed,
2029 % though 3.1.2.1.3(e) requires it. Each part is now kept separately, and the
2030 % optional first argument carries the institution.
2031 % CoppeExaminer: is retained with its old meaning for anything reading it.
2032 \newcommand\examiner[4][\{%
2033   \global\@namedef{CoppeExaminerInst:\expandafter\the\@examiner}{#1}
2034   \global\@namedef{CoppeExaminerTitle:\expandafter\the\@examiner}{#2}
2035   \global\@namedef{CoppeExaminerName:\expandafter\the\@examiner}{#3}
2036   \global\@namedef{CoppeExaminerDegree:\expandafter\the\@examiner}{#4}
2037   \global\@namedef{CoppeExaminer:\expandafter\the\@examiner}{#2\ #3}
2038   \global\advance\@examiner by 1
2039 }

```

\author It was redefined to allow the identification of the author's first names and surname.

```

2040 \renewcommand\author[2]{%
2041   \global\def\@authname{#1}
2042   \global\def\@authsurn{#2}
2043 }

```

```

\date Sets the document's nominal month and year (used by the cover, the folha
de rosto, and the catalog). Stores them in private slots (\coppe@docmonth /
\coppe@docyear) rather than mutating TeX's \month / \year primitives, which
would clobber \today.
2044 \renewcommand\date[2]{%
2045   \def\coppe@docmonth{#1}%
2046   \def\coppe@docyear{#2}}

\local@monthname
2047 \newcommand\local@monthname{\ifcase\coppe@docmonth\or
2048   Janeiro\or Fevereiro\or Mar{\c c}o\or Abril\or Maio\or Junho\or
2049   Julho\or Agosto\or Setembro\or Outubro\or Novembro\or Dezembro\fi}

\foreign@monthname
2050 \newcommand\foreign@monthname{\ifcase\coppe@docmonth\or
2051   January\or February\or March\or April\or May\or June\or
2052   July\or August\or September\or October\or November\or December\fi}

\keyword
2053 \newcounter{keywords}
2054 \newcommand\keyword[1]{%
2055   \global\@namedef{CoppeKeyword:\expandafter\the\c@keywords}{#1}
2056   \global\addtocounter{keywords}{1}
2057 }
2058 % Keywords for the foreign abstract. Annex F of the manual shows the English
2059 % abstract closing with English keywords, and \keyword alone cannot serve both
2060 % pages. When no \foreignkeyword is given the foreign abstract falls back to
2061 % the \keyword list, which is what a document written before v4.1 expects.
2062 \newcounter{foreignkeywords}
2063 \newcommand\foreignkeyword[1]{%
2064   \global\@namedef{CoppeForeignKeyword:\expandafter\the\c@foreignkeywords}{#1}
2065   \global\addtocounter{foreignkeywords}{1}
2066 }
2067 % Keywords for the optional third abstract (Portuguese, for the banca). Same
2068 % shape and same fallback as \foreignkeyword: without it the Portuguese page
2069 % would close with the main-language keywords under a Portuguese label, which
2070 % is what a Spanish-, French- or Italian-main work would produce.
2071 \newcounter{braziliankeywords}
2072 \newcommand\braziliankeyword[1]{%
2073   \global\@namedef{CoppeBrazilianKeyword:\expandafter\the\c@braziliankeywords}{#1}
2074   \global\addtocounter{braziliankeywords}{1}
2075 }
2076 % 3.1.2.1.4: the keywords close the resumo, preceded by the term, a colon, and
2077 % separated from one another by semicolons.
2078 % #1 label, #2 csname prefix, #3 how many.
2079 \newcommand\coppe@printkeywords[3]{%
2080   \ifnum#3>0
2081     \par\vspace*{7mm}%
2082     \noindent{\singlespacing\textbf{#1:}}\enspace
2083     \count1=0
2084     \@whilenum\count1<#3\do{%
2085       \ifnum\count1>0 ;\space\fi
2086       \csname #2\the\count1\endcsname

```

```

2087     \advance\count1 by 1}%
2088     .\par}%
2089     \fi}

\areaconcentracao Data for the folha adicional (Coleta CAPES) and for the natureza block of the
\linhapesquisa folha de rosto and folha de aprovacao. Every one of them defaults to empty;
\tipoproducao \makefolhaadicional prints an empty rule where a value is missing, so the pro-
\projetovinculado gramme can complete the sheet by hand if need be.
\nomeprojeto 2090 \newcommand\coppe@areaconc{}
\agenciafomento 2091 \newcommand\areaconcentracao[1]{\gdef\coppe@areaconc{#1}}
2092 \newcommand\coppe@dataaprov{}
2093 \newcommand\dataaprovacao[1]{\gdef\coppe@dataaprov{#1}}
2094 \newcommand\coppe@linhapesq{}
2095 \newcommand\linhapesquisa[1]{\gdef\coppe@linhapesq{#1}}
2096 \newcommand\coppe@tipoproducao{}
2097 \newcommand\tipoproducao[1]{\gdef\coppe@tipoproducao{#1}}
2098 \newcommand\coppe@projetovinc{}
2099 \newcommand\projetovinculado[1]{\gdef\coppe@projetovinc{#1}}
2100 \newcommand\coppe@nomeprojeto{}
2101 \newcommand\nomeprojeto[1]{\gdef\coppe@nomeprojeto{#1}}
2102 \newcounter{coppeagencias}
2103 \newcommand\agenciafomento[2]{%
2104     \global\@namedef{CoppeAgencia:\expandafter\the\c@coppeagencias}{#1}%
2105     \global\@namedef{CoppeAgenciaSigla:\expandafter\the\c@coppeagencias}{#2}%
2106     \global\addtocounter{coppeagencias}{1}}

```

`\freeconfig` This command allows easy changing of core class parameters.

```

2107 \newcommand\freeconfig[6]{%
2108     \providecommand\@degree{}
2109     \renewcommand\@degree{#1}
2110     \providecommand\@degreename{}
2111     \renewcommand\@degreename{#2}
2112     \providecommand\local@degname{}
2113     \renewcommand\local@degname{#3}
2114     \providecommand\foreign@degname{}
2115     \renewcommand\foreign@degname{#4}
2116     \providecommand\local@doctype{}
2117     \renewcommand\local@doctype{#5}
2118     \providecommand\foreign@doctype{}
2119     \renewcommand\foreign@doctype{#6}%
2120 }
2121 % \end{macrocode}
2122 % \end{macro}
2123 %
2124 % \begin{macro}{\frontmatter}
2125 % The number of pages for both frontmatter and mainmatter printed
2126 % in the cataloging details page is computed by means of simple
2127 % LaTeX labels.
2128 % \begin{macrocode}
2129 \renewcommand\frontmatter{%
2130     \cleardoublepage
2131     \@mainmatterfalse
2132 % The count already started at the folha de rosto (see \maketitle) and simply
2133 % continues here: 2.7 requires every folha from the folha de rosto onwards to

```



```

2134 % be counted sequentially, printing arabic numerals only from the first textual
2135 % folha. Resetting the counter here -- as this class did until v4.1 -- made the
2136 % Introducao come out as page 1, which the norm does not allow.
2137 %
2138 % The numeraisromanos option keeps the legacy behaviour (lowercase roman in the
2139 % pre-textual part, arabic restarting at 1 in the textual part). That departs
2140 % from 2.7, which has the pre-textual folhas counted but NOT numbered, so the
2141 % option warns.
2142 \if@copperoman
2143   \ClassWarningNoLine{coppe}{Option ‘numeraisromanos’ prints roman numerals
2144     on the pre-textual folhas and restarts the arabic count at 1 in the
2145     textual part. Section 2.7 of the SiBI manual (9th rev. ed., 2026) has
2146     those folhas counted but not numbered, and the arabic numbering
2147     continuing the count. Drop the option to comply}%
2148   \pagenumbering{roman}%
2149 \fi
2150 % \thispagestyle vale por UMA pagina. Se a folha de aprovacao transbordar --
2151 % uma banca grande com titulo longo consegue --, a pagina seguinte pegava o
2152 % estilo corrente e saia com folio, na parte pre-textual, onde a 2.7 nao
2153 % admite. O grupo com \pagestyle{empty} cobre quantas paginas forem, e o
2154 % \endgroup devolve o estilo anterior.
2155 \begingroup
2156   \pagestyle{empty}\thispagestyle{empty}
2157   \singlespacing
2158   \makefrontpage
2159   \clearpage
2160 \endgroup
2161 \if@copperoman
2162   \fancypagestyle{plain}{\fancyhf{}\fancyhead[R0,LE]{\coppe@folio}}%
2163   \renewcommand{\headrulewidth}{0pt}}%
2164   \pagestyle{coppe}%
2165 \else
2166   \fancypagestyle{plain}{\fancyhf{}\renewcommand{\headrulewidth}{0pt}}%
2167   \pagestyle{empty}%
2168 \fi
2169 }

\mainmatter
2170 \renewcommand\mainmatter{%
2171   \coppe@mainBegin
2172   \cleardoublepage
2173   \@mainmattertrue
2174   \fancypagestyle{plain}{\fancyhf{}\fancyhead[R0,LE]{\coppe@folio}}%
2175   \renewcommand{\headrulewidth}{0pt}}%
2176   \pagestyle{coppe}%
2177 % No \pagenumbering{arabic} in the default path: it would reset the counter to
2178 % 1 and make the Introducao folha 1, contrary to 2.7. The counter has been
2179 % arabic and running since the folha de rosto; all that changes here is that
2180 % the folio starts being PRINTED. Only the legacy numeraisromanos path, which
2181 % switched to roman in the front matter, has to switch back -- and there the
2182 % restart at 1 is intrinsic to that (non-conforming) scheme.
2183 \if@copperoman\pagenumbering{arabic}\fi}

\backmatter

```

```

2184 \renewcommand\backmatter{%
2185   \if@openright
2186     \cleardoublepage
2187   \else
2188     \clearpage
2189   \fi}
2190 %

```

\coppetexfinalpage Optional colophon-style closing page intended to face the rear cover. The macro forces a verso (even) page so that, in a two-sided print, the reader sees the colophon on the left and the inside of the rear cover on the right; pushes the text to the bottom of the page with `\vfill`; and types it in Portuguese unconditionally (the colophon is institutional metadata, not body content, so it is rendered in pt-BR regardless of the thesis main language). Three slot macros are user-overridable: `\coppefinalengine` (engine name; auto-detected via `iftex`), `\coppefinalfont` (text body font; defaults to Computer Modern, the L^AT_EX default), and `\coppefinalmanual` (the reference manual title; defaults to the current COPPE/UFRJ norm). Override them before `\coppetexfinalpage` with `\renewcommand` if the document uses a non-default font or follows an earlier edition of the norm.

```

2191 \providecommand{\coppefinalengine}{%
2192   \ifPDFTeX\hologo{pdfLaTeX}\fi
2193   \ifLuaTeX\hologo{LuaLaTeX}\fi
2194   \ifXeTeX\hologo{XeLaTeX}\fi}
2195 % A classe carrega lmodern (as bitmap da Computer Modern saiam em Type 3 e
2196 % inviabilizavam o PDF/A), entao o colofao diz Latin Modern -- quem trocar a
2197 % fonte redefine este comando.
2198 \providecommand{\coppefinalfont}{Latin Modern (\texttt{lmodern})}
2199 \providecommand{\coppefinalmanual}{%
2200   \emph{Norma para a Elabora\c c\~ao Gr\'afica de Teses e Disserta\c c\~oes
2201     da COPPE/UFRJ}}
2202 \newcommand{\coppetexfinalpage}{%
2203   \clearpage
2204   % Force the colophon onto a verso (even) page so it faces the rear cover.
2205   % If the current page is odd (recto), fill it with an empty verso first.
2206   \if@twoside
2207     \ifodd\c@page
2208       \hbox{}\thispagestyle{empty}\newpage
2209     \fi
2210   \fi
2211   \thispagestyle{empty}%
2212   \null\vfill
2213   \begin{otherlanguage}{brazilian}%
2214     \begin{center}
2215       {\footnotesize\setstretch{1.0}}%
2216       \textsc{Colof\~ao}\par
2217       \vspace{0.7em}
2218       \begin{minipage}{0.75\textwidth}\centering
2219         Este documento foi composto em \LaTeX{} com \coppefinalengine{}
2220         em \today, na fonte \coppefinalfont. A bibliografia foi processada
2221         por Bib\LaTeX{} e Biber. Foi utilizada a classe \CoppeTeX,
2222         vers\~ao\~\coppe@version, dispon\~i vel no reposit\~orio GitHub
2223         \mbox{\url{https://github.com/COPPE-UFRJ/CoppeTeX}}, em

```

```

2224         conformidade com a \coppefinalmanual.
2225         \end{minipage}\par}
2226     \end{center}
2227 \end{otherlanguage}%
2228 \vspace*{2cm}%
2229 \clearpage}

```

`\makecover` The ABNT cover (capa, NBR 14724) suggested in the manual's Anexo A. It is the institution block — the university, the full COPPE name, and the graduate program taken from `\department` — followed by the author, the title, and the city and year, all centred and uppercased. The optional UFRJ and COPPE logos head the page. The capa is not counted: it precedes the (roman) front matter and is produced before the `titlepage` (folha de rosto).

```

2230 \newcommand\makecover{%
2231   \begin{titlepage}%
2232   % A capa e um modelo institucional, e sai em espaco simples qualquer que seja o
2233   % espacamento do corpo. Sob a opcao doublespacing ela transbordava para uma
2234   % segunda e ate uma terceira folha -- que ainda por cima levavam folio, porque
2235   % o \thispagestyle do titlepage vale por uma pagina so. 0 \pagestyle{empty} e
2236   % local ao ambiente e cobre quantas paginas forem.
2237   \pagestyle{empty}%
2238   \singlespacing
2239   \centering
2240   \makebox[\linewidth]{%
2241     \includegraphics[height=2cm]{ufrj-logo}\hfill
2242     \includegraphics[height=2cm]{coppe-logo}}\par
2243   \vspace{1.5cm}
2244   {\singlespacing
2245   \MakeUppercase{\local@universityname}\par
2246     \nohyphens{\MakeUppercase{Instituto Alberto Luiz Coimbra de \
2247       P{\ ' o}s-Gradua{\ c c}{\ ^ a}o e Pesquisa de Engenharia}}\par
2248     \MakeUppercase{Programa de \local@deptname}\par}
2249   \vspace{\stretch{2}}
2250   \textbf{\MakeUppercase{\textbf{\@authname\ \@authsur}}}\par
2251   \vspace{\stretch{2}}
2252   \nohyphens{\coppe@titleblock}\par
2253   \coppe@volumeline
2254   \vspace{\stretch{3}}
2255   \MakeUppercase{\local@cityname}\par
2256   \coppe@docyear
2257   \end{titlepage}%
2258 }

```

`\maketitle`

```

2259 \renewcommand\maketitle{%
2260 % The capa is neither counted nor numbered (2.7), so it lives in its own
2261 % throwaway numbering scheme; \pagenumbering{arabic} right after it restarts
2262 % the counter, making the FOLHA DE ROSTO page 1 -- which is where 2.7 says the
2263 % count begins. Nothing is printed here: \frontmatter keeps \pagestyle{empty}
2264 % until the textual part starts.
2265   \pagenumbering{alph}
2266   \ltx@ifpackageloaded{hyperref}{\coppe@hypersetup}{}%
2267   \makecover

```

```

2268 \pagenumbering{arabic}
2269 \begin{titlepage}
2270 % Mesma razao da capa: modelo institucional, espaco simples, e sem folio ainda
2271 % que transborde.
2272 \pagestyle{empty}%
2273 \singlespacing
2274 \begin{flushleft}
2275 \vspace*{1.5mm}
2276 \setlength\baselineskip{0pt}
2277 \setlength\parskip{1mm}
2278 \makebox[\linewidth]{%
2279   \includegraphics[height=2cm]{ufrj-logo}\hfill
2280   \includegraphics[height=2cm]{coppe-logo}}
2281 \end{flushleft}
2282 \vspace{1.05cm}
2283 \begin{center}
2284   \nohyphens{\@authname\ \@authsur}\par
2285   \vspace*{3cm}
2286   \nohyphens{\coppe@titleblock}\par
2287   \coppe@volumeline
2288 \end{center}
2289 \vspace*{2.1cm}
2290 \begin{flushright}
2291 % 2.4: on the folha de rosto the natureza/objetivo/instituicao block is set
2292 % SINGLE spaced, and aligned from the middle of the mancha grafica to the right
2293 % margin. 0.5\textwidth tracks the mancha, so the block follows automatically if
2294 % the margins ever change -- the old fixed 8.45cm did not.
2295 \begin{minipage}{0.5\textwidth}
2296 \singlespacing\frontcover@maintext
2297 \end{minipage}\par
2298 \end{flushright}
2299 % Only the natureza block is indented to the right (2.4). The linha de pesquisa
2300 % and the orientadores return to the LEFT MARGIN, as the CAPES model of the
2301 % folha de rosto and Annex B of the manual show. Until v4.1 they sat inside the
2302 % right-hand block, which is what a recent review flagged.
2303 \vspace*{12mm}
2304 {\singlespacing
2305 \ifthenelse{\equal{\coppe@linhapesq}{}}{\%
2306   \noindent Linha de pesquisa: \coppe@linhapesq\par
2307   \vspace*{5mm}}{\%
2308 \nohyphens{\%
2309 \noindent\begin{tabularx}{\textwidth}[b]{@{}l@{ }>\raggedright\arraybackslash}X@{}}
2310   \local@advisorstring: &
2311   \count1=0
2312   \toks@={}
2313   \@whilenum \count1<\@advisor \do{\%
2314     \ifcase\count1 % same as \ifnum0=\count1
2315       \toks@=\expandafter{\csname CoppeAdvisorName:\the\count1%
2316         \expandafter\endcsname\expandafter\space%
2317         \csname CoppeAdvisorSurname:\the\count1\endcsname\\}
2318     \else
2319       \toks@=\expandafter\expandafter\expandafter{\%
2320         \expandafter\the\expandafter\toks@%
2321         \expandafter&\expandafter\space%

```

```

2322         \csname CoppeAdvisorName:\the\count1\expandafter\endcsname%
2323         \expandafter\space\csname CoppeAdvisorSurname:\the\count1\endcsname\\
2324     }%
2325     \fi
2326     \advance\count1 by 1}
2327     \the\toks@
2328 \end{tabularx}}}%
2329 \ifnum\@coadvisor>0
2330     \par\vspace*{2mm}%
2331     \nohyphens{%
2332     \noindent\begin{tabularx}{\textwidth}[b]{\@{}}l@{ }>{\raggedright\arraybackslash}X@{}}
2333     \local@coadvisorstring: &
2334     \count1=0
2335     \toks@={ }
2336     \@whilenum \count1<\@coadvisor \do{%
2337     \ifcase\count1
2338         \toks@=\expandafter{\csname CoppeCoadvisorName:\the\count1%
2339         \expandafter\endcsname\expandafter\space%
2340         \csname CoppeCoadvisorSurname:\the\count1\endcsname\\}
2341     \else
2342         \toks@=\expandafter\expandafter\expandafter{%
2343         \expandafter\the\expandafter\toks@%
2344         \expandafter&\expandafter\space%
2345         \csname CoppeCoadvisorName:\the\count1\expandafter\endcsname%
2346         \expandafter\space\csname CoppeCoadvisorSurname:\the\count1\endcsname\\
2347     }%
2348     \fi
2349     \advance\count1 by 1}
2350     \the\toks@
2351 \end{tabularx}}}%
2352 \fi
2353 \par}
2354 \vspace*{\fill}
2355 % 3.1.2.1.1(g): the folha de rosto carries the ANO DE DEPOSITO alone -- the year
2356 % the revised final version is handed to the Programa. The month was never asked
2357 % for here (the capa, Annex A, also shows only the year), and printing it
2358 % invited confusion with the defence date. \date still takes month and year:
2359 % the month is what the abstract pages and the colophon use.
2360 \begin{center}
2361 \local@cityname\par
2362 \coppe@docyear
2363 \end{center}
2364 \end{titlepage}
2365 % Ficha catalográfica on the verso of the folha de rosto (NBR 14724, R57):
2366 % the title page is a recto, so this lands on the immediately following verso,
2367 % ahead of the approval sheet produced by \frontmatter.
2368 % LaTeX's titlepage environment ends with \setcounter{page}{1} whenever the
2369 % document is one-sided (see \endtitlepage in classes.dtx). That silently ate
2370 % the folha de rosto from the count, putting the folha de aprovacao at 1 instead
2371 % of 2. The count is therefore restated here: the folha de rosto WAS folha 1, so
2372 % whatever follows it is folha 2.
2373 \setcounter{page}{2}%
2374 % The page carrying the ficha catalografica is explicitly excluded from the
2375 % count -- 2.7 of the 2026 manual ("a folha adicional que contem a ficha

```

```

2376 % catalografica nao pode ser contada ou numerada"), and the 2025 edition said
2377 % the same of the verso of the folha de rosto. Hence the counter is stepped
2378 % back after it.
2379 \ifthenelse{\boolean{maledoc}}{\}%
2380 \makefolhaadicional\clearpage\addtocounter{page}{-1}}%
2381 \global\let\maketitle\relax%
2382 \global\let\and\relax}

2383 % 3.1.2.1.3 lists the elements of the folha de aprovacao in order: (a) the
2384 % AUTHOR, (b) the title in full, (c) the natureza/objetivo/instituicao/area de
2385 % concentracao, (d) the date of approval, (e) the examining board. Until v4.1
2386 % this page opened with the title in caps and the author under it, and set the
2387 % natureza as a wall of uppercase (\frontpage@maintext). Annex D shows neither.
2388 % One board member. \coppe@membroid is the identification line required by
2389 % 3.1.2.1.3(e) -- name, titulacao and institution -- with the last two omitted
2390 % when not given, so a document that never supplied them looks as it did before.
2391 \newcommand\coppe@membroid[4]{%
2392 #1\ #2%
2393 \ifthenelse{\equal{#3}{}}{\},{ #3}%
2394 \ifthenelse{\equal{#4}{}}{\},{ #4}%
2395 % The gap above each signature rule is \coppe@membrogap, set from the size of
2396 % the board just before the list is assembled (see \makefrontpage). A fixed 9mm
2397 % fitted five members and overflowed at seven -- the sheet silently continued
2398 % onto a second page, which also printed a folio in the pre-textual part, where
2399 % 2.7 forbids one. The shrink lives in the length itself -- \newlength makes a
2400 % skip register, so "9mm minus 2mm" survives \setlength and reaches \vspace as
2401 % real glue, and a title long enough to wrap an extra line is absorbed instead
2402 % of costing a page. It has to go through \vspace and not a bare \vskip: the
2403 % board list is emitted inside \nohyphens, i.e. in horizontal mode, where
2404 % \vskip is an error and TeX then typesets the "plus ... minus ..." after it as
2405 % ordinary text -- which is exactly what landed on the page the first time.
2406 \newlength\coppe@membrogap
2407 \setlength\coppe@membrogap{9mm minus 2mm}
2408 \newcount\coppe@bancasize
2409 \newcommand\coppe@membro[4]{%
2410 \if@coppeassin
2411 \vspace{\coppe@membrogap}%
2412 \begin{center}
2413 \rule{95mm}{0.4pt}\par
2414 \nopagebreak{\footnotesize\coppe@membroid{#1}{#2}{#3}{#4}}\par
2415 \end{center}%
2416 \else
2417 \noindent\hspace*{8mm}%
2418 \begin{minipage}{\dimexpr\textwidth-8mm\relax}
2419 \coppe@membroid{#1}{#2}{#3}{#4}
2420 \end{minipage}\par\vspace{2mm}%
2421 \fi}
2422 \newcommand\makefrontpage{%
2423 \begin{center}
2424 \nohyphens{\@authname\ \@authsur}\par
2425 \vspace*{7mm}
2426 \sloppy\nohyphens{\coppe@selecttitle
2427 \@ifundefined{coppe@subtitle@\coppe@mainlang}{\{:
2428 \coppe@selectsubtitle}}\par

```

```

2429 \end{center}\par
2430 \vspace*{8mm}
2431 % Same block, same indentation and same single spacing as the folha de rosto:
2432 % 2.4 names the two pages together.
2433 \begin{flushright}
2434 \begin{minipage}{0.5\textwidth}
2435 \singlespacing\frontcover@maintext
2436 \end{minipage}
2437 \end{flushright}
2438 \vspace*{12mm}
2439 % (d) date of approval. Annex D prints "Aprovada em:" followed by a blank to be
2440 % filled in; \dataaprovacao supplies it when the date is already known.
2441 \noindent\local@approvedonname:\enspace
2442 \ifthenelse{\equal{\coppe@dataaprov}{}}{\hrulefill}{\coppe@dataaprov}\par
2443 \vspace*{12mm}
2444 % (e) the examining board, with the ORIENTADOR FIRST, as president of the board
2445 % (3.1.2.1.3(e)). Until v4.1 the advisors were printed in a block of their own
2446 % above a second list headed "Aprovada por:", which split the board in two and
2447 % did not mark the advisor as its president. They are now one list.
2448 %
2449 % The list is assembled into \coppe@bancalist with \g@addto@macro instead of
2450 % being built inside a tabularx: each entry now carries four fields (treatment,
2451 % name, titulacao, institution) and the old \toks@/\expandafter accumulation
2452 % could not carry them without becoming unreadable. \protected@edef keeps
2453 % accents and other protected commands in the names intact.
2454 % Size of the board decides the spacing above. The thresholds are measured, not
2455 % guessed: with the 9mm gap a seven-member board pushed three members onto a
2456 % second page, 7mm brought six back onto one sheet, 5mm did the same for seven,
2457 % and eight needed 3.5mm -- eight came up in the adversarial suite, in a
2458 % Spanish thesis whose longer natureza block leaves less room than the
2459 % Portuguese one. Boards of five or fewer keep the original 9mm, so every
2460 % document written before this change renders exactly as it did.
2461 \coppe@bancasize=\@advisor
2462 \advance\coppe@bancasize by \@coadvisor
2463 \advance\coppe@bancasize by \@examiner
2464 \ifnum\coppe@bancasize>7
2465 \setlength\coppe@membrogap{3.5mm minus 1.5mm}%
2466 \else\ifnum\coppe@bancasize>6
2467 \setlength\coppe@membrogap{5mm minus 2mm}%
2468 \else\ifnum\coppe@bancasize>5
2469 \setlength\coppe@membrogap{7mm minus 2mm}%
2470 \fi\fi\fi
2471 \gdef\coppe@bancalist{}%
2472 \count1=0
2473 \@whilenum\count1<\@advisor\do{%
2474 \protected@edef\coppe@tmp{\noexpand\coppe@membro
2475 {\csname CoppeAdvisorTitle:\the\count1\endcsname}%
2476 {\csname CoppeAdvisorName:\the\count1\endcsname}\space
2477 {\csname CoppeAdvisorSurname:\the\count1\endcsname}%
2478 {\csname CoppeAdvisorDegree:\the\count1\endcsname}%
2479 {\csname CoppeAdvisorInst:\the\count1\endcsname}}%
2480 \expandafter\g@addto@macro\expandafter\coppe@bancalist
2481 \expandafter{\coppe@tmp}%
2482 \advance\count1 by 1}%

```

```

2483 % Coorientadores come right after the orientadores and before the rest of the
2484 % board, keeping the order 3.1.2.1.3(e) prescribes.
2485 \count1=0
2486 \@whilenum\count1<\@coadvisor\do{%
2487   \protected@edef\coppe@tmp{\noexpand\coppe@membro
2488     {\csname CoppeCoadvisorTitle:\the\count1\endcsname}%
2489     {\csname CoppeCoadvisorName:\the\count1\endcsname}\space
2490     \csname CoppeCoadvisorSurname:\the\count1\endcsname}%
2491     {\csname CoppeCoadvisorDegree:\the\count1\endcsname}%
2492     {\csname CoppeCoadvisorInst:\the\count1\endcsname}}%
2493   \expandafter\g@addto@macro\expandafter\coppe@bancalist
2494   \expandafter{\coppe@tmp}%
2495   \advance\count1 by 1}%
2496 \count1=0
2497 \@whilenum\count1<\@examiner\do{%
2498   \protected@edef\coppe@tmp{\noexpand\coppe@membro
2499     {\csname CoppeExaminerTitle:\the\count1\endcsname}%
2500     {\csname CoppeExaminerName:\the\count1\endcsname}%
2501     {\csname CoppeExaminerDegree:\the\count1\endcsname}%
2502     {\csname CoppeExaminerInst:\the\count1\endcsname}}%
2503   \expandafter\g@addto@macro\expandafter\coppe@bancalist
2504   \expandafter{\coppe@tmp}%
2505   \advance\count1 by 1}%
2506 \noindent\local@approvedname:\par
2507 \nohyphens{\singlespacing\coppe@bancalist}\par
2508 % No city/date block at the foot: 3.1.2.1.3 ends the folha de aprovacao at the
2509 % examining board, and Annex D shows nothing below the signature lines. The date
2510 % that belongs on this page is the date of APPROVAL, printed above.
2511 % \frontpage@bottomtext stays defined for anyone calling it directly.
2512 \vspace*{\fill}}

2513 \newcommand\coppe@hypersetup{%
2514 \begingroup
2515 % changes to \toks@ and \count@ are kept local;
2516 % it's not necessary for them, but it is usually the case
2517 % for \count1, because the first ten counters are written
2518 % to the DVI file, thus you got lucky because of PDF output
2519 \toks@={}% in this special case not necessary
2520 \count@=0 %
2521 \@whilenum\count@<\value{keywords}\do{%
2522   % * a keyword separator is not necessary,
2523   %   if there is just one keyword
2524   % * \csname CoppeKeyword:\the\count@\endcsname must be expanded
2525   %   at least once, to get rid of the loop depended \count@
2526   \ifcase\count@ % same as \ifnum0=\count@
2527     \toks@=\expandafter{\csname CoppeKeyword:\the\count@\endcsname}%
2528   \else
2529     \toks@=\expandafter\expandafter\expandafter{%
2530       \expandafter\the\expandafter\toks@
2531       \expandafter;\expandafter\space
2532       \csname CoppeKeyword:\the\count@\endcsname
2533     }%
2534   \fi
2535   \advance\count@ by 1 %

```



```

2536 }%
2537 \edef\x{\endgroup
2538   \noexpand\hypersetup{%
2539     pdfkeywords={\the\toks0}%
2540   }%
2541 }%
2542 \x
2543 \hypersetup{%
2544   pdfauthor={\@authname\ \@authsur},
2545   pdftitle={\local@title},
2546   pdfsubject={\local@doctype\ de \@degreename\ em \local@deptname\ da COPPE/UFRJ},
2547   pdfcreator={LaTeX with CoppeTeX toolkit},
2548   breaklinks={true},
2549   raiselinks={true},
2550   pageanchor={true},
2551 }}

```

`\makecatalog` When the document has illustrations, it is required to insert “: il;” between the number of pages of the textual part and the page dimension. We have created a label to flag the existence of lists of figures. It is checked to be undefined using the plain T_EX command `\isundefined` (cf. the T_EX FAQ).

```

2552 \newcommand\makecatalog{%
2553   \vspace*{\fill}
2554   \begin{center}
2555     \setlength{\fboxsep}{5mm}
2556     \framebox[120mm][c]{\makebox[5mm][c]{}%
2557       \begin{minipage}[c]{105mm}
2558         \setlength{\parindent}{5mm}
2559         \noindent\sloppy\nohyphens\@authsur,
2560         \nohyphens\@authname\par
2561         \nohyphens{%
2562           \coppe@selecttitle/\@authname\ \@authsur. -- \local@cityname:
2563           UFRJ/COPPE, \coppe@docyear.}\par
2564           \pageref{front:pageno},
2565           \pageref{LastPage}
2566           p.\@ifundefined{r@cat:lofflag}{\pageref{cat:lofflag}}{$29,7$cm.\par
2567           % There is an issue here. When the last entry must be split between lines,
2568           % the spacing between it and the next paragraph becomes smaller.
2569           % Should we manually introduce a fixed space? But how could we know that
2570           % a name was split? Is this happening yet?
2571           \nohyphens{%
2572             \begin{tabularx}{100mm}[b]{@{}l@{ }>{\raggedright\arraybackslash}X@{}}
2573               \local@advisorstring: &
2574               \count1=0
2575               \toks@={}
2576               \@whilenum \count1<\@advisor \do{%
2577                 \ifcase\count1 % same as \ifnum0=\count1
2578                   \toks@=\expandafter{\csname CoppeAdvisorName:\the\count1%
2579                     \expandafter\endcsname\expandafter\space%
2580                     \csname CoppeAdvisorSurname:\the\count1\endcsname\\}
2581                 \else
2582                   \toks@=\expandafter\expandafter\expandafter{%
2583                     \expandafter\the\expandafter\toks@
2584                     \expandafter&\expandafter\space

```

```

2585         \csname CoppeAdvisorName:\the\count1\expandafter\endcsname%
2586         \expandafter\space\csname CoppeAdvisorSurname:\the\count1\endcsname\
2587     }%
2588     \fi
2589     \advance\count1 by 1}
2590     \the\toks@
2591 \end{tabularx}}\par
2592 \nohyphens{\local@doctype\ ({\MakeLowercase\@degreename}) --
2593 UFRJ/COPPE/Programa de \local@deptname, \coppe@docyear.}\par
2594 Refer{\^ e}ncias Bibliogr{\' a}ficas: p. \pageref{bib:begin} -- \pageref{bib:end}.\par
2595 \count1=0
2596 \count2=1
2597 \nohyphens{\@whilenum \count1<\value{keywords} \do {%
2598     \number\count2. \csname CoppeKeyword:\the\count1 \endcsname.
2599     \advance\count1 by 1
2600     \advance\count2 by 1}
2601 I. \csname CoppeAdvisorSurname:0\endcsname,%
2602 \ \csname CoppeAdvisorName:0\endcsname%
2603 \ifthenelse{\@advisor>1}{\ \emph{et~al.}}{}}.
2604 II. \local@universityname, COPPE, Programa de \local@deptname.
2605 III. T{\' i}tulo.}
2606 \end{minipage}}
2607 \end{center}
2608 \vspace*{\fill}}

```

\fichacatalografica The folha adicional required from August 2026 (SiBI manual, 9th rev. ed.,
\makefolhaaditional 3.1.2.1.2 and Annex H): the CAPES collection fields plus the ficha cata-
lografica, sitting immediately after the folha de rosto and counted by no-
body. \fichacatalografica takes the PDF produced by the SiBI generator at
<<http://fichacatalografica.sibi.ufrj.br/>>, optionally with a page number.

The sheet is institutional, so it is always in Portuguese whatever the main lan-
guage of the work – and deliberately without \selectlanguage, since the brazilian
babel language need not be loaded in, say, a Spanish-main document.

```

2609 \newcommand\coppe@fichafile{}
2610 \newcommand\coppe@fichapage{1}
2611 \newcommand\fichacatalografica[2][1]{%
2612     \gdef\coppe@fichapage{#1}\gdef\coppe@fichafile{#2}}
2613 \newcommand\coppe@marcatipo[1]{%
2614     \ifthenelse{\equal{\coppe@tipoproducao}{#1}}{$\times$}{\phantom{$\times$}}}
2615 \newcommand\coppe@marcavinc[1]{%
2616     \ifthenelse{\equal{\coppe@projetovinc}{#1}}{$\times$}{\phantom{$\times$}}}
2617 \newcommand\coppe@campo[1]{%
2618     \ifthenelse{\equal{#1}{}}{\hrulefill}{#1\enspace\hrulefill}}
2619 \newcommand\coppe@checatipo{%
2620     \ifthenelse{\equal{\coppe@tipoproducao}{}}{}}{%
2621         \ifthenelse{\equal{\coppe@tipoproducao}{bibliografica}}\OR
2622             \equal{\coppe@tipoproducao}{artistica}\OR
2623             \equal{\coppe@tipoproducao}{tecnologica}\OR
2624             \equal{\coppe@tipoproducao}{tecnica}}{}}{%
2625     \ClassWarning{coppe}{Unknown \string\coppe@tipoproducao{ } value
2626     '\coppe@tipoproducao'. Annex H of the SiBI manual allows only
2627     bibliografica, artistica, tecnologica or tecnica; no box will be
2628     ticked}}}}

```

```

2629 \newcommand\makefolhaadicional{%
2630 \clearpage
2631 \thispagestyle{empty}%
2632 \coppe@checativo
2633 {\singlespacing
2634 \begin{center}
2635 {\large\bfseries Informa\c c\~oes Coleta CAPES}
2636 \end{center}
2637 \vspace{7mm}
2638 \noindent Tipo de produ\c c\~ao intelectual:\par
2639 \vspace{1mm}
2640 \hspace*{10mm}(\coppe@marcatipo{bibliografica})\enspace Bibliogr\'afica\par
2641 \hspace*{10mm}(\coppe@marcatipo{artistica})\enspace Art'\i{stica}\par
2642 \hspace*{10mm}(\coppe@marcatipo{tecnologica})\enspace Tecnol\'ogica\par
2643 \hspace*{10mm}(\coppe@marcatipo{tecnica})\enspace T\'ecnica\par
2644 \vspace{5mm}
2645 \noindent Projeto de pesquisa vinculado \'a produ\c c\~ao intelectual?\par
2646 \vspace{1mm}
2647 \hspace*{10mm}(\coppe@marcavinc{sim})\enspace Sim\par
2648 \hspace*{10mm}(\coppe@marcavinc{nao})\enspace N\~ao\par
2649 \vspace{5mm}
2650 \noindent Nome do projeto de pesquisa:\par
2651 \vspace{1mm}
2652 \noindent\coppe@campo{\coppe@nomeprojeto}\par
2653 \vspace{5mm}
2654 \noindent \'Area de concentra\c c\~ao da produ\c c\~ao intelectual:\par
2655 \vspace{1mm}
2656 \noindent\coppe@campo{\coppe@areaconc}\par
2657 \vspace{5mm}
2658 \noindent Ag\~encia(s) de fomento financiadora(s) da produ\c c\~ao
2659 intelectual:\par
2660 \vspace{1mm}
2661 \ifthenelse{\value{coppeagencias}=0}{%
2662 \noindent\hrulefill\par}{%
2663 \count1=0
2664 \@whilenum\count1<\value{coppeagencias}\do{%
2665 \noindent\hspace*{10mm}\csname CoppeAgencia:\the\count1\endcsname
2666 \ (\csname CoppeAgenciaSigla:\the\count1\endcsname)\par
2667 \advance\count1 by 1}}%
2668 \vspace{9mm}
2669 \begin{center}
2670 {\large\bfseries Ficha catalogr\'afica}\par
2671 \vspace{4mm}
2672 \ifthenelse{\equal{\coppe@fichafile}{}}{%
2673 \if@copperascunho
2674 \makecatalog
2675 \else
2676 \framebox[125mm][c]{\parbox[c][62mm][c]{115mm}{\centering\footnotesize
2677 Espa\c co reservado \'a ficha catalogr\'afica.\[1ex]
2678 Gere-a em \texttt{fichacatalografica.sibi.ufrj.br} e inclua o
2679 arquivo com \texttt{\string\fichacatalografica\{arquivo.pdf\}},
2680 ou solicite-a \'a biblioteca do seu Programa.}}%
2681 \fi}{%
2682 \includegraphics[page=\coppe@fichapage,width=125mm]{\coppe@fichafile}}%

```

```

2683 \end{center}
2684 \par}}

```

\dedication

```

2685 \newcommand\dedication[1]{
2686 \gdef\@dedic{#1}
2687 \cleardoublepage
2688 \vspace*{\fill}
2689 \begin{flushright}
2690 \begin{minipage}{60mm}
2691 \raggedleft \it \normalsize \@dedic
2692 \end{minipage}
2693 \end{flushright}}

```

\coppe@coadvisorrows The coadvisor block of the abstract pages, in the shape the folha de rosto already uses. It prints nothing at all unless the `coorientador` class option is given AND the document actually declared a `\coadvisor`, so the default output is byte for byte what it was. The argument is the label macro, which differs per page: the main-language pages pass `\local@coadvisorstring` and the foreign one passes `\foreign@coadvisorstring`.

```

2694 \newcommand\coppe@coadvisorrows[1]{%
2695 \if@coppecoadvresumo
2696 \ifnum\@coadvisor>0
2697 \par\vspace*{2mm}%
2698 \noindent\begin{tabularx}{\textwidth}[b]{@{}l@{ }>{\raggedright\arraybackslash}X@{}}
2699 #1: &
2700 \count1=0
2701 \toks@={}
2702 \@whilenum \count1<\@coadvisor \do{%
2703 \ifcase\count1
2704 \toks@=\expandafter{\csname CoppeCoadvisorName:\the\count1%
2705 \expandafter\endcsname\expandafter\space%
2706 \csname CoppeCoadvisorSurname:\the\count1\endcsname\\}
2707 \else
2708 \toks@=\expandafter\expandafter\expandafter{%
2709 \expandafter\the\expandafter\toks@%
2710 \expandafter&\expandafter\space%
2711 \csname CoppeCoadvisorName:\the\count1\expandafter\endcsname%
2712 \expandafter\space\csname CoppeCoadvisorSurname:\the\count1\endcsname\\}
2713 }%
2714 \fi
2715 \advance\count1 by 1}
2716 \the\toks@
2717 \end{tabularx}%
2718 \fi
2719 \fi}

```

abstract (*env.*) This is a specialization of the abstract in the article standard class.

```

2720 \newenvironment{abstract}{%
2721 \cleardoublepage
2722 \thispagestyle{plain}
2723 \abstract@toptext\par
2724 \vspace*{8.6mm}

```

```

2725 \begin{center}
2726 \sloppy\nohyphens{\MakeUppercase\local@title}\par
2727 \vspace*{13.2mm}
2728 \@authname\ \@authsur \par
2729 \vspace*{7mm}
2730 \local@monthname/\coppe@docyear
2731 \end{center}\par
2732 \vspace*{1.5cm}
2733 \noindent%
2734 \begin{tabularx}{\textwidth}[b]{@{}l@{ }>{\raggedright\arraybackslash}X@{}}
2735 \local@advisorstring: &
2736 \count1=0
2737 \toks@={}
2738 \@whilenum \count1<\@advisor \do{%
2739 \ifcase\count1 % same as \ifnum0=\count1
2740 \toks@=\expandafter{\csname CoppeAdvisorName:\the\count1%
2741 \expandafter\endcsname\expandafter\space%
2742 \csname CoppeAdvisorSurname:\the\count1\endcsname\\}
2743 \else
2744 \toks@=\expandafter\expandafter\expandafter{%
2745 \expandafter\the\expandafter\toks@
2746 \expandafter\&\expandafter\space
2747 \csname CoppeAdvisorName:\the\count1\expandafter\endcsname%
2748 \expandafter\space\csname CoppeAdvisorSurname:\the\count1\endcsname\\}
2749 }%
2750 \fi
2751 \advance\count1 by 1}
2752 \the\toks@
2753 \end{tabularx}%
2754 \coppe@coadvisorrows{\local@coadvisorstring}\par
2755 \vspace*{2mm}
2756 \noindent\local@deptstring: \local@deptname\par
2757 \vspace*{7mm}}{%
2758 \coppe@printkeywords{\coppemainstring{keywordsname}}%
2759 {CoppeKeyword:}{\value{keywords}}}%
2760 \vspace*{\fill}}

```

foreignabstract (env.)

```

2761 \newenvironment{foreignabstract}{%
2762 \cleardoublepage
2763 \thispagestyle{plain}
2764 \begin{otherlanguage}{\coppe@foreignlang}
2765 \foreignabstract@toptext\par
2766 \vspace*{8.6mm}
2767 \begin{center}
2768 \sloppy\nohyphens{\MakeUppercase\foreign@title}\par
2769 \vspace*{13.2mm}
2770 \@authname\ \@authsur \par
2771 \vspace*{7mm}
2772 \foreign@monthname/\coppe@docyear
2773 \end{center}\par
2774 \vspace*{1.5cm}
2775 \noindent%
2776 \begin{tabularx}{\textwidth}[b]{@{}l@{ }>{\raggedright\arraybackslash}X@{}}

```

```

2777 \foreign@advisorstring: &
2778 \count1=0
2779 \toks@={}
2780 \@whilenum \count1<\@advisor \do{%
2781 \ifcase\count1 % same as \ifnum0=\count1
2782 \toks@=\expandafter{\csname CoppeAdvisorName:\the\count1%
2783 \expandafter\endcsname\expandafter\space%
2784 \csname CoppeAdvisorSurname:\the\count1\endcsname\\}
2785 \else
2786 \toks@=\expandafter\expandafter\expandafter{%
2787 \expandafter\the\expandafter\toks@
2788 \expandafter&\expandafter\space
2789 \csname CoppeAdvisorName:\the\count1\expandafter\endcsname%
2790 \expandafter\space\csname CoppeAdvisorSurname:\the\count1\endcsname\\
2791 }%
2792 \fi
2793 \advance\count1 by 1}
2794 \the\toks@
2795 \end{tabularx}%
2796 \coppe@coadvisorrows{\foreign@coadvisorstring}\par
2797 \vspace*{2mm}
2798 \noindent\foreign@deptstring: \foreign@deptname\par
2799 \vspace*{7mm}}{%
2800 \ifnum\value{foreignkeywords}>0
2801 \coppe@printkeywords{\coppeforeignstring{keywordsname}}%
2802 {CoppeForeignKeyword:}{\value{foreignkeywords}}%
2803 \else
2804 \coppe@printkeywords{\coppeforeignstring{keywordsname}}%
2805 {CoppeKeyword:}{\value{keywords}}%
2806 \fi
2807 \end{otherlanguage}
2808 \vspace*{\fill}
2809 \global\let\@author\@empty
2810 \global\let\@date\@empty
2811 \global\let\foreign@title\@empty
2812 \global\let\foreign@title\relax
2813 \global\let\local@title\@empty
2814 \global\let\local@title\relax
2815 \global\let\author\relax
2816 \global\let\author\relax
2817 \global\let\date\relax}

```

brazilianabstract (*env.*) Optional third abstract for non-pt main languages: a Brazilian Portuguese abstract printed in addition to **abstract** (main language) and **foreignabstract** (English by default). Wraps its body in `\begin{otherlanguage}{brazilian}` so the babel typography (hyphenation, quotation marks) is correct regardless of the current main language. Uses the same `\local@...` Portuguese strings the **abstract** env uses today. Meaningful only when main is neither **brazilian** nor **english**; harmless otherwise (the document just won't include it).

```

2818 \newenvironment{brazilianabstract}{%
2819 \cleardoublepage
2820 \thispagestyle{plain}
2821 \begin{otherlanguage}{brazilian}

```

```

2822 \abstract@top text\par
2823 \vspace*{8.6mm}
2824 \begin{center}
2825 \sloppy\nohyphens{\MakeUppercase\local@title}\par
2826 \vspace*{13.2mm}
2827 \@authname\ \@authsur \par
2828 \vspace*{7mm}
2829 \local@monthname/\coppe@docyear
2830 \end{center}\par
2831 \vspace*{1.5cm}
2832 \noindent%
2833 \begin{tabularx}{\textwidth}[b]{@{}l@{ }>{\raggedright\arraybackslash}X@{}}
2834 \local@advisorstring: &
2835 \count1=0
2836 \toks@={}
2837 \@whilenum \count1<\@advisor \do{%
2838 \ifcase\count1
2839 \toks@=\expandafter{\csname CoppeAdvisorName:\the\count1%
2840 \expandafter\endcsname\expandafter\space%
2841 \csname CoppeAdvisorSurname:\the\count1\endcsname\\}
2842 \else
2843 \toks@=\expandafter\expandafter\expandafter{%
2844 \expandafter\the\expandafter\toks@
2845 \expandafter&\expandafter\space
2846 \csname CoppeAdvisorName:\the\count1\expandafter\endcsname%
2847 \expandafter\space\csname CoppeAdvisorSurname:\the\count1\endcsname\\}
2848 }%
2849 \fi
2850 \advance\count1 by 1}
2851 \the\toks@
2852 \end{tabularx}%
2853 \coppe@coadvisorrows{\local@coadvisorstring}\par
2854 \vspace*{2mm}
2855 \noindent\local@deptstring: \local@deptname\par
2856 \vspace*{7mm}}{%
2857 % \coppestring takes ONE argument and resolves against the current language;
2858 % this runs before \end{otherlanguage}, so \language is already brazilian.
2859 \ifnum\value{braziliankeywords}>0
2860 \coppe@printkeywords{\coppestring[keywordsname]}%
2861 {CoppeBrazilianKeyword:}{\value{braziliankeywords}}%
2862 \else
2863 \coppe@printkeywords{\coppestring[keywordsname]}%
2864 {CoppeKeyword:}{\value{keywords}}%
2865 \fi
2866 \end{otherlanguage}
2867 \vspace*{\fill}}

```

\listoffigures

```

2868 \renewcommand\listoffigures{%
2869 \coppe@hasLoF
2870 \if@twocolumn
2871 \@restonecoltrue\onecolumn
2872 \else
2873 \@restonecolfalse

```

```

2874 \fi
2875 \chapter*{\listfigurename}%
2876 \coppe@listtoc{\listfigurename}%
2877 \mkboth{\MakeUppercase\listfigurename}%
2878 {\MakeUppercase\listfigurename}%
2879 \@starttoc{lof}%
2880 \if@restonecol\twocolumn\fi
2881 }

\listoftables
2882 \renewcommand\listoftables{%
2883 \if@twocolumn
2884 \@restonecoltrue\onecolumn
2885 \else
2886 \@restonecolfalse
2887 \fi
2888 \chapter*{\listtablename}%
2889 \coppe@listtoc{\listtablename}%
2890 \mkboth{%
2891 \MakeUppercase\listtablename}%
2892 {\MakeUppercase\listtablename}%
2893 \@starttoc{lot}%
2894 \if@restonecol\twocolumn\fi
2895 }

\printlosymbols
2896 \newcommand\printlosymbols{%
2897 \renewcommand\glossaryname{\listsymbolname}%
2898 \begingroup\@coppeglosspretrue
2899 \@input@{\jobname.los}\endgroup}

\makelosymbols
2900 \def\makelosymbols{%
2901 \newwrite\@losfile
2902 \immediate\openout\@losfile=\jobname.syx
2903 \newcommand\syml[3][\@bsphack\begingroup
2904 \ifstrempy{##1}{\def\@tempsymb1{##2}}{\def\@tempsymb1{##1}}}%
2905 \@sanitize%
2906 \@wrtlos{\@tempsymb1}{##2}{##3}}\typeout%
2907 {Writing index of symbols file \jobname.syx}%
2908 \let\makelosymbols\@empty%
2909 }%
2910 \@onlypreamble\makelosymbols

2911 \AtBeginDocument{%
2912 \@ifpackageloaded{hyperref}{%
2913 \newcommand\@wrtlos[3]{%
2914 \protected@write\@losfile{%
2915 {\string\indexentry{#1[#2] #3|hyperpage}{\thepage}}%
2916 \endgroup%
2917 \@esphack}}}%
2918 \newcommand\@wrtlos[3]{%
2919 \protected@write\@losfile{%
2920 {\string\indexentry{#1[#2] #3}{\thepage}}}%

```



```

2921 \endgroup%
2922 \@esphack}}}%

\printloabbreviations
2923 \newcommand\printloabbreviations{%
2924 \renewcommand\glossaryname{\listabbreviationname}%
2925 \begingroup\@coppeglosspretrue
2926 \@input@{\jobname.lab}\endgroup}

\makeloabbreviations
2927 \def\makeloabbreviations{%
2928 \newwrite\@labfile
2929 \immediate\openout\@labfile=\jobname.abx
2930 \newcommand\abbrev[3][\@bsphack\begingroup
2931 \ifstrempy{##1}{\def\@tempsymb1{##2=}}{\def\@tempsymb1{##1=}}
2932 \@sanitize
2933 \@wrlab{\@tempsymb1}{##2}{##3}\typeout
2934 {Writing index of abbreviations file \jobname.abx}%
2935 \let\makeloabbreviations\@empty
2936 }
2937 \@onlypreamble\makeloabbreviations

2938 \AtBeginDocument{%
2939 \ifpackageloaded{hyperref}{%
2940 \newcommand\@wrlab[3]{%
2941 \protected@write\@labfile{%
2942 {\string\indexentry{#1[#2] #3|hyperpage}}{\thepage}}%
2943 \endgroup%
2944 \@esphack}}{%
2945 \newcommand\@wrlab[3]{%
2946 \protected@write\@labfile{%
2947 {\string\indexentry{#1[#2] #3}{\thepage}}%
2948 \endgroup%
2949 \@esphack}}}%

\newsigla Acronyms (siglas), manual2024 2.8 / R43. \newsigla{key}{SIGLA}{full form}
\sigla declares one; \sigla{key} prints “full form (SIGLA)” on first use and registers it in
the list of abbreviations and acronyms (through \abbrev, so \makeloabbreviations
must be active), then just “SIGLA” on later uses. \sigla*{key} forces the full
form. The first-use flag is set globally so it survives groups (floats, footnotes).

2950 \newcommand\newsigla[3]{%
2951 \@namedef{coppe@sigla@s@#1}{#2}%
2952 \@namedef{coppe@sigla@l@#1}{#3}%
2953 \newcommand\coppe@sigla@reg[1]{%
2954 \@ifundefined{abbrev}{%
2955 {\abbrev{\@nameuse{coppe@sigla@s@#1}}{\@nameuse{coppe@sigla@l@#1}}}}
2956 \newcommand\coppe@sigla@full[1]{%
2957 \@nameuse{coppe@sigla@l@#1}\space(\@nameuse{coppe@sigla@s@#1})}
2958 \newcommand\coppe@sigla@undef[1]{%
2959 \PackageWarning{coppe}{Sigla '#1' nao declarada com \string\newsigla}%
2960 \textbf{??#1??}}
2961 \newcommand\coppe@sigla@first[1]{%
2962 \global\@namedef{coppe@sigla@done@#1}{\coppe@sigla@reg{#1}\coppe@sigla@full{#1}}
2963 \newcommand\coppe@sigla@norm[1]{%

```

```

2964 \ifundefined{coppe@sigla@s#1}{\coppe@sigla@undef{#1}}%
2965 {\ifundefined{coppe@sigla@done@#1}{\coppe@sigla@first{#1}}%
2966 {\@nameuse{coppe@sigla@s#1}}}%
2967 \newcommand\coppe@sigla@star[1]{%
2968 \ifundefined{coppe@sigla@s#1}{\coppe@sigla@undef{#1}}%
2969 {\global\@namedef{coppe@sigla@done@#1}{\coppe@sigla@full{#1}}}%
2970 \newcommand\sigla{\ifstar\coppe@sigla@star\coppe@sigla@norm}

2971 %%% \AtBeginDocument{%
2972 %%% \ifpackageloaded{hyperref}{%
2973 %%% \def\@wrlab#1#2{%
2974 %%% \protected@write\@labfile{%
2975 %%% {\string\indexentry{[#1] #2|hyperpage}{\thepage}}%
2976 %%% \endgroup
2977 %%% \@esphack}}{%
2978 %%% \def\@wrlab#1#2{%
2979 %%% \protected@write\@labfile{%
2980 %%% {\string\indexentry{[#1] #2}{\arabic{page}}}%
2981 %%% \endgroup
2982 %%% \@esphack}}}

2983 % Catalog-label writers. Two language-related traps must be avoided here:
2984 % 1. The label NAMES contain ':' and '.', which babel makes active in
2985 % some languages (e.g. Brazilian's "shorthand" set). If the write
2986 % argument is tokenised while one of those is active, pdfTeX aborts
2987 % with "<inserted text> }\endwrite" because the shorthand expansion
2988 % injects unbalanced braces. \detokenize emits its argument as
2989 % catcode-12 characters regardless of the current activation, so the
2990 % label NAMES survive whatever language is current at writer time.
2991 % 2. Spanish babel redefines \@roman (and therefore \roman{...}) to
2992 % produce small-caps roman numerals via \es@scroman, whose expansion
2993 % contains an \uppercase group that is unbalanced inside \write. The
2994 % same crash signature appears. The fix is to bypass LaTeX's
2995 % \roman/\arabic counter wrappers entirely and use the raw e-TeX
2996 % primitives \romannumeral and \number directly on the \c@page count
2997 % register --- both are fully expandable, language-agnostic, and
2998 % always produce plain ASCII digits or lowercase roman letters.
2999 % \Roman{page} is left in place because babel-spanish only redefines
3000 % \@Roman under the (non-default) ucroman option.
3001 \AtBeginDocument{%
3002 \ltx@ifpackageloaded{hyperref}{%
3003 \def\coppe@bibEnd{%
3004 \immediate\write\@auxout{%
3005 \string\newlabel{\detokenize{bib:end}}%
3006 {\number\c@page}{\detokenize{page.}\number\c@page}}}%
3007 \def\coppe@bibBegin{%
3008 \immediate\write\@auxout{%
3009 \string\newlabel{\detokenize{bib:begin}}%
3010 {\number\c@page}{\detokenize{page.}\number\c@page}}}%
3011 \def\coppe@mainBegin{%
3012 \immediate\write\@auxout{%
3013 \string\newlabel{\detokenize{front:pageno}}%
3014 {\Roman{page}}{\detokenize{page.}\romannumeral\c@page}}}%
3015 \def\coppe@hasLof{%
3016 \immediate\write\@auxout{%

```

```

3017         \string\newlabel{\detokenize{cat:lofflag}}%
3018         {{{\detokenize{:~il.;}}}{\detokenize{page.}\romannumeral\c@page}}}}}%
3019 }{%
3020     \def\coppe@bibEnd{%
3021         \immediate\write\@auxout{%
3022             \string\newlabel{\detokenize{bib:end}}>{{\number\c@page}}}}}%
3023     \def\coppe@bibBegin{%
3024         \immediate\write\@auxout{%
3025             \string\newlabel{\detokenize{bib:begin}}>{{\number\c@page}}}}}%
3026     \def\coppe@mainBegin{%
3027         \immediate\write\@auxout{%
3028             \string\newlabel{\detokenize{front:pageno}}>{{\Roman{page}}}}}%
3029     \def\coppe@hasLof{%
3030         \immediate\write\@auxout{%
3031             \string\newlabel{\detokenize{cat:lofflag}}>{{\detokenize{:~il.;}}}}}%
3032 }%
3033 }
3034 % The reference list is now produced by biblatex's \printbibliography (see the
3035 % bibliography setup after babel, above). The former thebibliography
3036 % redefinition and its \bibindent are no longer used and have been removed; the
3037 % bib:begin/bib:end catalog labels are now written by \AtBeginBibliography /
3038 % \AtEndBibliography via \coppe@bibBegin and \coppe@bibEnd, defined just above.

```

longquote (*env.*)

```

3039 \newlength{\recuolongquote}
3040 \setlength{\recuolongquote}{4cm}
3041
3042 \newenvironment*{longquote}[1][default]{%
3043     \begin{list}{}{%
3044         \setlength{\leftmargin}{\recuolongquote}%
3045         \setlength{\rightmargin}{0pt}%
3046         \setlength{\itemindent}{0pt}%
3047         \setlength{\listparindent}{0pt}%
3048         \setlength{\parsep}{0pt}%
3049         \setlength{\topsep}{\baselineskip}%
3050     }%
3051     \item\relax
3052     \footnotesize
3053     \singlespacing
3054     \ifthenelse{\not\equal{#1}{default}}{%
3055         {\itshape\selectlanguage{#1}}%
3056     }%
3057     \ignorespaces
3058 }{%
3059     \end{list}%
3060     \ignorespacesafterend
3061 }

3062 \newenvironment{theglossary}{%
3063     \if@twocolumn%
3064         \@restonecoltrue\onecolumn%
3065     \else%
3066         \@restonecolfalse%
3067     \fi%

```

```

3068 \@mkboth{\MakeUppercase\glossaryname}%
3069 {\MakeUppercase\glossaryname}%
3070 \chapter*{\glossaryname}%
3071 \if@coppeglosspre
3072   \coppe@listtoc{\glossaryname}%
3073 \else
3074   \addcontentsline{toc}{chapter}{\glossaryname}%
3075 \fi
3076 \list{}
3077 {\setlength{\listparindent}{0in}%
3078  \setlength{\labelwidth}{1.0in}%
3079  \setlength{\leftmargin}{1.5in}%
3080  \setlength{\labelsep}{0.5in}%
3081  \setlength{\itemindent}{0in}}%
3082 \sloppy}%
3083 {\if@restonecol\twocolumn\fi%
3084 \endlist}
3085 %
3086 \renewenvironment{theindex}{%
3087   \if@twocolumn
3088     \@restonecolfalse
3089   \else
3090     \@restonecoltrue
3091   \fi
3092   \twocolumn[\@makeschapterhead{\indexname}]%
3093   \@mkboth{\MakeUppercase\indexname}%
3094   {\MakeUppercase\indexname}%
3095   \thispagestyle{plain}\parindent\z@
3096   \addcontentsline{toc}{chapter}{\indexname}
3097   \parskip\z@ \@plus .3\p@\relax
3098   \columnseprule \z@
3099   \columnsep 35\p@
3100   \let\item\@idxitem}
3101   {\if@restonecol\onecolumn\else\clearpage\fi}
3102 \newcommand\listabbreviationname{\coppemainstring{listabbreviation}}
3103 \newcommand\listsymbolname{\coppemainstring{listsymbol}}
3104 \newcommand\glossaryname{\coppemainstring{glossary}}
3105 %
3106 \newcommand\local@advisorstring{Orientador}
3107 \newcommand\foreign@advisorstring{Advisor}
3108 % Fed by the string dispatcher instead of a hard-coded Portuguese literal.
3109 % Expansion is lazy (it happens inside \makefrontpage), so \coppe@mainlang is
3110 % already set by the time this is used.
3111 \ifthenelse{\boolean{maledoc}}{%
3112   \newcommand\local@approvedname{\coppemainstring{approvedmale}}%
3113 }{%
3114   \newcommand\local@approvedname{\coppemainstring{approvedfemale}}%
3115 }
3116 \ifthenelse{\boolean{maledoc}}{%
3117   \newcommand\local@approvedonname{\coppemainstring{approvedonmale}}%
3118 }{%
3119   \newcommand\local@approvedonname{\coppemainstring{approvedonfemale}}%
3120 }
3121 \newcommand\local@universityname{Universidade Federal do Rio de Janeiro}

```

```

3122 \newcommand\local@deptstring{Programa}
3123 \newcommand\foreign@deptstring{Department}
3124 \newcommand\local@cityname{Rio de Janeiro}
3125 \newcommand\local@statename{RJ}
3126 \newcommand\local@countryname{Brasil}
3127 %
3128 % 3.1.2.1.1(e) of the 2026 manual requires the natureza, the objetivo, the
3129 % name of the institution AND the area de concentracao in this block. The area
3130 % was missing; it is appended as its own sentence when \areaconcentracao was
3131 % given, and simply omitted otherwise.
3132 \newcommand\frontcover@maintext{
3133   \sloppy\nohyphens{\local@doctype\ de \@degreename\
3134   \ifthenelse{\boolean{maledoc}}{apresentado}{apresentada}
3135   ao Programa de P{\' o}s-gradua{\c c}{\~ a}o em \local@deptname,
3136   COPPE, da \local@universityname, como parte dos requisitos
3137   necess{\' a}rios {\' a} obten{\c c}{\~ a}o do t{\' i}tulo de
3138   \local@degname\ em \local@deptname.}%
3139   \ifthenelse{\equal{\coppe@areaconc}{}}{ }{\%
3140     \par\vspace{1ex}%
3141     \sloppy\nohyphens{\' Area de concentra{\c c}{\~ a}o: \coppe@areaconc.}}}%
3142 }
3143 %
3144 \newcommand\frontpage@maintext{
3145   \noindent {\MakeUppercase\local@doctype}
3146   \ifthenelse{\boolean{maledoc}}{SUBMETIDO}{SUBMETIDA}
3147   \sloppy\nohyphens{AO CORPO DOCENTE DO INSTITUTO ALBERTO LUIZ COIMBRA
3148   DE P{\' O}S-GRADUA{\c C}{\~ A}O E PESQUISA DE ENGENHARIA DA
3149   UNIVERSIDADE FEDERAL DO RIO DE JANEIRO COMO PARTE DOS REQUISITOS
3150   NECESS{\' A}RIOS PARA A OBTEN{\c C}{\~ A}O DO GRAU DE
3151   {\MakeUppercase\local@degname} EM CI{\^E}NCIAS EM
3152   {\MakeUppercase\local@deptname.\par}}}%
3153 }
3154 %
3155 \newcommand\frontpage@bottomtext{%
3156   \begin{center}
3157   {\MakeUppercase{\local@cityname, \local@statename\ -- \local@countryname}}\par
3158   {\MakeUppercase\local@monthname\ DE \coppe@docyear}
3159   \end{center}}%
3160 }
3161 %
3162 \newcommand\abstract@toptext{%
3163   \noindent Resumo \ifthenelse{\boolean{maledoc}}{do}{da}
3164   \local@doctype\ \ifthenelse{\boolean{maledoc}}{apresentado}{apresentada}
3165   \sloppy\nohyphens{{\\' a} COPPE/UFRJ como parte dos requisitos
3166   necess{\' a}rios para a obten{\c c}{\~ a}o do grau de
3167   \local@degname\ em Ci{\^ e}ncias (\@degree)}
3168 }
3169 \newcommand\foreignabstract@toptext{%
3170   \noindent \sloppy\nohyphens{Abstract of \foreign@doctype\ presented to
3171   COPPE/UFRJ as a partial fulfillment of the requirements for the
3172   degree of \foreign@degname\ of Science (\@degree)}
3173 }
3174 %

```

Late-load packages: setspace, hyperref, biblatex

These three packages *must* be the last things the class loads, and in this exact order:

- **setspace** – supplies `\singlespacing`, `\onehalfspacing` and friends. It is loaded here so the line-spacing state it carries around is settled before **hyperref** patches all the cross-reference and TOC plumbing.
- **hyperref** – patches an enormous number of LaTeX internals (`\section`, `\label`, `\ref`, `\bibcite`, list-of-* commands, ...). Loading it as late as possible is the standard rule of thumb so it sees the final version of those macros, including the ones the **coppe** class redefines (e.g. `\@chapter`, `\listofquados`).
- **biblatex** – must come after **hyperref** so its citation links and back-references hook into **hyperref** correctly; otherwise the `\cite` commands print as plain (non-clickable) text.

```
3175 \RequirePackage{setspace}
3176 % Apply the default line spacing now that setspace is available. The flag was
3177 % set above by \DeclareOption{doublespacing}; absent the option, one-and-a-half.
3178 \if@coppedoublespace\doublespacing\else\onehalfspacing\fi
3179 % pdfx MUST precede hyperref -- it loads hyperref itself, with the options
3180 % PDF/A needs, and cannot fix up an already-loaded one. Level a-2b rather than
3181 % a-1b (the "ISO 19005-1" of Annex I): a-1b forbids transparency, which
3182 % tcolorbox and several graphics packages produce, so a-1b would fail on
3183 % documents that are otherwise perfectly fine. a-2b is ISO 19005-2, accepted by
3184 % Pantheon and by the SiBI's own instructions, which also list PDF/A-2b.
3185 \if@coppepdfa
3186   \RequirePackage[a-2b]{pdfx}
3187 % pdfTeX stamps /PTEX.Fullbanner into the document information dictionary.
3188 % It is not one of the keys ISO 19005-2 6.6.2.3.2 maps to an XMP property, and
3189 % no extension schema declares it, so it is a DocInfo entry with no XMP
3190 % counterpart -- a finding a PDF/A checker has every reason to raise, and one
3191 % that costs nothing to remove. -1 sets every suppression bit.
3192 %
3193 % The two engines spell this differently, and not just in the name. pdfTeX has
3194 % \pdfsuppressptexinfo, whose bits only ever govern PTEX.* entries, so -1 (all
3195 % of them) is safe. LuaTeX moved the pdfTeX integer parameters behind
3196 % \pdfvariable in 0.85 AND renamed this one to suppressoptionalinfo, whose bits
3197 % reach further: besides the four PTEX.* entries (1, 2, 4, 8) it can also
3198 % suppress Creator, the dates, Producer, Trapped and -- the dangerous one --
3199 % the trailer /ID, which PDF/A requires. So -1 is exactly what must NOT be
3200 % written there; 15 is the four PTEX.* bits and nothing else. Checked on a real
3201 % LuaTeX run: /ID survives.
3202 %
3203 % Getting the name wrong is not a quiet failure. \pdfvariable with an unknown
3204 % variable name makes LuaTeX typeset the name, and the document stops with
3205 % "Missing \begin{document}" -- which is how the first attempt at this was
3206 % caught.
3207 \ifdefined\pdfsuppressptexinfo
3208   \pdfsuppressptexinfo=-1
3209 \else
3210   \ifdefined\pdfvariable
```

```

3211     \pdfvariable suppressoptionalinfo=15
3212     \fi
3213 \fi
3214 % pdfx takes the XMP metadata from \jobname.xmpdata, which it reads when it is
3215 % LOADED -- before the preamble has run, so before \title, \author and
3216 % \keyword exist. The file is therefore WRITTEN by the class and picked up on
3217 % the NEXT run. The build already runs pdflatex three times for cross-
3218 % references, so the deposited PDF always carries current metadata; only a
3219 % first-ever compilation of a brand-new file emits pdfx's "no \jobname.xmpdata"
3220 % warning and lacks dc:title.
3221 %
3222 % The write happens at the END of the document, not at \begin{document}. This
3223 % class has always documented -- and every example follows it -- that \title,
3224 % \author, \department and \keyword go AFTER \begin{document}, next to
3225 % \maketitle. A write at \begin{document} therefore ran before any of them
3226 % existed and died with "Undefined control sequence" on \@authname, so the
3227 % pdfa option was unusable in exactly the style the manual teaches. It only
3228 % ever worked for a document that set its metadata in the preamble. At the end
3229 % everything is set whichever style the author chose, and \immediate still puts
3230 % the file on disk in time for the next run.
3231 %
3232 % The hook is enddocument/afteraux, and the stream is \@mainaux rather than one
3233 % of our own. TeX has sixteen output streams and example.tex was already using
3234 % all of them (toc, lof, lot, loa, lol, loq, los, idx, glo, aux, bcf, ...), so
3235 % \newwrite here failed with "No room for a new \write" on precisely the
3236 % document that most needs validating. enddocument/afteraux fires after LaTeX
3237 % has closed the .aux, which frees that stream for the one \immediate\openout
3238 % we need and costs the document nothing. Kernels older than 2020-10-01 have no
3239 % such hook; there the class falls back to \AtEndDocument with a stream of its
3240 % own, which is the pre-existing behaviour.
3241 %
3242 % Every field is guarded, and the guards are not cosmetic. \coppe@selecttitle
3243 % is \@nameuse{coppe@title@<lang>}, and \csname on a name that does not exist
3244 % yet MAKES it \relax: unexpandable, so \write puts the control sequence NAME
3245 % into the file instead of the title. The .xmpdata then carried the literal
3246 % \coppe@title@brazilian, pdfx read it back on the following run, and the
3247 % document stopped compiling at all -- a single bad run bricked the file until
3248 % someone deleted the .xmpdata by hand. A field that is not set is now simply
3249 % omitted, which pdfx handles, instead of poisoning the next run.
3250 %
3251 % The writes must be \immediate. \protected@write defers to the shipout, and
3252 % the stream is closed here and now, so deferred writes would land nowhere --
3253 % which is exactly what happened first time round: a 0-byte .xmpdata.
3254 % \let\protect\string inside the group gives the protection that
3255 % \protected@write would have given: accents and other robust commands are
3256 % written literally instead of being expanded. pdfx reads the .xmpdata as
3257 % LaTeX, so \a arrives there intact.
3258 \ifdefined\AddToHook
3259     \let\coppe@xmpout\@mainaux
3260     \def\coppe@xmphook{\AddToHook{enddocument/afteraux}}%
3261 \else
3262     \newwrite\coppe@xmpout
3263     \let\coppe@xmphook\AtEndDocument
3264 \fi

```

```

3265 \coppe@xmphook{%
3266   \gdef\coppe@xmpkeys{%
3267     \count@=0
3268     \@whilenum\count@<\value{keywords}\do{%
3269       \ifnum\count@>0
3270         \g@addto@macro\coppe@xmpkeys{\string\sep\space}%
3271       \fi
3272       \protected@edef\coppe@tmp{\csname CoppeKeyword:\the\count@\endcsname}%
3273       \expandafter\g@addto@macro\expandafter\coppe@xmpkeys
3274       \expandafter{\coppe@tmp}%
3275       \advance\count@ by 1}%
3276 \immediate\openout\coppe@xmpout=\jobname.xmpdata
3277 \begingroup
3278   \let\protect\string
3279   \@ifundefined{coppe@title@\coppe@mainlang}{%{}%
3280     \immediate\write\coppe@xmpout{\string\Title{\coppe@selecttitle}}}%
3281   \@ifundefined{@authname}{%{}%
3282     \immediate\write\coppe@xmpout{%
3283       \string\Author{\@authname\space\@authsur}}}%
3284   \immediate\write\coppe@xmpout{\string\Keywords{\coppe@xmpkeys}}%
3285   \ifthenelse{equal{\coppe@deptcode}{}}{%{}%
3286     \immediate\write\coppe@xmpout{\string\Subject{\local@doctype\space de
3287       \@degreename. Programa de \meta@deptname\space(\coppe@deptcode),
3288       COPPE/UFRJ}}}%
3289   \immediate\write\coppe@xmpout{\string\Publisher{\local@universityname}}}%
3290   \@ifundefined{coppe@docyear}{%{}%
3291     \immediate\write\coppe@xmpout{\string\Date{\coppe@docyear}}}%
3292 \endgroup
3293 \immediate\closeout\coppe@xmpout}%
3294 \fi
3295 \RequirePackage{hyperref}
3296 % Bibliography engine: the full coppe BibLaTeX/biber ABNT style, replacing the
3297 % former natbib+bibtex setup. The numbers class option selects the numeric
3298 % citation style; otherwise the author-date default is used. natbib=true keeps
3299 % \citel/\citep available for users migrating from natbib.
3300 \if@coppenumeric
3301   \RequirePackage[backend=biber,database=coppe,%
3302     bibstyle=coppe-numeric,citestyle=coppe-numeric,natbib=true]{bibtex}
3303 \else
3304   \RequirePackage[backend=biber,database=coppe,%
3305     bibstyle=coppe,citestyle=coppe,natbib=true]{bibtex}
3306 \fi
3307 % Reference-list heading "Referencias" (pt) / "References" (en) -- manual2024
3308 % 3.1.4.1 requires the post-textual list to be titled "Referencias", not
3309 % "Bibliografia" (babel's default \bibname). The title is rendered through
3310 % \coppestring so it follows the current main language in BOTH the chapter
3311 % heading and the .toc entry (bibtex's \bibstring proved unreliable across
3312 % the .toc write, and babel's \bibname caption override was equally fragile).
3313 \defbibheading{bibliography}{%
3314   \chapter*{\coppestring{references}}%
3315   \addcontentsline{toc}{chapter}{\coppestring{references}}
3316 % The catalog page range (bib:begin/bib:end) was written by the old
3317 % thebibliography redefinition, which bibtex's \printbibliography does not
3318 % use. bibtex offers \AtBeginBibliography but no end counterpart, so the

```



```

3319 % begin label uses that hook and the end label is emitted by wrapping
3320 % \printbibliography to write it just after the reference list.
3321 \AtBeginBibliography{\coppe@bibBegin}
3322 \let\coppe@orig@printbibliography\printbibliography
3323 \renewcommand\printbibliography[1][]{\%
3324   \coppe@orig@printbibliography[#1]\%
3325   \coppe@bibEnd}
3326 \</class>

3327 \<*glossary>
3328 actual '='
3329 quote '!'
3330 level '>'
3331 %%%% delim_0 " , p. "
3332 delim_0 "\\dotfill "
3333 lethead_flag 0
3334 headings_flag 0
3335 preamble
3336 "\n\begin{theglossary}\n \\makeatletter"
3337 postamble
3338 "\n \\end{theglossary}\n"
3339 \</glossary>

3340 \<*class>
3341 \newcommand{\annex}{
3342   \renewcommand{\appendixname}{\coppestring{annex}}\%
3343   \appendix
3344   \@coppeannextrue
3345 % \appendix has already re-issued the centred chapter format, but it baked in
3346 % the \appendixname of that moment. Re-issue it now that the name is "Anexo".
3347   \coppe@appendixheadings
3348 }
3349 \</class>

```

BibLaTeX style files

Generated from this .dtx by coppe.ins (one docstrip module each).

```

3350 \<*dbx>
3351 \ProvidesFile{coppe.dbx}[2026/09/09 v4.1 CoppeTeX ABNT datamodel]
3352 % Custom entry types beyond biblatex's standard datamodel (games, TV, maps).
3353 % map = cartographic document (NBR 6023 4.2.12); biblatex has no native @map.
3354 % The remaining ABNT "exotic" types (legislation, jurisdiction, legal, patent,
3355 % standard, music, audio, image, artwork, video, dataset, letter, periodical,
3356 % performance) are already in biblatex's standard datamodel.
3357 \DeclareDatamodelEntrytypes{game,videogame,boardgame,tvshow,map}
3358 % Custom fields.
3359 \DeclareDatamodelFields[type=field,datatype=literal]{course} % "(... em <course>)"
3360 \DeclareDatamodelFields[type=field,datatype=literal]{coppedegree} % degree word
3361 \DeclareDatamodelFields[type=field,datatype=literal]{platform} % videogame platform
3362 \DeclareDatamodelFields[type=field,datatype=literal]{scale} % map scale (escala)
3363 \DeclareDatamodelFields[type=list,datatype=literal]{artist} % game/illustration art
3364 \DeclareDatamodelFields[type=list,datatype=literal]{director} % movie/tvshow/video
3365 \DeclareDatamodelFields[type=list,datatype=literal]{producer}
3366 \DeclareDatamodelFields[type=field,datatype=literal]{caderno} % newspaper section

```

```

3367 \DeclareDatamodelFields[type=field,datatype=literal]{relator}      % jurisprudence rapporteur
3368 \DeclareDatamodelEntryfields{course,coppedegree}
3369 \DeclareDatamodelEntryfields[game,videogame,boardgame,software,movie,tvshow,video]%
3370   {artist,platform,director,producer,version}
3371 \DeclareDatamodelEntryfields[map]{scale}
3372 \DeclareDatamodelEntryfields[article]{caderno}
3373 \DeclareDatamodelEntryfields[jurisdiction]{relator}
3374 </dbx>
3375 <*bbx>
3376 \ProvidesFile{coppe.bbx}[2026/09/09 v4.1 CoppeTeX ABNT bibliography style]
3377 % The reference list is single-spaced (ABNT 2.4) via \singlespacing in
3378 % \AtBeginBibliography. coppe.cls already loads setspace, but the style must work
3379 % standalone too (e.g. \usepackage[bibstyle=coppe]{biblatex} in a plain article),
3380 % so load setspace here as well (idempotent).
3381 \RequirePackage{setspace}
3382 % ABNT reference list keeps the date at the end (imprenta), same for author-date
3383 % and numeric modes, so we derive from the standard style, not authoryear.
3384 \RequireBibliographyStyle{standard}
3385
3386 % Note: standard style never dashes repeated authors, which already matches the
3387 % post-2020 ABNT rule (repeat the entry in full), so no 'dashed' option is needed.
3388 \ExecuteBibliographyOptions{%
3389   maxbibnames=99,      % ABNT 4.3.1.3: list all authors by default
3390   giveninits=false,
3391   uniquename=false,
3392   uniquelist=false,
3393   labeldateparts=true, % needed for author-date extradate (1999a/1999b)
3394   sorting=nyt,
3395 }
3396
3397 % Language strings (Disponível em / Acesso em / In, etc.)
3398 \DeclareLanguageMapping{brazilian}{brazilian-coppe}
3399 \DeclareLanguageMapping{english}{english-coppe}
3400
3401 % --- Names: all reversed "FAMILY, Given", separated by ";" in the list ---
3402 \DeclareNameAlias{author}{family-given}
3403 \DeclareNameAlias{editor}{family-given}
3404 \DeclareNameAlias{translator}{family-given}
3405 \DeclareNameAlias{default}{family-given}
3406 \DeclareDelimFormat[bib,biblist]{multinamedelim}{\addsemicolon\space}
3407 \DeclareDelimFormat[bib,biblist]{finalnamedelim}{\addsemicolon\space}
3408
3409 % ABNT (NBR 6023 8.1.1.4): a collection entered under the name of its
3410 % organizer prints the kind of responsibility in parentheses after the
3411 % name, abbreviated, lowercase and in the singular even for several
3412 % organizers -- e.g. "SOUSA, A. M. de; OLIVEIRA, L. da C. (org.)". The
3413 % standard style prints ", Org." (comma, no parentheses) when the editor
3414 % takes the author slot, so override editor+others to wrap the role string
3415 % in "(...)". The host position ("In: NAME (org.)") already parentheseses
3416 % the role via coppe:bookeditor below. The role word itself is the
3417 % lowercase, singular "org." (and "ed.", "cur.", ...) set in the language
3418 % packs (brazilian-coppe.lbx et al.).
3419 \renewbibmacro*{editor+others}{%
3420   \ifboolexpr{ test \ifuseeditor and not test {\ifnameundef{editor}} }

```

```

3421     {\printnames{editor}%
3422      \setunit*{\addspace}%
3423      \printtext{\mkbibparens{\usebibmacro{editor+othersstrg}}}%
3424      \clearname{editor}}
3425     {}
3426 % Same parenthesised role for drivers that take the editor through the plain
3427 % editor macro instead of editor+others (e.g. @manual).
3428 \renewbibmacro*{editor}{%
3429   \ifboolexpr{ test \ifuseeditor and not test {\ifnameundef{editor}} }
3430     {\printnames{editor}%
3431      \setunit*{\addspace}%
3432      \printtext{\mkbibparens{\usebibmacro{editorstrg}}}%
3433      \clearname{editor}}
3434     {}
3435
3436 % Uppercase the surname (family + prefix) only inside the bibliography list;
3437 % citations keep normal caps (post-2020 rule).
3438 % ABNT date (NBR 6023 §4.3.5.5.2): "dia mês. ano" -- without the brazilian "de"
3439 % between the parts (e.g. "4 fev. 1992", "abr. 1996"). #1=year, #2=month, #3=day
3440 % field names; the month stays abbreviated via \mkbibmonth.
3441 \protected\def\coppe@abntdate#1#2#3{%
3442   \iffieldundef{#3}{\stripzeros{\thefield{#3}}\nobreakspace}%
3443   \iffieldundef{#2}{\mkbibmonth{\thefield{#2}}\iffieldundef{#1}{\space}}%
3444   \iffieldundef{#1}{\stripzeros{\thefield{#1}}}}
3445 \AtBeginBibliography{%
3446   \singlespacing % entries single-spaced internally (2.4)
3447   \setlength{\bibitemsep}{\baselineskip}% one blank line between entries (2.4)
3448   \renewcommand*{\mkbibnamefamily}[1]{\MakeUppercase{#1}}%
3449   \renewcommand*{\mkbibnameprefix}[1]{\MakeUppercase{#1}}%
3450   \let\mkbibdatelong\coppe@abntdate % ABNT date format (drop the "de")
3451   \let\mkbibdateshort\coppe@abntdate
3452 }
3453
3454 % --- Titles: bold for whole-document types; plain for parts; entry-by-title
3455 % (no author/editor) -> not bold, first word in CAPS (ABNT 4.2). ---
3456 % \coppeUCtitle uppercases the first SIGNIFICANT word of the title (ABNT 4.2).
3457 % If the title starts with an article (o/a/os/as/um/uma/uns/umas, the/an), the
3458 % article AND the next word are uppercased; otherwise just the first word. The
3459 % delimited macros split off words; \coppeUCkeep strips the trailing-space
3460 % sentinel so no spurious space is left before the following punctuation.
3461 \protected\def\coppeUCtitle#1{\coppeUCtitleA#1 \coppeUCstop}
3462 \protected\def\coppeUCtitleA#1 #2\coppeUCstop{%
3463   \coppeUCifart{#1}
3464     {\MakeUppercase{#1}\ifblank{#2}{\space\coppeUCtitleB#2\coppeUCstop}}%
3465     {\MakeUppercase{#1}\ifblank{#2}{\space\coppeUCkeep#2\coppeUCstop}}%
3466   \protected\def\coppeUCtitleB#1 #2\coppeUCstop{%
3467     \MakeUppercase{#1}\ifblank{#2}{\space\coppeUCkeep#2\coppeUCstop}}
3468   \protected\def\coppeUCkeep#1 \coppeUCstop{#1}
3469 % \coppeUCifart{word}{if-article}{if-not}: case-insensitive membership test.
3470 \protected\def\coppeUCifart#1{%
3471   \begingroup
3472   \lowercase{\def\coppeUC@w{#1}}%
3473   \edef\coppeUC@w{\coppeUC@w}%
3474   \expandafter\in@\expandafter{\coppeUC@w}{a,o,as,os,um,uma,uns,umas,the,an,}%

```

```

3475 \ifin@aftergroup\@firstoftwo\else\aftergroup\@secondoftwo\fi
3476 \endgroup}
3477 \DeclareFieldFormat{title}{%
3478 \ifboolexpr{ test {\ifnameundef{labelname}} }
3479 {#1}%
3480 {\mkbibbold{#1}}}
3481 % First word of the title in CAPS for entry-by-title (ABNT 4.2). The titlecase
3482 % format receives the RAW title text (the title format above only sees the
3483 % field-printing commands), so \coppeUCtitle can split off and uppercase the
3484 % first word here.
3485 \DeclareFieldFormat{titlecase}{%
3486 \ifboolexpr{ test {\ifnameundef{labelname}} }
3487 {\coppeUCtitle{#1}}%
3488 {#1}}
3489 \DeclareFieldFormat[article,incollection,inbook,inproceedings,suppperiodical,review]{title}{%
3490 \DeclareFieldFormat[thesis]{title}{\mkbibbold{#1}}
3491 % biblatex's standard style quotes patent (and unpublished) titles; ABNT bolds
3492 % the title of a whole document, so override those back to bold.
3493 \DeclareFieldFormat[patent]{title}{\mkbibbold{#1}}
3494 % Legal documents (NBR 6023 4.2.6): the law's identification/ementa is NOT
3495 % highlighted; the entry leads with the jurisdiction (entity author, CAPS).
3496 \DeclareFieldFormat[legislation,jurisdiction,legal]{title}{#1}
3497 \DeclareFieldFormat[journaltitle]{\mkbibbold{#1}}
3498 \DeclareFieldFormat{booktitle}{\mkbibbold{#1}}
3499 \DeclareFieldFormat{maintitle}{\mkbibbold{#1}}
3500
3501 % --- Pages: ABNT uses "p." (also for ranges); theses use "f." for pagetotal ---
3502 \DeclareFieldFormat{pagetotal}{#1\addnbspace p\adddot}
3503 \DeclareFieldFormat[thesis]{pagetotal}{#1\addnbspace f\adddot}
3504 \DeclareFieldFormat{pages}{p\adddot\addnbspace#1}
3505
3506 % --- Periodical volume / number prefixes ---
3507 \DeclareFieldFormat[article]{volume}{v\adddot\addnbspace#1}
3508 \DeclareFieldFormat[article]{number}{n\adddot\addnbspace#1}
3509
3510 % --- "In:" before host title (incollection/inproceedings/inbook) ---
3511 \renewbibmacro*{in:}{\printtext{\bibstring{in}\addcolon\space}}
3512
3513 % --- Online: "Disponível em: <url>." + "Acesso em: <date>." via lbx strings ---
3514 \DeclareFieldFormat{url}{\bibstring{availablefrom}\addcolon\space\url{#1}}
3515
3516 % --- ABNT article driver: no "In:"; Journal, Local, v., n., p., date, DOI/URL.
3517 % Newspapers (entrysubtype=newspaper, set by the @artigodejornal source map)
3518 % differ: the newspaper name (journaltitle) and city, then either
3519 % "<date>. <caderno>, p. <pages>" (when a caderno/section is given), or
3520 % "p. <pages>, <date>" (page before date, when none),
3521 % per NBR 6023 §4.2.3.6. ---
3522 \DeclareBibliographyDriver{article}{%
3523 \usebibmacro{bibindex}%
3524 \usebibmacro{begentry}%
3525 \usebibmacro{author/translator+others}%
3526 \setunit{\labelnamepunct}\newblock
3527 \usebibmacro{title}%
3528 \newunit\newblock

```

```

3529 \printfield{journaltitle}%
3530 \setunit{\addcomma\space}\printlist{location}%
3531 \iffieldequalstr{entrysubtype}{newspaper}
3532   {\iffieldundef{caderno}
3533     {\setunit{\addcomma\space}\printfield{pages}%
3534       \setunit{\addcomma\space}\printdate}%
3535     {\setunit{\addcomma\space}\printdate%
3536       \setunit{\adddot\space}\printfield{caderno}%
3537       \setunit{\addcomma\space}\printfield{pages}}}%
3538   {\setunit{\addcomma\space}\printfield{volume}%
3539     \setunit{\addcomma\space}\printfield{number}%
3540     \setunit{\addcomma\space}\printfield{pages}%
3541     \setunit{\addcomma\space}\printdate}%
3542 \newunit\newblock
3543 \printfield{doi}%
3544 \newunit\newblock
3545 \usebibmacro{url+urldate}%
3546 \usebibmacro{finentry}}
3547
3548 % --- incollection: ABNT puts the host organizer before the booktitle, as
3549 % "AUTOR. Parte. In: ORGANIZADOR (Org.). Booktitle. Local: Editora, ano,
3550 % p. X-Y." The organizer role string is "Org./Orgs." (see the .lbx). ---
3551 \newbibmacro*{coppe:bookeditor}{%
3552   \ifnameundef{editor}
3553     {}
3554     {\printnames{editor}%
3555       \setunit*{\addspace}\printtext{\mkbibparens{\usebibmacro{editorstrg}}}%
3556       \clearname{editor}%
3557       \setunit{\adddot\space}}
3558 \newbibmacro*{coppe:partdriver}{%
3559   \usebibmacro{bibindex}\usebibmacro{begentry}%
3560   \usebibmacro{author/translator+others}%
3561   \setunit{\labelnamepunct}\newblock
3562   \usebibmacro{title}%
3563   \newunit\newblock
3564   \usebibmacro{in:}%
3565   \usebibmacro{coppe:bookeditor}%
3566   \printfield{booktitle}%
3567   \newunit\newblock
3568   \printfield{edition}%
3569   \newunit\newblock
3570   \printlist{location}%
3571   \setunit*{\addcolon\space}\printlist{publisher}%
3572   \setunit*{\addcomma\space}\printdate%
3573   \newunit\newblock
3574   \printfield{pages}%
3575   \setunit{\addcomma\space}\printfield{chapter}%
3576   \newunit\newblock
3577   \usebibmacro{url+urldate}%
3578   \usebibmacro{finentry}}
3579 \DeclareBibliographyDriver{incollection}{\usebibmacro{coppe:partdriver}}
3580 \DeclareBibliographyDriver{inbook}{\usebibmacro{coppe:partdriver}}
3581
3582 % --- type field: resolve a known bibliography string (babel-aware), else literal ---

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```

3583 \DeclareFieldFormat{type}{\ifbibstring{#1}{\bibstring{#1}}{#1}}
3584
3585 % --- Thesis: "<type> (em <course>)" using the custom course field (ABNT). ---
3586 \DeclareFieldFormat{course}{\mkbibparens{\coppestring{in:}\addspace#1}}
3587 \DeclareBibliographyDriver{thesis}{%
3588   \usebibmacro{bibindex}\usebibmacro{begentry}%
3589   \usebibmacro{author}%
3590   \setunit{\labelnamepunct}\newblock
3591   \usebibmacro{title}%
3592   \newunit\newblock
3593   \printfield{type}\setunit*\addspace\printfield{course}%
3594   \newunit\newblock
3595   \printlist{institution}%
3596   \setunit*\addcomma\space\printlist{location}%
3597   \setunit*\addcomma\space\printdate%
3598   \newunit\newblock
3599   \printfield{pagetotal}%
3600   \newunit\newblock
3601   \printfield{note}%
3602   \newunit\newblock
3603   \usebibmacro{url+urldate}%
3604   \usebibmacro{finentry}}
3605 % --- Games / audiovisual: render the custom fields (artist, platform,
3606 %       director, producer, version) the generic misc driver ignores. ---
3607 \newbibmacro*{coppe:media}{%
3608   \iflistundef{artist}{\setunit{\addperiod\space}%
3609     \printtext{\coppestring{art}\addcolon\space}\printlist{artist}}%
3610   \iffieldundef{platform}{\setunit{\addperiod\space}%
3611     \printtext{\coppestring{platform}\addcolon\space}\printfield{platform}}%
3612   \iflistundef{director}{\setunit{\addperiod\space}%
3613     \printtext{\coppestring{directedby}\addcolon\space}\printlist{director}}%
3614   \iflistundef{producer}{\setunit{\addperiod\space}%
3615     \printtext{\coppestring{producedby}\addcolon\space}\printlist{producer}}%
3616 \newbibmacro*{coppe:mediadriver}{%
3617   \usebibmacro{bibindex}\usebibmacro{begentry}%
3618   \usebibmacro{author/translator+others}%
3619   \setunit{\labelnamepunct}\newblock
3620   \usebibmacro{title}%
3621   \newunit\newblock
3622   \usebibmacro{coppe:media}%
3623   \newunit\newblock
3624   \printlist{publisher}%
3625   \setunit*\addcomma\space\printlist{location}%
3626   \setunit*\addcomma\space\printdate%
3627   \newunit\newblock
3628   \printfield{version}%
3629   \newunit\newblock
3630   \printfield{note}%
3631   \newunit\newblock
3632   \usebibmacro{url+urldate}%
3633   \usebibmacro{finentry}}
3634 \DeclareBibliographyDriver{game}{\usebibmacro{coppe:mediadriver}}
3635 \DeclareBibliographyDriver{videogame}{\usebibmacro{coppe:mediadriver}}
3636 \DeclareBibliographyDriver{boardgame}{\usebibmacro{coppe:mediadriver}}

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3637 \DeclareBibliographyDriver{movie}{\usebibmacro{coppe:mediadriver}}
3638 \DeclareBibliographyDriver{tvshow}{\usebibmacro{coppe:mediadriver}}
3639 % Cartographic document (map, NBR 6023 4.2.12): like a book (author/title/
3640 % imprenta) plus the mandatory scale ("Escala 1:N").
3641 \DeclareFieldFormat{scale}{\bibstring{coppescale}\addnbspace#1}
3642 \DeclareBibliographyDriver{map}{%
3643   \usebibmacro{bibindex}\usebibmacro{begentry}%
3644   \usebibmacro{author/translator+others}%
3645   \setunit{\labelnamepunct}\newblock
3646   \usebibmacro{title}%
3647   \newunit\newblock
3648   \printlist{location}%
3649   \setunit*{\addcolon\space}\printlist{publisher}%
3650   \setunit*{\addcomma\space}\printdate%
3651   \newunit\newblock
3652   \printfield{scale}%
3653   \newunit\newblock
3654   \usebibmacro{url+urldate}%
3655   \usebibmacro{finentry}}
3656 % Patent (NBR 6023 4.2.5): inventor. \textbf{title}. Depositante: holder. type
3657 % number. date. The holder (depositante) is printed in normal case, not the
3658 % ALL-CAPS family treatment the bibliography gives the leading author.
3659 \newbibmacro*{coppe:depositor}{%
3660   \ifnameundef{holder}
3661   {}
3662   {\printtext{\bibstring{depositor}\addcolon\space}%
3663     \begingroup
3664       \def\mkbibnamefamily##1{##1}%
3665       \def\mkbibnameprefix##1{##1}%
3666       \printnames{holder}%
3667     \endgroup}}
3668 \DeclareBibliographyDriver{patent}{%
3669   \usebibmacro{bibindex}%
3670   \usebibmacro{begentry}%
3671   \usebibmacro{author/translator+others}%
3672   \setunit{\labelnamepunct}\newblock
3673   \usebibmacro{title}%
3674   \newunit\newblock
3675   \usebibmacro{coppe:depositor}%
3676   \newunit\newblock
3677   \printfield{type}%
3678   \setunit*{\addspace}\printfield{number}%
3679   \newunit\newblock
3680   \printdate%
3681   \newunit\newblock
3682   \usebibmacro{url+urldate}%
3683   \usebibmacro{finentry}}
3684 % Exotic ABNT types that biblatex's standard style leaves without a driver:
3685 % give each a baseline (book-like for published docs: norma/partitura; misc for
3686 % the rest) so they print. Per-type ABNT refinements can follow later.
3687 \DeclareBibliographyDriver{standard}{\usedriver{}{book}}
3688 \DeclareBibliographyDriver{music}{\usedriver{}{book}}
3689 % Legal documents (NBR 6023 4.2.6-4.2.8): jurisdiction/entity (CAPS author),
3690 % the law id + ementa (plain, not bold), then location and date.

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```

3691 \newbibmacro*{coppe:legaldriver}{%
3692   \usebibmacro{bibindex}\usebibmacro{begentry}%
3693   \usebibmacro{author/translator+others}%
3694   \setunit{\labelnamepunct}\newblock
3695   \usebibmacro{title}%
3696   \newunit\newblock
3697   % Jurisprudência: "Relator: <nome>." after the decision title (NBR 6023).
3698   \iffieldundef{relator}{\}\{\printtext{Relator\addcolon\space}\printfield{relator}\newunit\ne
3699   \printlist{location}%
3700   \setunit{\addcomma\space}\printdate%
3701   \newunit\newblock
3702   \usebibmacro{url+urldate}%
3703   \usebibmacro{finentry}}
3704 \DeclareBibliographyDriver{legislation}{\usebibmacro{coppe:legaldriver}}
3705 \DeclareBibliographyDriver{jurisdiction}{\usebibmacro{coppe:legaldriver}}
3706 \DeclareBibliographyDriver{legal}{\usebibmacro{coppe:legaldriver}}
3707 \DeclareBibliographyDriver{audio}{\usedriver{}{misc}}
3708 \DeclareBibliographyDriver{video}{\usedriver{}{misc}}
3709 \DeclareBibliographyDriver{software}{\usedriver{}{misc}}
3710 \DeclareBibliographyDriver{image}{\usedriver{}{misc}}
3711 \DeclareBibliographyDriver{artwork}{\usedriver{}{misc}}
3712 \DeclareBibliographyDriver{performance}{\usedriver{}{misc}}
3713 \DeclareBibliographyDriver{dataset}{\usedriver{}{misc}}
3714
3715 % --- biber source map: Portuguese synonyms (entry types + fields) + thesis family ---
3716 \DeclareSourcemap{%
3717   \maps[datatype=bibtex]{%
3718     % entry-type PT synonyms -> canonical English
3719     \map{step[typesource=livro,typetarget=book]}
3720     \map{step[typesource=artigo,typetarget=article]}
3721     \map{step[typesource=artigodejornal,typetarget=article,final]\step[fieldset=entrysubtyp
3722     \map{step[typesource=partedelivro,typetarget=incollection]}
3723     \map{step[typesource=emlivro,typetarget=inbook]}
3724     \map{step[typesource=verbete,typetarget=inreference]}
3725     \map{step[typesource=anais,typetarget=proceedings]}
3726     \map{step[typesource=trabalhodeevento,typetarget=inproceedings]}
3727     \map{step[typesource=periodico,typetarget=periodical]}
3728     \map{step[typesource=relatorio,typetarget=report]}
3729     \map{step[typesource=relatoriotecnico,typetarget=report]}
3730     \map{step[typesource>manual,typetarget>manual]}
3731     \map{step[typesource=correspondencia,typetarget=letter]}
3732     \map{step[typesource=patente,typetarget=patent]}
3733     \map{step[typesource=diverso,typetarget=misc]}
3734     \map{step[typesource=jogo,typetarget=game]}
3735     \map{step[typesource=jogoeletronico,typetarget=videogame]}
3736     \map{step[typesource=jogodetabuleiro,typetarget=boardgame]}
3737     \map{step[typesource=filme,typetarget=movie]}
3738     \map{step[typesource=serietv,typetarget=tvshow]}
3739     \map{step[typesource=programadetelevisao,typetarget=tvshow]}
3740     % Exotic ABNT types (NBR 6023 4.2.5-4.2.14); targets are biblatex-native
3741     % except map (declared in coppe.dbx).
3742     \map{step[typesource=legislacao,typetarget=legislation]}
3743     \map{step[typesource=atonomativo,typetarget=legislation]}
3744     \map{step[typesource=jurisprudencia,typetarget=jurisdiction]}

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3745 \map{\step[typesource=documentojuridico,typetarget=legal]}
3746 \map{\step[typesource=documentocivil,typetarget=legal]}
3747 \map{\step[typesource=norma,typetarget=standard]}
3748 \map{\step[typesource=partitura,typetarget=music]}
3749 \map{\step[typesource=iconografico,typetarget=image]}
3750 \map{\step[typesource=obradearte,typetarget=artwork]}
3751 \map{\step[typesource=tridimensional,typetarget=artwork]}
3752 \map{\step[typesource=cartografico,typetarget=map]}
3753 \map{\step[typesource=mapa,typetarget=map]}
3754 \map{\step[typesource=audiovisual,typetarget=video]}
3755 \map{\step[typesource=conjuntodedados,typetarget=dataset]}
3756 \map{\step[typesource=fasciculo,typetarget=supperiodical]}
3757 \map{\step[typesource=trabalhoapresentado,typetarget=unpublished]}
3758 % thesis family: retype to thesis and preset the (babel-aware) type word
3759 \map{\step[typesource=tese,typetarget=thesis]}
3760 \map{\step[typesource=dissertacao,typetarget=thesis]}
3761 \map{\step[typesource=masterdissertation,typetarget=thesis,final]\step[fieldset=type,fiel
3762 \map{\step[typesource=mscdissertation,typetarget=thesis,final]\step[fieldset=type,fiel
3763 \map{\step[typesource=dissertacaomestrado,typetarget=thesis,final]\step[fieldset=type,fi
3764 \map{\step[typesource=dscthesis,typetarget=thesis,final]\step[fieldset=type,fieldvalue=d
3765 \map{\step[typesource=tesedoutorado,typetarget=thesis,final]\step[fieldset=type,fieldval
3766 \map{\step[typesource=phdthesis,typetarget=thesis,final]\step[fieldset=type,fieldvalue=p
3767 \map{\step[typesource=tesephd,typetarget=thesis,final]\step[fieldset=type,fieldvalue=phd
3768 \map{\step[typesource=tcc,typetarget=thesis,final]\step[fieldset=type,fieldvalue=tcc]}
3769 % field PT synonyms (unaccented) -> canonical English
3770 \map{\step[fieldsource=periodico,fieldtarget=journaltitle]}
3771 \map{\step[fieldsource=revista,fieldtarget=journaltitle]}
3772 \map{\step[fieldsource=jornal,fieldtarget=journaltitle]}
3773 \map{\step[fieldsource=titulo,fieldtarget=title]}
3774 \map{\step[fieldsource=subtitulo,fieldtarget=subtitle]}
3775 \map{\step[fieldsource=autor,fieldtarget=author]}
3776 \map{\step[fieldsource=editora,fieldtarget=publisher]}
3777 \map{\step[fieldsource=local,fieldtarget=location]}
3778 \map{\step[fieldsource=ano,fieldtarget=year]}
3779 \map{\step[fieldsource=data,fieldtarget=date]}
3780 \map{\step[fieldsource=edicao,fieldtarget=edition]}
3781 \map{\step[fieldsource=paginas,fieldtarget=pages]}
3782 \map{\step[fieldsource=numero,fieldtarget=number]}
3783 \map{\step[fieldsource=tradutor,fieldtarget=translator]}
3784 \map{\step[fieldsource=organizador,fieldtarget=editor]}
3785 \map{\step[fieldsource=instituicao,fieldtarget=institution]}
3786 \map{\step[fieldsource=tipo,fieldtarget=type]}
3787 \map{\step[fieldsource=nota,fieldtarget=note]}
3788 \map{\step[fieldsource=dataacesso,fieldtarget=urldate]}
3789 \map{\step[fieldsource=curso,fieldtarget=course]}
3790 \map{\step[fieldsource=grau,fieldtarget=coppedegree]}
3791 \map{\step[fieldsource=artista,fieldtarget=artist]}
3792 \map{\step[fieldsource=plataforma,fieldtarget=platform]}
3793 \map{\step[fieldsource=diretor,fieldtarget=director]}
3794 \map{\step[fieldsource=produtor,fieldtarget=producer]}
3795 \map{\step[fieldsource=versao,fieldtarget=version]}
3796 \map{\step[fieldsource=escala,fieldtarget=scale]}
3797 }%
3798 }

```

```

3799
3800 % --- ABNT list layout: flush left, single spacing, blank line between entries ---
3801 \setlength{\bibhang}{0pt}
3802 \setlength{\bibitemsep}{\baselineskip}
3803
3804 </bbx>
3805 <*cbx>
3806 \ProvidesFile{coppe.cbx}[2026/09/09 v4.1 CoppeTeX ABNT author-date citations]
3807 % Author-date is the CoppeTeX 4.0 default. authoryear-comp gives compressed
3808 % author-year cites with normal-caps names (post-2020 rule) out of the box.
3809 \RequireCitationStyle{authoryear-comp}
3810
3811 % --- NBR 10520:2023 author separator -----
3812 % Parenthetical citations separate authors with ";" -> (Silva; Souza, 2020).
3813 % Narrative citations (\textcite, natbib \citet) keep comma + "e/and" ->
3814 % Silva, Souza e Oliveira (2020). The reference list keeps its own ";" via
3815 % coppe.bbx ([bib,biblist]); these contextless formats cover the cite side.
3816 \DeclareDelimFormat{multinamedelim}{\addsemicolon\space}
3817 \DeclareDelimFormat{finalnamedelim}{\addsemicolon\space}
3818 \DeclareDelimFormat[textcite]{multinamedelim}{\addcomma\space}
3819 \DeclareDelimFormat[textcite]{finalnamedelim}{\addspace\bibstring{and}\space}
3820
3821 % --- apud: citation of a citation (ABNT NBR 6023 §4.1.2.2 / NBR 10520) -----
3822 % ABNT lists only the work actually CONSULTED. So only the consulted work (#4, a
3823 % .bib key) is cited and enters the reference list; the ORIGINAL work (#2 author,
3824 % #3 year) was not consulted and is given as plain text -- it is never a cite key,
3825 % so it is never added to the bibliography.
3826 % \citetapud[<page>]{<orig author>}{<orig year>}{<consulted key>}
3827 % -> Orig Author (year) apud Consulted (yr[, page]) [narrative]
3828 % \citepapud[<page>]{<orig author>}{<orig year>}{<consulted key>}
3829 % -> (Orig Author, year apud CONSULTED, yr[, page]) [parenthetical]
3830 % #1 = optional postnote (page/loc) on the consulted work; "apud" is latin -> italic.
3831 \DeclareRobustCommand{\citetapud}[4][]{\%
3832 #2\space(#3)\space\mkbibemph{apud}\space\textcite[#1]{#4}}
3833 \DeclareRobustCommand{\citepapud}[4][]{\%
3834 \mkbibparens{#2\addcomma\space#3\space\mkbibemph{apud}\space
3835 \citeauthor{#4}\addcomma\space\citeyear{#4}}\%
3836 \ifblank{#1}{\}\{\addcomma\space\mkpageprefix[pagination]{#1}}\}}
3837 </cbx>
3838 <*lbxbr>
3839 \ProvidesFile{brazilian-coppe.lbx}[2026/09/09 v4.1 CoppeTeX brazilian strings]
3840 \InheritBibliographyExtras{brazilian}
3841 \NewBibliographyString{availablefrom,mscdiss,dscthes, tcc,depositor,coppescale}
3842 \DeclareBibliographyStrings{%
3843 inherit = {brazilian},
3844 availablefrom = {{Dispon'\i vel em}{Dispon'\i vel em}},
3845 urlseen = {{Acesso em}{Acesso em}},
3846 in = {{In}{In}},
3847 editor = {{organizador}{org\adddot}},
3848 editors = {{organizadores}{org\adddot}},
3849 depositor = {{Depositante}{Depositante}},
3850 coppescale = {{Escala}{Escala}},
3851 mscdiss = {{Disserta\c{c}\~{a}o de Mestrado}{Disserta\c{c}\~{a}o de Mestrado}},
3852 dscthes = {{Tese de Doutorado}{Tese de Doutorado}},

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3853 phdthesis      = {{Tese de Doutorado}{Tese de Doutorado}},
3854 tcc              = {{Trabalho de Conclus~{a}o de Curso}{Trabalho de Conclus~{a}o de Curso}}
3855 }
3856 </lbxbr>
3857 <*lbxen>
3858 \ProvidesFile{english-coppe.lbx}[2026/09/09 v4.1 CoppeTeX english strings]
3859 \InheritBibliographyExtras{english}
3860 \NewBibliographyString{availablefrom,mscdiss,dscthesi,tcc,depositor,coppescale}
3861 \DeclareBibliographyStrings{%
3862   inherit        = {english},
3863   availablefrom = {{Available from}{Available from}},
3864   urlseen        = {{Access on}{Access on}},
3865   in              = {{In}{In}},
3866   editor          = {{editor}{ed\addot}},
3867   editors         = {{editors}{ed\addot}},
3868   depositor      = {{Depositor}{Depositor}},
3869   coppescale     = {{Scale}{Scale}},
3870   mscdiss        = {{M\addot Sc\addot\ Dissertation}{M\addot Sc\addot\ Diss\addot}},
3871   dscthesi       = {{D\addot Sc\addot\ Thesis}{D\addot Sc\addot\ Thesis}},
3872   phdthesis      = {{Ph\addot D\addot\ Thesis}{Ph\addot D\addot\ Thesis}},
3873   tcc            = {{Undergraduate Thesis}{Undergraduate Thesis}},
3874 }
3875 </lbxen>
3876 <*lbxes>
3877 \ProvidesFile{spanish-coppe.lbx}[2026/09/09 v4.1 CoppeTeX spanish strings]
3878 \InheritBibliographyExtras{spanish}
3879 \NewBibliographyString{availablefrom,mscdiss,dscthesi,tcc,depositor,coppescale}
3880 \DeclareBibliographyStrings{%
3881   inherit        = {spanish},
3882   availablefrom = {{Disponible en}{Disponible en}},
3883   urlseen        = {{Consultado el}{Consultado el}},
3884   in              = {{En}{En}},
3885   editor          = {{editor}{ed\addot}},
3886   editors         = {{editores}{ed\addot}},
3887   depositor      = {{Solicitante}{Solicitante}},
3888   coppescale     = {{Escala}{Escala}},
3889   mscdiss        = {{Tesis de Maestr'\{i\}a}{Tesis de Maestr'\{i\}a}},
3890   dscthesi       = {{Tesis Doctoral}{Tesis Doctoral}},
3891   phdthesis      = {{Tesis Doctoral}{Tesis Doctoral}},
3892   tcc            = {{Trabajo de Fin de Carrera}{Trabajo de Fin de Carrera}},
3893 }
3894 </lbxes>
3895 <*lbxfr>
3896 \ProvidesFile{french-coppe.lbx}[2026/09/09 v4.1 CoppeTeX french strings]
3897 \InheritBibliographyExtras{french}
3898 \NewBibliographyString{availablefrom,mscdiss,dscthesi,tcc,depositor,coppescale}
3899 \DeclareBibliographyStrings{%
3900   inherit        = {french},
3901   availablefrom = {{Disponible \{a\}}{Disponible \{a\}}},
3902   urlseen        = {{Consult\{e\} le}{Consult\{e\} le}},
3903   in              = {{Dans}{Dans}},
3904   editor          = {{\{e\}diteur}{\{e\}d\addot}},
3905   editors         = {{\{e\}diteurs}{\{e\}d\addot}},
3906   depositor      = {{D\{e\}posant}{D\{e\}posant}},

```

```

3907 coppescale = {{\Echelle}{\Echelle}},
3908 msdiss = {{M\emoire de Master}{M\emoire de Master}},
3909 dscthesi = {{Th\ese de Doctorat}{Th\ese de Doctorat}},
3910 phdthesis = {{Th\ese de Doctorat}{Th\ese de Doctorat}},
3911 tcc = {{M\emoire de fin d'\etudes}{M\emoire de fin d'\etudes}},
3912 }
3913 </lboxfr>
3914 <*lboxit>
3915 \ProvidesFile{italian-coppe.lbx}[2026/09/09 v4.1 CoppeTeX italian strings]
3916 \InheritBibliographyExtras{italian}
3917 \NewBibliographyString{availablefrom,msdiss,dscthesi,tcc,depositor,coppescale}
3918 \DeclareBibliographyStrings{%
3919 inherit = {italian},
3920 availablefrom = {{Disponibile da}{Disponibile da}},
3921 urlseen = {{Consultato il}{Consultato il}},
3922 in = {{In}{In}},
3923 editor = {{curatore}{cur\adddot}},
3924 editors = {{curatori}{cur\adddot}},
3925 depositor = {{Depositante}{Depositante}},
3926 coppescale = {{Scala}{Scala}},
3927 msdiss = {{Tesi di Laurea Magistrale}{Tesi di Laurea Magistrale}},
3928 dscthesi = {{Tesi di Dottorato}{Tesi di Dottorato}},
3929 phdthesis = {{Tesi di Dottorato}{Tesi di Dottorato}},
3930 tcc = {{Tesi di Laurea Triennale}{Tesi di Laurea Triennale}},
3931 }
3932 </lboxit>
3933 <*langes>
3934 %% coppe-lang-spanish.def -- CoppeTeX 4.x Spanish class-string pack.
3935 %% Loaded by coppe.cls when \documentclass[spanish]{coppe} (or any other
3936 %% spanish slot) is active. Populates the dispatcher table for the
3937 %% spanish language and registers the biblatex language mapping.
3938 %% Institutional names (universityname/cityname/statename/countryname)
3939 %% are deliberately kept Portuguese: the COPPE/UFRJ cover and folha de
3940 %% rosto are a Brazilian institutional template that does not translate.
3941 \ProvidesFile{coppe-lang-spanish.def}[2026/09/09 v4.1 CoppeTeX spanish class strings]
3942 \copperdefstring{spanish}{appendix} {Ap\endice}
3943 \copperdefstring{spanish}{annex} {Anexo}
3944 \copperdefstring{spanish}{frame} {Cuadro}
3945 \copperdefstring{spanish}{listframe} {Lista de Cuadros}
3946 \copperdefstring{spanish}{source} {Fuente}
3947 \copperdefstring{spanish}{program} {Programa}
3948 \copperdefstring{spanish}{listprogram} {Lista de Programas}
3949 \copperdefstring{spanish}{references} {Referencias}
3950 \copperdefstring{spanish}{in:} {en}
3951 \copperdefstring{spanish}{art} {Arte}
3952 \copperdefstring{spanish}{platform} {Plataforma}
3953 \copperdefstring{spanish}{directedby} {Direcci\on}
3954 \copperdefstring{spanish}{producedby} {Producci\on}
3955 \copperdefstring{spanish}{listabbreviation} {Lista de Abreviaturas y Siglas}
3956 \copperdefstring{spanish}{listsymbol} {Lista de S\imbolos}
3957 \copperdefstring{spanish}{glossary} {Glosario}
3958 \copperdefstring{spanish}{advisor} {Director}
3959 \copperdefstring{spanish}{advisors} {Directores}
3960 \copperdefstring{spanish}{dept} {Programa}

```

```

3961 \copperdefstring{spanish}{approvedmale}    {Aprobado por}
3962 \copperdefstring{spanish}{approvedfemale}    {Aprobada por}
3963 \copperdefstring{spanish}{approvedonmale}    {Aprobado el}
3964 \copperdefstring{spanish}{approvedonfemale}{Aprobada el}
3965 \copperdefstring{spanish}{keywordsname} {Palabras clave}
3966 \copperdefstring{spanish}{volumename} {Volumen}
3967 \copperdefstring{spanish}{coadvisor} {Codirector}
3968 \copperdefstring{spanish}{coadvisors} {Codirectores}
3969 \copperdefstring{spanish}{ofname} {de}
3970 \copperdefstring{spanish}{universityname}{Universidade Federal do Rio de Janeiro}
3971 \copperdefstring{spanish}{cityname} {Rio de Janeiro}
3972 \copperdefstring{spanish}{statename} {RJ}
3973 \copperdefstring{spanish}{countryname} {Brasil}
3974 \copperdefstring{spanish}{monthname1} {enero}
3975 \copperdefstring{spanish}{monthname2} {febrero}
3976 \copperdefstring{spanish}{monthname3} {marzo}
3977 \copperdefstring{spanish}{monthname4} {abril}
3978 \copperdefstring{spanish}{monthname5} {mayo}
3979 \copperdefstring{spanish}{monthname6} {junio}
3980 \copperdefstring{spanish}{monthname7} {julio}
3981 \copperdefstring{spanish}{monthname8} {agosto}
3982 \copperdefstring{spanish}{monthname9} {septiembre}
3983 \copperdefstring{spanish}{monthname10}{octubre}
3984 \copperdefstring{spanish}{monthname11}{noviembre}
3985 \copperdefstring{spanish}{monthname12}{diciembre}
3986 \copperdefstring{spanish}{algorithm2eopt}{spanish}
3987 \copperdefstring{spanish}{babelname} {spanish}
3988 %% biblatex is loaded by coppe.cls AFTER this .def file is read; defer the
3989 %% language-mapping registration until \begin{document} so biblatex's
3990 %% \DeclareLanguageMapping is available.
3991 \AtBeginDocument{\DeclareLanguageMapping{spanish}{spanish-coppe}}
3992 \end{langes}
3993 \end{*langfr}
3994 %% coppe-lang-french.def -- CoppeTeX 4.x French class-string pack.
3995 %% See coppe-lang-spanish.def for the structure. Institutional names
3996 %% remain Portuguese; only document-content strings translate.
3997 \ProvidesFile{coppe-lang-french.def}[2026/09/09 v4.1 CoppeTeX french class strings]
3998 \copperdefstring{french}{appendix} {Annexe}
3999 \copperdefstring{french}{annex} {Annexe}
4000 \copperdefstring{french}{frame} {Encadr\'e}
4001 \copperdefstring{french}{listframe} {Liste des encadr\'es}
4002 \copperdefstring{french}{source} {Source}
4003 \copperdefstring{french}{program} {Programme}
4004 \copperdefstring{french}{listprogram} {Liste des programmes}
4005 \copperdefstring{french}{references} {R\'ef\'erences}
4006 \copperdefstring{french}{in:} {dans}
4007 \copperdefstring{french}{art} {Illustration}
4008 \copperdefstring{french}{platform} {Plateforme}
4009 \copperdefstring{french}{directedby} {R\'ealis\'e par}
4010 \copperdefstring{french}{producedby} {Produit par}
4011 \copperdefstring{french}{listabbreviation}{Liste des abr\'eviations et acronymes}
4012 \copperdefstring{french}{listsymbol} {Liste des symboles}
4013 \copperdefstring{french}{glossary} {Glossaire}
4014 \copperdefstring{french}{advisor} {Directeur}

```

```

4015 \copperdefstring{french}{advisors}      {Directeurs}
4016 \copperdefstring{french}{dept}          {Programme}
4017 \copperdefstring{french}{approvedmale}  {Approuv\'e par}
4018 \copperdefstring{french}{approvedfemale}{Approuv\'ee par}
4019 \copperdefstring{french}{approvedonmale}{Approuv\'e le}
4020 \copperdefstring{french}{approvedonfemale}{Approuv\'ee le}
4021 \copperdefstring{french}{keywordsname} {Mots-cl\'es}
4022 \copperdefstring{french}{volumename} {Volume}
4023 \copperdefstring{french}{coadvisor} {Co-directeur}
4024 \copperdefstring{french}{coadvisors} {Co-directeurs}
4025 \copperdefstring{french}{ofname} {de}
4026 \copperdefstring{french}{universityname}{Universidade Federal do Rio de Janeiro}
4027 \copperdefstring{french}{cityname} {Rio de Janeiro}
4028 \copperdefstring{french}{statename} {RJ}
4029 \copperdefstring{french}{countryname} {Brasil}
4030 \copperdefstring{french}{monthname1} {janvier}
4031 \copperdefstring{french}{monthname2} {f\'evrier}
4032 \copperdefstring{french}{monthname3} {mars}
4033 \copperdefstring{french}{monthname4} {avril}
4034 \copperdefstring{french}{monthname5} {mai}
4035 \copperdefstring{french}{monthname6} {juin}
4036 \copperdefstring{french}{monthname7} {juillet}
4037 \copperdefstring{french}{monthname8} {ao\^ut}
4038 \copperdefstring{french}{monthname9} {septembre}
4039 \copperdefstring{french}{monthname10}{octobre}
4040 \copperdefstring{french}{monthname11}{novembre}
4041 \copperdefstring{french}{monthname12}{d\'ecembre}
4042 \copperdefstring{french}{algorithm2eopt}{french}
4043 \copperdefstring{french}{babelname} {french}
4044 \AtBeginDocument{\DeclareLanguageMapping{french}{french-coppe}}
4045 \</langfr>
4046 \<*langit>
4047 %% coppe-lang-italian.def -- CoppeTeX 4.x Italian class-string pack.
4048 %% See coppe-lang-spanish.def for the structure. The algorithm2e
4049 %% language option for Italian is spelled "italiano" -- that's the
4050 %% spelling algorithm2e ships, even though babel calls the language
4051 %% "italian".
4052 \ProvidesFile{coppe-lang-italian.def}[2026/09/09 v4.1 CoppeTeX italian class strings]
4053 \copperdefstring{italian}{appendix}      {Appendice}
4054 \copperdefstring{italian}{annex}        {Allegato}
4055 \copperdefstring{italian}{frame}        {Riquadro}
4056 \copperdefstring{italian}{listframe}    {Elenco dei riquadri}
4057 \copperdefstring{italian}{source}       {Fonte}
4058 \copperdefstring{italian}{program}      {Programma}
4059 \copperdefstring{italian}{listprogram}  {Elenco dei programmi}
4060 \copperdefstring{italian}{references}    {Riferimenti}
4061 \copperdefstring{italian}{in:}          {in}
4062 \copperdefstring{italian}{art}           {Illustrazione}
4063 \copperdefstring{italian}{platform}     {Piattaforma}
4064 \copperdefstring{italian}{directedby}   {Regia di}
4065 \copperdefstring{italian}{producedby}   {Prodotto da}
4066 \copperdefstring{italian}{listabbreviation}{Elenco di abbreviazioni e sigle}
4067 \copperdefstring{italian}{listsymbol}   {Elenco dei simboli}
4068 \copperdefstring{italian}{glossary}     {Glossario}

```

```

4069 \copperdefstring{italian}{advisor}      {Relatore}
4070 \copperdefstring{italian}{advisors}     {Relatori}
4071 \copperdefstring{italian}{dept}         {Programma}
4072 \copperdefstring{italian}{approvedmale} {Approvato da}
4073 \copperdefstring{italian}{approvedfemale} {Approvata da}
4074 \copperdefstring{italian}{approvedonmale} {Approvato il}
4075 \copperdefstring{italian}{approvedonfemale} {Approvata il}
4076 \copperdefstring{italian}{keywordsname} {Parole chiave}
4077 \copperdefstring{italian}{volumename} {Volume}
4078 \copperdefstring{italian}{coadvisor} {Correlatore}
4079 \copperdefstring{italian}{coadvisors} {Correlatori}
4080 \copperdefstring{italian}{ofname} {di}
4081 \copperdefstring{italian}{universityname} {Universidade Federal do Rio de Janeiro}
4082 \copperdefstring{italian}{cityname} {Rio de Janeiro}
4083 \copperdefstring{italian}{statename} {RJ}
4084 \copperdefstring{italian}{countryname} {Brasil}
4085 \copperdefstring{italian}{monthname1} {gennaio}
4086 \copperdefstring{italian}{monthname2} {febbraio}
4087 \copperdefstring{italian}{monthname3} {marzo}
4088 \copperdefstring{italian}{monthname4} {aprile}
4089 \copperdefstring{italian}{monthname5} {maggio}
4090 \copperdefstring{italian}{monthname6} {giugno}
4091 \copperdefstring{italian}{monthname7} {luglio}
4092 \copperdefstring{italian}{monthname8} {agosto}
4093 \copperdefstring{italian}{monthname9} {settembre}
4094 \copperdefstring{italian}{monthname10} {ottobre}
4095 \copperdefstring{italian}{monthname11} {novembre}
4096 \copperdefstring{italian}{monthname12} {dicembre}
4097 \copperdefstring{italian}{algorithm2eopt}{italiano}
4098 \copperdefstring{italian}{babelname} {italian}
4099 \AtBeginDocument{\DeclareLanguageMapping{italian}{italian-coppe}}
4100 </langit>
4101 <*numbbx>
4102 \ProvidesFile{coppe-numeric.bbx}[2026/09/09 v4.1 CoppeTeX ABNT numeric bibliography]
4103 % Numeric reference list for the ABNT numeric system (NBR 10520). It reuses the
4104 % full coppe ABNT style (drivers, CAPS surnames, bold titles, "In:", p./f.,
4105 % the PT->EN source map and the language mappings) and only adds the [n] list
4106 % label and the numeric sorting on top.
4107 \RequireBibliographyStyle{coppe}
4108
4109 % ABNT numeric system lists references in the order they are first cited
4110 % (NBR 10520), i.e. unsorted. Override with sorting=nty for an alphabetical
4111 % numbered list. 'labelnumber' generates the [n] field.
4112 \ExecuteBibliographyOptions{labelnumber,sorting=none}
4113
4114 % Bracketed labels, both in the list and for any shorthand entries.
4115 \DeclareFieldFormat{labelnumberwidth}{\mkbibbrackets{#1}}
4116 \DeclareFieldFormat{shorthandwidth}{\mkbibbrackets{#1}}
4117
4118 % Reference list driven by the [n] label (copied from biblatex's numeric.bbx),
4119 % keeping the ABNT blank line between entries via \bibitemsep set in coppe.bbx.
4120 \defbibenvironment{bibliography}
4121 {list
4122 {\printtext[labelnumberwidth]{%

```

```

4123         \printfield{labelprefix}%
4124         \printfield{labelnumber}}}%
4125     {\setlength{\labelwidth}{\labelnumberwidth}%
4126      \setlength{\leftmargin}{\labelwidth}%
4127      \setlength{\labelsep}{\biblabelsep}%
4128      \addtolength{\leftmargin}{\labelsep}%
4129      \setlength{\itemsep}{\bibitemsep}%
4130      \setlength{\parsep}{\bibparsep}}}%
4131     \renewcommand*{\makelabel}[1]{\hss#1}}
4132 {\endlist}
4133 {\item}
4134
4135 </numbbx>
4136 <*numcbx>
4137 \ProvidesFile{coppe-numeric.cbx}[2026/09/09 v4.1 CoppeTeX ABNT numeric citations]
4138 % Numeric citation system (ABNT NBR 10520): in-text references are bracketed
4139 % numbers [1], compressed and sorted within a single cite group. This is the
4140 % opt-in alternative to the author-date default (coppe.cbx); it pairs with
4141 % bibstyle=coppe-numeric. With the biblatex natbib option, \citep -> [1] and
4142 % \citet -> Author [1].
4143 \RequireCitationStyle{numeric-comp}
4144 </numcbx>

```

10.2 The bibliographic databases

The two BibLaTeX/biber databases shipped with the class — `example.bib` (the sample database cited by the example monograph) and `coppe.bib` (the references of this manual) — are generated from this source too, so the whole distribution still has a single editable origin. They live inside an `\iffalse` block below: the documentation driver never typesets them, and only `coppe.ins` extracts them, through the `examplebib` and `coppebib` docstrip targets. Edit the entries here, never the generated `.bib` files.

10.3 The generated companion files

Everything the distribution ships is extracted from this file: the class, the BibLaTeX styles, the language packs, the databases, the `.ist`, the demonstration documents and the `latexmkrc`. One `pdflatex coppe.ins` rebuilds all of it and there is no second place to edit. Each one sits in an `\iffalse` block: the documentation driver never typesets them, and only `coppe.ins` extracts them.

The line is drawn at what is DISTRIBUTED. What exists only to prove the class works stays outside, as ordinary files under version control:

`tests/` the regression suite. Nobody installing CoppeTeX needs it, and generating it from here would put proof and product in the same box.

`adversativa/` twelve documents — four kinds of work by three languages — each firing everything the class offers at once.

`tools/*.ps1` the build harness itself.

`NORMA_COPPE_2026.tex`, `futuremanual2026.tex` documents about the norm that live in this repository with their own life cycle.

Both suites are run by one command, `tools\prova.ps1`, and that is what has to come out clean before a version is tagged.

10.3.1 example_pt.tex -- pt-main demo

`example_pt.tex` – CoppeTeX 4.0 multilingual demo, main language: brazilian. Short demonstration document (cover + folha + abstracts + 1 chapter) used by the CPGP submission package. The full pt-main example with bibliography and lists is the historical `example.tex`.

10.3.2 example_en.tex -- en-main demo

`example_en.tex` – CoppeTeX 4.0 multilingual demo, main language: english.

10.3.3 example_es.tex -- es-main demo

`example_es.tex` – CoppeTeX 4.0 multilingual demo, main language: spanish. Requires the language pack files `coppe-lang-spanish.def` and `spanish-coppe.lbx` to be installed alongside `coppe.cls`.

10.3.4 example_fr.tex -- fr-main demo

`example_fr.tex` – CoppeTeX 4.0 multilingual demo, main language: french. Requires the language pack files `coppe-lang-french.def` and `french-coppe.lbx` to be installed alongside `coppe.cls`.

10.3.5 example_it.tex -- it-main demo

`example_it.tex` – CoppeTeX 4.0 multilingual demo, main language: italian. Requires the language pack files `coppe-lang-italian.def` and `italian-coppe.lbx` to be installed alongside `coppe.cls`.

10.3.6 example_pdfa.tex -- the full example as PDF/A

`example_pdfa.tex` – o `example.tex` completo, compilado com a opção ‘pdfa’.

Existe para dar ao veraPDF o documento mais exigente que a classe produz: o exemplo inteiro, com bibliografia, listas, figuras, tabelas, algoritmos e as caixas do `tcolorbox` – ou seja, transparência, que é justamente o que o a-1b proibiria e o a-2b admite. Validar só um documento curto não diria nada sobre o que vai de fato para o depósito.

`\PassOptionsToClass` tem de vir ANTES do `\documentclass` que o `example.tex` traz dentro de si; é o único jeito de acrescentar uma opção de classe a um arquivo gerado pelo docstrip sem duplicá-lo aqui e deixar as duas cópias divergirem com o tempo.

10.3.7 covers_5languages.tex -- the five covers side by side

`covers_5languages.tex` – One-page visual demonstration of the CoppeTeX 4.0 multilingual model: the five built-in main languages side by side, all sharing the same Portuguese institutional template, differing only in the title (and the body language, not visible on the cover).

This file uses the article class deliberately – it is a meta-document about the coppe class, not a thesis. It expects the five cover PNGs `cover_pt.png`, `cover_en.png`, `cover_es.png`, `cover_fr.png`, `cover_it.png` to live in the same directory (generated by `pdftoppm` from the `example_<lang>.pdf` demos).

10.3.8 latexmkrc -- latexmk/Overleaf build configuration

latexmkrc – CoppeTeX build configuration (works locally and on Overleaf).

Run with: `latexmk example` (or `latexmk futuremanual2026`, your thesis...) `latexmk` reads this file automatically when it is in the build directory. On Overleaf, place this file at the project root; Overleaf runs `latexmk`.

Engine: pdfLaTeX by default (fast). For LuaLaTeX (recommended, full Unicode) set `$pdf_mode = 4` below, or use the `"% !TeX program = lualatex"` magic comment / the Overleaf "Menu > Compiler" setting.

Acknowledgments

Thanks to all COPPETEX users who have reported their experience with this class. We also thank to professor Fernando Lizarralde and Heiko Oberdiek for their helpful comments. The authors would like to thank the National Council for Scientific and Technological Development (CNPq) of Brazil.

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Change History

v0.0		v3.1	
General: Creation Date.	1	<code>\department</code> : Included a sort key in <code>symb</code>	52
v0.1		v3.2	
General: Documentation:		General: Fixed version problem between <code>cls</code> and <code>dtx</code>	1
bibliography fixed, title translation.	1	v3.3	
Sourceforge submission.	1	General: Extend <code>abrev</code> to work like <code>symb</code> and accept a sorting key .	1
v0.2		v3.4	
General: Unification of the code for the list of symbols and abbreviations.	1	General: Some examples for figures, tables, longtables, etc. .	1
v0.3		v3.4.1	
General: Added 'draft' option. . . .	1	General: Fix english option to consider brazilian a second language, describe it in document	51
<code>\maketitle</code> : Added number of examiners test.	62	v3.5	
Generalization.	62	General: New command <code>annex</code> . .	98
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General: Beta documentation. . . .	1	General: New command <code>annex</code> now supports brazilian or english .	98
v0.5		v4.0	
<code>abstract</code> : Changed from macro to environment.	69	General: Catalog-label writers (<code>coppe@mainBegin</code> , <code>coppe@bibBegin</code> , <code>coppe@bibEnd</code> , <code>coppe@hasLof</code>) made language-agnostic: <code>detokenize</code> wrap on label names protects ':' and '.' from babel shorthand activation, and the raw <code>etex</code> primitives <code>romannumeral</code> / <code>number</code> replace LaTeX's <code>roman</code> / <code>arabic</code> to bypass <code>spanish-babel</code> 's <code>small-caps @roman</code> redefinition that would otherwise crash <code>pdftex</code> with "Unbalanced write command" on every <code>es-main</code> document.	1
<code>\backmatter</code> : Added <code>mainmatter</code> pages counter.	58	Class-wide string dispatcher (<code>copperdefstring</code> , <code>coppestring</code> , <code>coppemainstring</code> , <code>coppeforeignstring</code>) replaces scattered <code>iflanguage</code> and if english branches; <code>pt/en</code> outputs remain byte-identical. .	1
<code>foreignabstract</code> : Changed from macro to environment.	70	Multilingual model: three fixed slots (<code>main</code> , <code>foreign</code> , <code>optional</code> <code>third</code>); five built-in language options (<code>brazilian</code> , <code>english</code> ,	
v1.0			
General: First COPPETeX release. .	1		
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General: COPPETeX release 2.0. . .	1		
v2.1			
General: COPPETeX release 2.1:			
Matching the new rules.	1		
<code>\advisor</code> : Advisors, co-advisors, co-co-advisors, etc., all of them are simply considered advisors. .	54		
v2.2			
General: Matching new guidelines, including new logo.	1		
v2.2.2			
General: Fixed some text constants in <code>.bib</code> and documented it here. .	1		
v3.0			
General: Added support for monographs in English.	51		
New approval page layout. . . .	64		
<code>\department</code> : Added new course on Nanotechnology.	52		
<code>\examiner</code> : Examiners expansion without degree.	54		

spanish, french, italian); plug-in mechanism for any other Babel language via coppe-lang-lang.def plus lang-coppe.lbx.	1	doublespacing, no folio is printed on a pre-textual sheet when something does overflow, the approval sheet holds a board of eight, and es-nosectiondot removes the period babel-spanish put in the sumario indicative.	1
New user API: titlein, usecoppelanguage, brazilianabstract environment. .	1		
\coppetexfinalpage: Initial coppetexfinalpage colophon page.	59	New options: pdfa (PDF/A-2b via pdfx, XMP built from the document), assinaturas, coorientador, rascunhoficha, listasnosumario, morewrites/semmorewrites. New command coadvisor. Under pdfTeX the class loads morewrites by itself when installed, because a work that asks for every list needs seventeen of TeX's sixteen output streams.	1
v4.1 General: Conformance with the revised 9th edition (2026) of the UFRJ/SiBI manual: continuous pagination from the folha de rosto, folio in 10pt at 2cm from the top and right edges, one-sided layout with a 3cm left margin, additional sheet with the catalog card and the CAPES collection fields, approval sheet of 3.1.2.1.3, keywords closing the three abstracts, pre-textual lists out of the sumario, Apendice and Anexo headings centred, fifth heading level, and Latin Modern instead of the bitmap fonts.	1	Single source: pdflatex coppe.ins generates everything distributed – class, biblatex styles, language packs, .bib bases, .ist, the five per-language examples, example_pdfa, the cover montage and latexmkrc. The regression suite, the adversarial documents and the harness prove the class and are NOT distributed.	1
Fixes found by the adversarial review: cover and folha de rosto compose in single spacing so they do not overflow under			